

Snapper Documentation

Comprehensive project guide

Generated on 2026-06-07

Table of Contents

[Snapper](#)

[AI Integration](#)

[API](#)

[Architecture](#)

[Backtesting](#)

[CLI](#)

[Configuration](#)

[Development](#)

[Messaging \(ZeroMQ\)](#)

[Observability](#)

[Operations runbook — multi-instance trade coordinator](#)

[snapper-egress sidecar — operator runbook](#)

[Trading Strategies](#)

Snapper

Trading platform with market data collection, trade runtime, and backtester. Supports Kraken (WebSocket; spot + futures + equities), Waluomat (REST polling), and Polygon.io (REST) market data.

Quick Steps

```
# One-time: install frontend deps and build the static UI so the
# FastAPI server can serve the dashboard at `/.`.
make ui-setup
make ui-build

# Initialize the database + seed dev data, then refresh the verified
# symbol mappings and market snapshots so the dashboard has data to show.
make migrate-dev run-static

# Start the server
make run-server
```

Open <http://localhost:8000/> and log in with the dev seed credentials — the default values are defined in the bundled seed file `src/snapper/data/seed/dev.toml` (admin / change-me-after-first-login); maintainers with the proprietary submodule get `proprietary/data/seed/dev.toml` via the three-tier seed lookup. Override per-environment by editing the seed file before running `make migrate-dev`, or rotate after first login via `POST /api/auth/users/{user_id}/change-password` (self-service) or `POST /api/auth/users/{user_id}/admin-reset-password` (admin reset).

Features

- **Market data collection** — Kraken (WebSocket; spot + futures + equities), Waluomat (REST polling), Polygon.io (REST)
- **Symbol correlation** — Underlying asset model linking instruments across exchanges (e.g., SPY, ESM6-CME, SPYX-USD-PERP all map to S&P 500). YAML-driven pattern matching, front-month rollover, contract ladder API
- **Trade runtime** — Facts-canonical live and paper trading with canonical Order and Execution facts, rebuildable Position and Balance projections, dispatched via a durable outbox
- **Strategies** — Framework for creating strategies based on RSI, MACD, cointegration, and TA-Lib indicators
- **ZeroMQ messaging** — Pub/sub architecture for market data and signals

- **Provenance and audit** — Every payload item carries `session_id` and `sequence_id` for gap detection. Control table (always-on) and telemetry table (toggleable) record all commands and operational events
- **Web dashboard** — FastAPI + React with real-time WebSocket
- **CLI** — Full system management via Typer CLI
- **Backtesting** — Strategy testing on historical data

Quick Start

Requirements

- Python 3.14+
- Poetry
- Node.js 26+ and pnpm 11+ (for frontend)
- Build tools for Python packages. The TA-Lib adapter uses TA-Lib when importable and falls back to pure Python indicators otherwise; install the native TA-Lib C library only if your local wheel build requires it.

Installation

```
# Clone repository
git clone https://github.com/mateusz-klatt/snapper.git
cd snapper

# Install system dependencies
make system-deps

# Install Python dependencies
make setup

# Install frontend dependencies
make ui-setup

# Copy and configure environment variables
cp .env.example .env
```

Configuration

Edit `.env` file:

```
# Database (SQLite dev, PostgreSQL prod)
DB_URL=sqlite+aiosqlite:///./data/snapper.db

# Settings encryption in database
MASTER_PASSWORD=your_master_password
```

```
# HTTP Server
SERVER_HOST=0.0.0.0
SERVER_PORT=8000
SERVER_RELOAD=false
SERVER_API_ONLY=false
SERVER_PROXY_HEADERS=true
SERVER_FORWARDED_ALLOW_IPS=127.0.0.1,172.17.0.1

# Telemetry recording (pings, heartbeats, GET reads)
TELEMETRY_RECORDING_ENABLED=false

# ZeroMQ broker
ZMQ_BROKER_XSUB=tcp://127.0.0.1:7500
ZMQ_BROKER_XPUB=tcp://127.0.0.1:7501
```

Running

```
# Initialize database and seed data
make migrate-dev

# Start server
snapper server
```

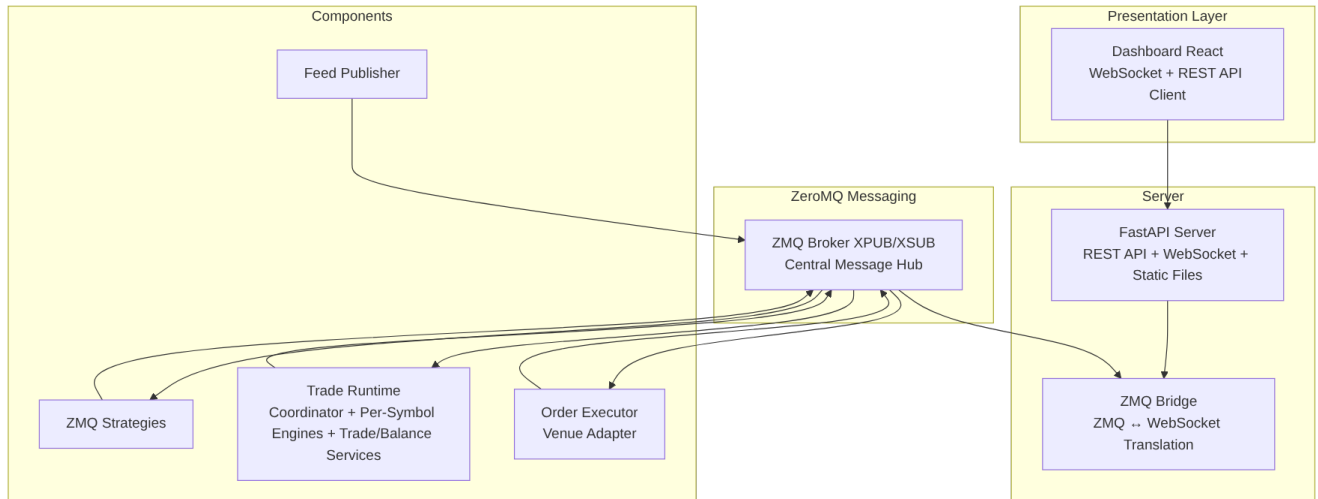
Two ways to bring up the dashboard:

- **Backend-served (production / single port).** Run `make ui-setup`
- `make ui-build` to produce `frontend/dist/`, then start the server. The FastAPI app mounts the static UI at `/` only when `frontend/dist/` exists; the dashboard is then at `http://localhost:8000/`.
- **Vite dev server (hot reload).** Run `make ui-dev` in a separate terminal — Vite serves the dashboard at `http://localhost:3000/` with a backend proxy for `/api` and `/api/ws`. This does NOT produce `frontend/dist/`, so `http://localhost:8000/` will still return 404 for the UI until the build step is run.

Trade Runtime Dispatch

The trade runtime always uses durable, outbox-driven dispatch: commands are persisted as `TradeCommand` rows and published by the outbox dispatcher. On the `accepted/fill` paths, executor `VenueEvent` persistence is fail-closed — a failed write raises before the corresponding `orders.events.*` publish.

System Overview



CLI Commands

Server and Infrastructure

```
snapper server          # Start FastAPI server
snapper broker          # Start ZMQ broker
snapper trade-zmq      # Trade runtime / coordinator (pass --instance-id +
--instance-count for N>=2, see docs/operations.md)
snapper executor       # Order executor
snapper feed           # Market data publisher
snapper feed-engine    # Dedicated feed container endpoint
snapper egress         # WireGuard + SOCKS5 egress sidecar endpoint
```

Database

```
snapper db-init        # Initialize schema
snapper db-upgrade     # Alembic migrations
snapper db-downgrade   # Rollback migrations
```

Users

```
snapper init-admin     # Create admin user
snapper list-users     # List users
snapper reset-password # Reset password
```

Local AI delegate PAT (for @mateusz-klatt/snapper-mcp bridge)

After make dev-backend is running and the seed has provisioned the admin user (per dev.toml), mint a long-lived AI delegate PAT into a JSON file the bridge consumes via --config=PATH:

```
snapper dev-mint-pat          # writes data/dev-pat.json (mode 0600)
```

Wire your ~/.claude.json mcpServers entry once with the file path:

```
"snapper-local": {
  "command": "node",
  "args": [
    "/path/to/snapper-mcp/dist/index.js",
    "--config=/path/to/snapper/data/dev-pat.json"
  ]
}
```

After running, in this order, (a) rm data/snapper.db, (b) make migrate-dev, (c) make dev-backend (in a separate terminal — it blocks the foreground with the live backend), and (d) snapper dev-mint-pat, the bridge picks up the refreshed token automatically — no manual edits to ~/.claude.json per DB wipe. CLI flags + environment overrides:

--base-url URL	SNAPPER_DEV_BASE_URL	(default http://localhost:8000)
--admin-username USER	SNAPPER_DEV_ADMIN_USERNAME	(default None → resolved from seed profiles, mcp then dev)
--admin-password PASS	SNAPPER_DEV_ADMIN_PASSWORD	(default None → resolved from seed profiles, mcp then dev)
--output PATH	SNAPPER_DEV_PAT_OUTPUT	(default data/dev-pat.json)
--label LABEL		(default "Local Dev MCP")

The script drives production REST endpoints (POST /api/auth/login + POST /api/ai-delegates) so the seeded delegate exercises the same code path the AI Integration UI uses. Tokens never appear in stderr — HTTP error bodies are passed through a JWT-shape redaction helper before printing.

Market Data

```
snapper update-kraken-symbols          # Sync Kraken symbols
snapper update-kraken-futures-symbols
snapper update-kraken-equities-symbols
snapper update-walutomat-symbols
snapper update-polygon-symbols         # Sync Polygon symbols
snapper update-underlyings             # Sync underlying asset mappings from YAML
snapper update-kraken-market-snapshot
snapper update-kraken-futures-market-snapshot
snapper update-kraken-equities-market-snapshot
snapper update-walutomat-market-snapshot
```

```

snapper polygon-backfill-aggregates
# Step 1: download history to CSV cache (no DB write)
snapper polygon-load-csv --all          # Step 2: load the CSV cache into the
database
snapper kraken-futures-backfill-candles
snapper kraken-equities-backfill-candles
snapper update-kraken-futures-funding-rates
snapper archive --day 2024-01-15      # Export candle cache to CSV
snapper archive --from 2024-01-01 --to 2024-01-31 --exchange polygon
snapper archive --table ticks --day 2024-01-15 --exchange polygon
snapper archive --table candles-audit --day 2024-01-15 --closed-only --purge

```

See [docs/cli.md](#) for full options and examples.

Strategies

Creating your own strategy:

```

from snapper.strategies.base import BaseStrategy, StrategySignal, StrategyConfig
from snapper.strategies.decorators import register_strategy,
create_strategy_process
from snapper.messaging.schemas.data import CandleData

@register_strategy("MyStrategy")
@create_strategy_process(
    process_name="my_strategy",
    default_config={
        "name": "my_strategy",
        "inputs": ["market.kraken.BTC-USD.candles.1h"],
        "outputs": ["BTC-USD"],
        "exchange": "paper",
        "params": {"threshold": 0.5},
    },
)
class MyStrategy(BaseStrategy):
    def __init__(self, config: StrategyConfig) -> None:
        super().__init__(config)
        self.threshold = self.params.get("threshold", 0.5)

    async def reset(self) -> None:
        """Clear any per-instrument indicator/state buffers."""

    async def on_candle(self, instrument: str, candle: CandleData) ->
StrategySignal | None:
        if some_condition:
            return StrategySignal(
                instrument=instrument,
                side="buy",
                strength=1.0,
                price=candle.close,
                reason="Condition met",
            )
        return None

```

reset() is abstract on BaseStrategy — every subclass must implement it so the runtime can recycle per-instrument state on restart / replay.

Development

Quality Gates

```
make check-all    # Full quality gate (backend + frontend + tests + 100%  
coverage)  
make fix-all      # Automatic formatting and linting fixes
```

Individual Steps

```
make fmt          # Check formatting  
make lint         # Linting (ruff)  
make typecheck    # Type checking (mypy)  
make test         # Unit tests  
make cov          # Tests with coverage (100% required)
```

Frontend

```
make ui-dev       # Vite development server  
make ui-build     # Production build  
make ui-lint      # ESLint  
make ui-format    # Prettier
```

Docker

```
make docker-build-dev    # Build dev image (backend + caddy binary baked in)  
make docker-build-prod   # Build production image  
make docker-migrate-dev  # Initialize and seed the Docker SQLite database  
make docker-run          # Run container  
make docker-stop         # Stop containers
```

Or with docker compose:

```
make docker-migrate-dev  
docker compose up -d
```

Compose topology

The compose stack runs four application services on the internal snapper-internal bridge network, all from a **single image** (klattm/snapper:latest) that bundles the python runtime + the Caddy binary. Each service overrides entrypoint / command to launch the right process. The optional postgres service is gated behind the dev compose profile.

- snapper — FastAPI backend, ZMQ broker, strategy/process control, and non-publisher runtime processes. PROCESS_AUTOSTART_PROFILE=api, command: ["server"], expose: 8000/7500/7501 internally.
- snapper-feed — dedicated market-data publisher container. PROCESS_AUTOSTART_PROFILE=feed, command: ["feed-engine"]; it connects to the backend broker via tcp://snapper:7500/7501 and uses reduced DB pool settings so publisher subprocesses do not overrun PostgreSQL connection limits.
- snapper-egress — WireGuard + SOCKS5 sidecar for outbound publisher traffic. command: ["egress"], cap_add: NET_ADMIN, kernel module bind.
- snapper-web — Caddy serving the React SPA from /srv/dist and reverse-proxying /api/*, /api/ws, /api/mcp, /docs, /redoc, /openapi.json to snapper:8000. entrypoint: ["/usr/bin/caddy"], command: ["run", "--config", "/etc/caddy/Caddyfile"]. Owns the host 127.0.0.1:8000:8000 bind.

All four application services are restart: unless-stopped so they come back automatically after a host reboot (assuming systemctl is-enabled docker returns enabled). Services explicitly stopped via docker compose stop stay stopped.

Targeted restarts (no tick-stream drop)

Restart targets are now strictly recreate-only — they use whatever image tag exists locally. Rebuild explicitly first if you need new code baked in:

```
# Build (explicit, when you want new code in the image):
make docker-build-dev      # caller UID (host user)
make docker-build-prod     # UID 888 (matches log file ownership)

# Recreate containers (uses current image, no rebuild):
make restart-frontend      # Recreate Caddy sidecar [does NOT touch backend]
make restart-backend       # Recreate backend [drops ticks 30-90s]
make restart-all          # Recreate full stack [drops ticks]

# Common idiom — build + restart in one shot:
make docker-build-prod restart-backend
```

make restart-frontend is the fast path for shipping UI changes without restarting publishers. Only the Caddy container is recreated; WS clients reconnect within ~3 s but the backend snapper container's PID is unchanged and the Kraken / Walutomat tick streams continue uninterrupted.

Footgun: a plain docker compose up -d (without --no-deps) follows depends_on edges and may recreate the backend too. Use the targeted Makefile commands for selective restarts, and reserve docker compose up -d for full-stack restarts when both should restart anyway.

Documentation

Detailed documentation in [docs/](#) directory:

- [Architecture](#) — System structure and layers
- [Configuration](#) — Environment variables and settings
- [CLI](#) — Full command documentation
- [Strategies](#) — Creating trading strategies
- [API](#) — REST API and WebSocket
- [Messaging](#) — ZeroMQ architecture
- [Development](#) — Developer guidelines (incl. [Internationalization](#))

License

MIT License — see [LICENSE](#) file.

AI Integration

Snapper exposes a **Model Context Protocol (MCP)** endpoint at `/api/mcp` so Claude Desktop, Cursor, Windsurf, or a plain curl client can authenticate with a long-lived AI delegate access JWT and call a narrow set of trading + market-data tools. The surface is vendor-neutral — no Anthropic or OpenAI SDK is bundled in Snapper core.

This doc covers:

1. [Feature flag](#)
 2. [Creating an AI delegate](#)
 3. [Token model](#)
 4. [Client configuration examples](#)
 5. [Available tools](#)
 6. [Safety caps](#)
 7. [Kill switch + deactivation](#)
 8. [Rate limits](#)
 9. [Error catalog](#)
 10. [Security model](#)
-

Feature flag

The `/api/mcp` sub-app is always mounted and gated by the `ai_integration_enabled` database setting.

1. The flag **defaults to true** — a fresh install exposes the MCP endpoint and the AI Integration navigation entry with no manual setup. Admins who need to disable the feature flip the setting to false via the Settings UI (admin) or directly:

```
curl -X POST http://localhost:8000/api/settings/ai_integration_enabled/set \
-H "Authorization: Bearer <admin-jwt>" \
-H "Content-Type: application/json" \
-d '{"session_id":"cli","sequence_id":1,"public_id":"$(uuidgen)","timestamp":"2026-04-20T00:00:00Z","payload":{"value":"false","category":"system"}}'
```

2. The frontend reads `GET /api/settings/features` on mount (no auth required) and surfaces the "AI Integration" navigation entry whenever the flag is on. The MCP endpoint returns 503 `feature_disabled` only when the flag is explicitly set to false.
 3. Whether the flag is on or off, the MCP endpoint requires every request to carry a valid `Authorization: Bearer <jwt>` header. Anonymous requests receive 401 `missing_bearer_token`.
-

Creating an AI delegate

An **AI delegate** is a dedicated AI_DELEGATE-role user an operator mints per MCP client. Delegates:

- Cannot log in via the web UI password form (the placeholder password hash is opaque to humans).
- Cannot see other delegates, operators, or wallets — their wallet scope comes from the operator they are bound to.
- Carry per-delegate safety caps independent of the operating operator's caps.

Create one via POST /api/ai-delegates:

```
curl -X POST http://localhost:8000/api/ai-delegates \
-H "Authorization: Bearer <operator-jwt>" \
-H "Content-Type: application/json" \
-H "X-CSRF-Token: <csrf>" \
-d '{
  "session_id": "cli",
  "sequence_id": 1,
  "public_id": "'$(uuidgen)'",
  "timestamp": "2026-04-20T00:00:00Z",
  "payload": {
    "label": "Claude Desktop",
    "operator_public_id": "<operator-public-id>",
    "caps": {
      "max_open_orders": 3,
      "max_daily_notional_usd": 1000.0,
      "max_cancels_per_minute": 10
    }
  }
}'
```

payload.operator_public_id is optional. When omitted, Snapper binds the delegate to the caller's primary_operator_public_id; when supplied, it must be one of the caller's authenticated operator claims. The minted delegate token inherits scope from that bound operator, so later scope grant changes for that operator control which wallets/instruments the delegate can act on.

The response is **one-shot**. Copy the tokens out of the HTTP session immediately — Snapper never re-serves them:

```
{
  "type": "delegate_created_response",
  "sequence_id": 1,
  "public_id": "<envelope-uuid7>",
  "timestamp": "2026-04-20T00:00:00Z",
  "session_id": "<server-session>",
  "topic": null,
  "payload": {
    "delegate": {
      "public_id": "019da9e...",

```

```
    "username": "ai-clausedesktop-a1b2c3",
    "label": "clausedesktop",
    "created_by_user_public_id": "<operator-id>",
    "created_at": "2026-04-20T00:00:00Z",
    "is_active": true,
    "caps": { "max_open_orders": 3, "max_daily_notional_usd": 1000.0,
"max_cancels_per_minute": 10 }
  },
  "access_token": "<jwt-with-90d-exp>",
  "expires_in": 7776000
}
```

Other endpoints on /api/ai-delegates:

- GET /api/ai-delegates — list the caller's delegates (no tokens re-served).
 - GET /api/ai-delegates/{id} — single delegate detail.
 - PATCH /api/ai-delegates/{id} — update caps (SCD2 close+insert). label/username are immutable post-mint.
 - POST /api/ai-delegates/{id}/deactivate — kill switch. Publishes admin.user_deactivated on the bus; every Snapper instance disconnects matching WebSocket sessions and evicts the token from every LRU within one bus round-trip.
-

Token model

Each AI delegate mints a single long-lived (~3-month) access JWT. The same token authenticates both the proxy MCP server and the optional push-wakeup watch monitor. Revocation works instantly: POST /api/ai-delegates/{id}/deactivate flips users.is_active=False, publishes admin.user_deactivated on the bus, and every Snapper instance evicts the delegate's user_active_tokens row from the verify-cache within one bus round-trip. The 90-day exp is a ceiling, not a commitment; operators are expected to rotate delegate tokens on the cadence that fits their key-management hygiene.

Client configuration examples

Claude Code plugin (recommended)

In any Claude Code session:

```
/plugin marketplace add mateusz-klatt/snapper-mcp
/plugin install snapper-mcp@mateusz-klatt-snapper-mcp
/reload-plugins
```

Claude Code prompts for two required values at install time (per the plugin's userConfig schema in `.claude-plugin/plugin.json`):

- **Snapper API URL** -- your backend's `/api/mcp` endpoint. The `@mateusz-klatt/snapper-mcp` bridge accepts the value with or without a trailing slash and normalizes it before connecting.
- **Access token** -- the `access_token` from the `delegate_created` response (or paste from the Settings -> AI Delegates config-snippet generator). Delegates are PAT-style: the JWT lifetime is ~3 months (`LONG_LIVED_TOKEN_EXPIRE_DAYS = 90`); operators are expected to rotate by minting a fresh delegate via the same flow before the existing one expires. Revocation is immediate: deactivating the delegate in Snapper invalidates the associated `user_active_tokens` row server-side within one bus round-trip.

Run `/mcp list` to confirm the snapper server is connected. The plugin pins the runtime to a specific `@mateusz-klatt/snapper-mcp` version — the manifest hardcodes the exact version string in `mcpServers.snapper.args` (currently `@0.11.0`), kept in lockstep with the plugin's own version field by the `integrations/snapper-mcp/test/plugin_manifest.test.ts` parity test so a bump to either side fails CI until both match. Future runtime publishes do not silently upgrade existing installs; `/plugin update snapper-mcp` opts in. Sensitive values land in the OS keychain (with `~/.claude/.credentials.json` fallback) -- never in `settings.json` or the plugin manifest.

Claude Desktop

Add to `~/Library/Application Support/Claude/claude_desktop_config.json`:

```
{
  "mcpServers": {
    "snapper": {
      "command": "npx",
      "args": ["-y", "@mateusz-klatt/snapper-mcp"],
      "env": {
        "SNAPPER_BASE_URL": "https://snapper.example.com/api/mcp",
        "SNAPPER_ACCESS_TOKEN": "<copied-from-create-response>"
      }
    }
  }
}
```

The `@mateusz-klatt/snapper-mcp` npm wrapper presents the access token as a Bearer header on every request. On 401 the bridge surfaces the auth failure to the MCP host once per session; recovery is via recreating the AI delegate in Snapper.

Cursor

In `~/.cursor/config.json`:

```
{
  "mcpServers": {
    "snapper": {
      "url": "https://snapper.example.com/api/mcp/",
      "headers": {
        "Authorization": "Bearer <access-token>"
      }
    }
  }
}
```

The delegate access token Cursor sends as a Bearer header is a long-lived PAT JWT (~3 months), so day-to-day operation does not require token rotation more often than the 90-day expiry. Revocation is by deactivating the delegate in Snapper (which invalidates the underlying user_active_tokens row); recovery is to recreate the delegate and update Cursor's Authorization header with the new token.

Windsurf

~/codeium/windsurf/mcp_config.json:

```
{
  "mcpServers": {
    "snapper": {
      "serverUrl": "https://snapper.example.com/api/mcp/",
      "auth": { "type": "bearer", "token": "<access-token>" }
    }
  }
}
```

curl (diagnostics)

Quick reachability check:

```
# Feature flag
curl http://localhost:8000/api/settings/features

# Hello-world MCP initialise
curl -X POST http://localhost:8000/api/mcp/ \
  -H "Authorization: Bearer <access-token>" \
  -H "Content-Type: application/json" \
  -H "Accept: application/json, text/event-stream" \
  -d '{"jsonrpc": "2.0", "id": 1, "method": "initialize", "params": {
    "protocolVersion": "2024-11-05", "capabilities": {}, "clientInfo": {
      "name": "curl", "version": "8"
    }
  }'}
```

Available tools

Read-only tools surface the delegate's order book, position state, and venue market data. Write tools (`submit_manual_order`, `cancel_order`) are gated by per-tool permissions; only `submit_manual_order` is unconditionally guarded by `TradingCapsEnforcer.guard` — `cancel_order` invokes the guard only on the cancel paths that affect open exposure, and `submit_ai_review_decision` is not wired to `caps_enforcer_getter` at registration.

- **`list_instruments(exchange: str)`** — returns sorted native symbols for the exchange inventory. It is not wallet/operator scoped; read-only access is gated by `READ_MARKET_DATA` permission (`AI_DELEGATE` role satisfies).
- **`list_orders(wallet_public_id?, status?, exchange?, instrument?, limit=50, offset=0)`** — paged read of the delegate's order history within accessible wallets. Requires `READ_ORDERS`. Surfaces `order_not_found` for inaccessible wallets (anti-enumeration).
- **`get_order_status(command_public_id: str)`** — single-order lookup keyed by the trade-command public id returned from `submit_manual_order`. Requires `READ_ORDERS`.
- **`list_positions(wallet_public_id?, exchange?, instrument?)`** — active positions across the caller's accessible wallets. Requires `READ_POSITIONS`.
- **`get_position_cycle(cycle_public_id: str)`** — full open→close audit trail for a position cycle. Requires `READ_POSITIONS`.
- **`get_ohlcv(exchange, instrument, timeframe, since?, until?, limit=200)`** — OHLCV candles for a venue + instrument. Range mode (both `since` + `until` ISO 8601 UTC) returns chronological candles in the closed window; latest-as-of mode (both omitted) returns the most recent limit candles in DESC order. Allowed timeframes: 1m, 5m, 15m, 1h, 4h, 1d. Requires `READ_MARKET_DATA`; market data is public so no wallet scope applies.
- **`list_recent_signals(since, instrument?, strategy?, exchange?, wallet_public_id?, limit=50)`** — strategy signals fired after a watermark, ordered by `fired_at` DESC. `since` is REQUIRED ISO 8601 UTC. Requires `READ_SIGNALS`; surfaces `signal_not_found` on wallet scope violation (anti-enumeration).
- **`submit_manual_order(exchange, instrument, instrument_public_id, side, order_type, quantity, wallet_public_id, idempotency_key, price?, operator_public_id?, ai_review_public_id?)`** — enqueues a trade command under the delegate's `user_public_id` with `source_surface='mcp'` + the caps check from `TradingCapsEnforcer.guard`. Fails closed on any cap violation. Wraps the REST `create_order` route.

- **cancel_order(plan_public_id, idempotency_key)** — cancels an active execution plan via the same PlansCancelService REST goes through. Idempotent: same idempotency_key on retry returns the current plan state without re-executing.
- **submit_ai_review_decision(review_id, decision, rationale?)** — REST mirror of the POST /api/ai-reviews/{review_public_id}/decision route for the in-process MCP surface; lets the delegate approve or reject a pending CONSULT review. review_id is the UUID7 of the ai_reviews row. Not gated by the caps enforcer (the underlying review-decision path does its own SCD2 close-and-insert + bus fanout).

The tool catalog is discoverable via the MCP tools/list JSON-RPC method:

```
curl -X POST http://localhost:8000/api/mcp/ \
-H "Authorization: Bearer <access-token>" \
-H "Content-Type: application/json" \
-H "Accept: application/json, text/event-stream" \
-d '{"jsonrpc": "2.0", "id": 2, "method": "tools/list", "params": {}}'
```

Safety caps

Every delegate has its own user_trading_caps row. All fields are optional; null means "unbounded on this axis".

- max_order_quantity_per_instrument — JSON dict {instrument: max_qty} OR a scalar applied to every instrument.
- max_open_orders — all-time count of the delegate's in-flight commands (statuses in created/dispatched/acked/accepted/ partially_filled).
- max_daily_notional_usd — rolling 24h sum of submit_quantity * submit_price_usd across non-rejected commands (submit-time commitment basis; partial fills don't change accounting).
- max_cancels_per_minute — sliding-window cancel rate.

Caps are enforced at trade-command insert sites via the TradingCapsEnforcer.guard() surface — submit_manual_order routes through it unconditionally; cancel_order invokes the guard only on the cancel paths that affect open exposure (the bare-cancel fast path skips it). The other MCP tools (submit_ai_review_decision, etc.) are not currently wired to caps_enforcer_getter at registration.

Update caps via PATCH /api/ai-delegates/{id} — the write is SCD2 close+insert, so cap history is auditable.

Kill switch + deactivation

POST /api/ai-delegates/{id}/deactivate runs the same code path as operator deactivation:

1. SCD2 close+insert on the users row flipping is_active=False.
2. TokenManager.revoke_user_sessions loads every user_active_tokens.jti for the delegate, flips revoked_at=NOW() in one SQL UPDATE, and seeds the in-memory JTI blacklist with a 10-second grace period.
3. UserService.deactivate_user publishes admin.user_deactivated on the bus (sole publisher).
4. Every Snapper instance's WebSocketAuthManager subscribes to the topic and closes matching WebSocket connections with code 4003.
5. Every Snapper instance's TokenManager subscribes to the topic and evicts matching verify-cache entries via invalidate_user_cache.

Latency: same-instance kill is immediate (JTI blacklist consulted before LRU in verify_token_with_db); cross-instance kill is bounded by one bus-message round-trip (sub-second on local ZMQ) instead of the 30-second LRU TTL.

In-flight MCP tool handlers are **not** force-cancelled. The guarantee is narrowly "subsequent MCP requests are rejected once the JTI blacklist propagates" — the in-memory blacklist carries a small grace period (so requests that raced the propagation window can still complete) before subsequent verifies fail.

Rate limits

REST rate limits apply globally via RestCallTracker — observed via GET /api/metrics/rest-rate. MCP-specific rate limits are enforced by a separate per-principal middleware on /api/mcp (default 60/minute); the per-delegate max_cancels_per_minute cap then applies on top inside cancel_order. Bursts are served best-effort; sustained abuse over published exchange limits (Walutomat 20 req/s, Kraken 15 req/s, Polygon 5/min) surfaces as rate_limited warnings at 80/95% utilisation.

Error catalog

Transport and middleware failures on /api/mcp use standard HTTP status codes plus a Snapper-specific error_code in the JSON body. Clients should branch on error_code, not on status text.

Envelope-based MCP tools return a normal MCP CallToolResult instead: the single TextContent.text entry contains JSON with success, error_code, message, and details, and CallToolResult.isError mirrors not success. A few older tools still return raw dictionaries on success or surface FastMCP ToolError/permission exceptions on failure; client code should handle both shapes until those tools are migrated.

Status	error_code	When it fires	Client action
503	feature_disabled	Flag explicitly set to false	Show "AI Integration disabled" banner
429	rate_limit_exceeded	Per-principal MCP middleware quota exhausted	Back off and retry after Retry-After seconds
503	mcp_unavailable	Repository dep unavailable (lifespan not ready)	Retry with backoff
401	missing_bearer_token	No Authorization: Bearer ... header	Prompt user to authenticate
401	invalid_bearer_token	JWT signature/expiry/blacklist/inventory failure	AI delegates have no refresh token (90-day PAT); deactivate + recreate the delegate in Snapper, then update the client's bearer token. Operator (cookie) sessions can fall back to POST /api/auth/refresh.
401	user_deactivated	Owner account deactivated	Prompt re-login; don't auto-refresh
401	Refresh token redeemed	Replay of a spent refresh JWT	Re-login

Status	error_code	When it fires	Client action
401	Account deactivated	Session cookie flow	Re-login
403	MCP write helpers: wallet_out_of_scope: (<i>prefixed message</i>) / REST: "Wallet not in accessible set" (<i>detail string</i>)	Mutating tool targets a wallet outside the caller's scope. The helper path raises <code>PermissionError(f"wallet_out_of_scope: ...")</code> lifted by FastMCP into a <code>ToolError</code> ; REST raises <code>HTTPException(403, detail="Wallet not in accessible set")</code> from <code>server/scoping.py</code> . Tool-level read/cancel paths may instead return structured anti-enumeration envelopes such as <code>order_not_found</code> , <code>position_not_found</code> , or <code>signal_not_found</code> .	Pick a wallet the caller still has a live grant on
403	MCP write helpers: operator_out_of_scope: (<i>prefixed message</i>) / REST: "Operator not in accessible set" (<i>detail string</i>)	Same write-helper shape as the wallet variant. Read/cancel tools can intentionally collapse out-of-scope and not-found cases into structured not-found envelopes to avoid leaking resource existence.	Pick an operator from the caller's authenticated set

Delegate CRUD:

Status	Detail	When it fires
401	Requires populated <code>user_public_id</code>	Principal has blank <code>user_public_id</code> (legacy token rollout)
403	<code>require_role(OPERATOR)</code>	AI_DELEGATE or VIEWER trying to manage delegates
404	Delegate not found	Unknown ID OR cross-tenant (no existence leak)
409	Could not derive a unique username ...	Label slug collides 8+ times (pathological)
422	Operator '<id>' is not in ...	Caller picked <code>operator_public_id</code> outside their claim set
422	Caller has no primary operator ...	No explicit operator and no primary → binding is ambiguous

Security model

Token lifetime

- **Operator** access tokens live **15 minutes** (configurable via `auth_access_token_expire_minutes`). **AI delegate** access tokens are PAT-style and live for `LONG_LIVED_TOKEN_EXPIRE_DAYS` (~3 months / 90 days); operators rotate them on the cadence that fits their key-management hygiene, and revocation is immediate via deactivating the delegate rather than waiting for expiry.
- Refresh tokens live **7 days** (30 days with `remember_me=true`).
- Refresh rotation is **atomic**: one DB transaction revokes the old refresh JTI AND persists the new access+refresh pair. Replay of a spent refresh JWT fails with 401 Refresh token already redeemed — the transaction rolls back cleanly.

Inventory-based revocation

`user_active_tokens` is the ground truth for "is this JWT still valid?". Every mint persists a row; every verify consults it via a 30-second LRU. Deactivation flips `revoked_at=NOW()` in one UPDATE and fans out a bus event so every cross-instance LRU evicts the matching entry on receipt.

No secrets leaked at rest

Delegate password hashes are bcrypt-derived from 32 bytes of cryptographically random data the operator never sees. An attacker with DB read rights cannot brute-force the hash.

Single-publisher invariant

`UserService.deactivate_user` is the SOLE publisher of `admin.user_deactivated` across the entire process. `TokenManager` and `WebSocketAuthManager` are pure subscribers — they evict + close on receipt but never emit the event themselves. This holds for operator deactivation AND delegate deactivation (the route reuses `UserService.deactivate_user`).

MCP transport = Streamable HTTP

The MCP endpoint uses Model Context Protocol's Streamable HTTP transport. No cookies, no CSRF on MCP calls (bearer header bypass). Every tool call opens a sub-request inside the same bearer-authenticated HTTP connection.

Related reading

- `docs/architecture.md` — repository + SCD2 + bus layering.

- docs/operations.md — multi-instance trade coordinator and static-hash shard partitioning.
- docs/api.md — full REST surface.

API

Snapper provides a REST API and WebSocket interface for platform interaction. The API accepts JWT authentication either through HTTP-only cookies or an Authorization: Bearer <jwt> header. When both are present, Bearer auth wins. Cookie-authenticated mutating requests require a valid X-CSRF-Token header; Bearer-authenticated requests skip CSRF because they are not ambient browser credentials.

Authentication

Browser sessions usually use HTTP-only cookies. After login, the server sets access_token, refresh_token, and csrf_token cookies automatically. API clients and MCP/AI delegates may send the access JWT as a Bearer token instead.

Roles and Permissions

Role	Access
viewer	Read-only market data, orders, positions, strategies, system status
ai_delegate	Scoped automation principal for MCP and AI-review workflows; can read market/orders/positions/strategies/signals/backtests/system status and create/cancel/manage scoped orders/positions, but cannot manage users, settings, processes, credentials, or scope grants
operator	Viewer permissions plus trade execution, process management
admin	Full access including user management, system configuration, wallet/credential management, scope grant management, operator impersonation

Multi-tenant permissions (ADMIN only): read:wallet_credentials, manage:wallet_credentials, manage:scope_grants, impersonate:operator.

POST /api/auth/login

Authenticate and create a session. Sets access_token, refresh_token, and csrf_token cookies on the response.

Request: the body is a LoginRequest envelope (PayloadRequest) wrapping a LoginBody payload — the client stamps its own provenance on the outer envelope.

```
POST /api/auth/login
Content-Type: application/json
```

```
{
  "type": "login_request",
  "sequence_id": 1,
  "public_id": "<client-uuid7>",
  "timestamp": "2026-01-18T12:00:00Z",
  "session_id": "<client-session>",
  "topic": null,
  "payload": {
    "username": "admin",
    "password": "password123",
    "remember_me": false
  }
}
```

Response (200):

```
{
  "type": "login_response",
  "sequence_id": 1,
  "public_id": "<uuid7>",
  "timestamp": "2026-01-18T12:00:00Z",
  "session_id": "<server-session>",
  "topic": null,
  "payload": {
    "type": "login",
    "sequence_id": 1,
    "public_id": "<uuid7>",
    "timestamp": "2026-01-18T12:00:00Z",
    "session_id": "<server-session>",
    "topic": null,
    "message": "Login successful",
    "expires_in": 900,
    "user": {
      "type": "user_profile",
      "sequence_id": 1,
      "public_id": "<uuid7>",
      "timestamp": "2026-01-18T12:00:00Z",
      "session_id": "<server-session>",
      "topic": null,
      "username": "admin",
      "email": "admin@example.com",
      "role": "admin",
      "is_active": true,
      "created_at": "2026-01-10T08:00:00Z",
      "operator_public_ids": [],
      "primary_operator_public_id": null,
      "active_wallet_public_id": null,
      "default_language": null
    },
    "access_token": null,
    "refresh_token": null
  }
}
```

The login handler returns the raw UserProfile from authenticate_user() and overlays principal.active_wallet_public_id onto the returned user via model_copy. On the initial login path the freshly-built principal has no wallet selected yet, so active_wallet_public_id

lands as null; the operator-membership fields are not enriched on this surface either (operator_public_ids = [], primary_operator_public_id = null). Call GET /api/auth/me after login to get a fully-enriched profile with operator memberships resolved.

The outer envelope (LoginResponse) and the inner payload (LoginData) both carry the standard provenance fields (type, sequence_id, public_id, timestamp, session_id, topic). access_token and refresh_token are null in the cookie flow; they are populated only when the caller passes ?return_tokens=true for headless integrations.

Cookies set:

Cookie	HttpOnly	Secure	SameSite	Path	Max-Age	Description
access_token	Yes	Yes (prod)	Lax/Strict	/	session	JWT access token (15 min)
refresh_token	Yes	Yes (prod)	Lax/Strict	/api/auth	7 days	JWT refresh token
csrf_token	No	Yes (prod)	Lax/Strict	/	session	CSRF protection token

POST /api/auth/refresh

Refresh session tokens. The refresh_token cookie is sent automatically by the browser. Returns new tokens in cookies plus a WebSocket authentication token in the response body. Long-running clients that already hold a valid access token and only need a fresh ws_token should call [POST /api/auth/ws_token](#) instead — that route does not rotate the refresh-token pair.

Request:

```
POST /api/auth/refresh
```

Response (200):

```
{
  "type": "refresh_response",
  "sequence_id": 2,
  "public_id": "<uuid7>",
  "timestamp": "2026-01-18T12:00:00Z",
  "session_id": "<server-session>",
  "topic": null,
  "payload": {
    "type": "refresh",
    "sequence_id": 2,
    "public_id": "<uuid7>",
    "timestamp": "2026-01-18T12:00:00Z",
    "session_id": "<server-session>",
```

```

"topic": null,
"message": "session refreshed",
"ws_token": "eyJhbGciOi...",
"ws_token_exp": "2026-01-18T12:30:00Z",
"csrf_token": "abc123def456...",
"user": {
  "type": "user_profile",
  "sequence_id": 2,
  "public_id": "<uuid7>",
  "timestamp": "2026-01-18T12:00:00Z",
  "session_id": "<server-session>",
  "topic": null,
  "username": "admin",
  "email": "admin@example.com",
  "role": "admin",
  "is_active": true,
  "created_at": "2026-01-10T08:00:00Z",
  "operator_public_ids": [],
  "primary_operator_public_id": null,
  "active_wallet_public_id": null,
  "default_language": null
},
"access_token": null,
"refresh_token": null
}

```

The refresh handler returns the UserProfile from `get_user_by_id()` and overlays the principal's `active_wallet_public_id` onto it — that value is seeded from `token_data.active_wallet_public_id` (and may be swapped via an optional RefreshTokenPayload wallet hint), so the refresh response generally carries the active wallet (it is only null when the refresh JWT itself carries no wallet claim and no hint is supplied). Operator memberships are still NOT resolved here (`operator_public_ids = []`, `primary_operator_public_id = null`); use `GET /api/auth/me` for the enriched view.

`access_token` and `refresh_token` are populated only when `?return_tokens=true` is set (headless integrations); the cookie flow leaves them null and writes the rotated JWTs as Set-Cookie headers. The RefreshTokenRequest body (optional) carries `active_wallet_public_id` / `clear_active_wallet` to atomically swap the caller's active wallet during the rotation; an `Authorization: Bearer <refresh-jwt>` header is accepted as a fallback for headless callers that don't have access to the `refresh_token` cookie.

POST /api/auth/ws_token

Mint a one-shot WebSocket authentication token without rotating the caller's refresh-token pair. Authenticates via the access bearer (header or cookie) and returns a fresh `ws_token` bound to the access JWT's session. Per-source-IP rate-limited so a reconnect storm cannot exhaust the token store.

Use this route when a long-running client (e.g. a push-wakeup monitor) needs to mint successive `ws_tokens` on its own cadence without rotating the refresh-token pair shared with sibling processes.

Request:

```
POST /api/auth/ws_token
Authorization: Bearer <access_token>
```

Response (200):

```
{
  "type": "ws_token_response",
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:15:00Z",
  "topic": null,
  "payload": {
    "type": "ws_token",
    "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5c",
    "session_id": "<server-session>",
    "sequence_id": 1,
    "timestamp": "2026-01-18T12:15:00Z",
    "topic": null,
    "message": "ws_token issued",
    "ws_token": "eyJhbGciOi...",
    "ws_token_exp": "2026-01-18T12:30:00Z",
    "expires_in": 900
  }
}
```

`expires_in` mirrors `ws_token_exp` as relative seconds for clients that prefer relative-deadline math.

Errors:

- 401 Unauthorized — access bearer is absent, expired, or invalid.
- 429 Too Many Requests — per-source-IP minute budget exhausted.

POST /api/auth/logout

Logout and invalidate the current session. Clears all authentication cookies and blacklists tokens.

Request:

```
POST /api/auth/logout
```

Response (200):

```
{
  "type": "message",
  "public_id": "<uuid7>",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "payload": "Logged out successfully"
}
```

GET /api/auth/me

Get the currently authenticated user's profile.

Request:

```
GET /api/auth/me
```

Response (200):

Handler returns a UserResponse envelope (PayloadResponse[user_response, UserProfile]) wrapping the full UserProfile payload, which carries the multi-tenant trio:

```
{
  "type": "user_response",
  "sequence_id": 3,
  "public_id": "<uuid7>",
  "timestamp": "2026-01-18T12:00:00Z",
  "session_id": "<server-session>",
  "topic": null,
  "payload": {
    "type": "user_profile",
    "sequence_id": 3,
    "public_id": "<uuid7>",
    "timestamp": "2026-01-18T12:00:00Z",
    "session_id": "<server-session>",
    "topic": null,
    "username": "admin",
    "email": "admin@example.com",
    "role": "admin",
    "is_active": true,
    "created_at": "2026-01-10T08:00:00Z",
    "operator_public_ids": ["019d6ca4-..."],
    "primary_operator_public_id": "019d6ca4-...",
    "active_wallet_public_id": "019d7e9a-...",
    "default_language": "pl"
  }
}
```

Only /api/auth/me enriches the full UserProfile. The operator_public_ids / primary_operator_public_id fields are populated from user_operator_memberships (ADMIN receives every active operator; OPERATOR/VIEWER receive only their explicit memberships); active_wallet_public_id reflects the active wallet selection. These fields

power the frontend OperatorPicker without a second round trip. The login / refresh responses return the UserProfile from authenticate_user() / get_user_by_id() with the active wallet overlaid from the request principal — operator memberships are NOT resolved on those two surfaces (operator_public_ids = [], primary_operator_public_id = null). Active-wallet semantics differ between the two: login lands active_wallet_public_id = null (no wallet selected at first auth); refresh propagates whatever wallet the refresh JWT carries (or the request hint). Clients should call /api/auth/me to get a fully-resolved profile.

POST /api/auth/me/update

Update the authenticated caller's self-service preferences. Currently exposes default_language only; additional preference fields may be added later without changing the endpoint contract. Mirrors the admin POST /api/auth/users/{user_id}/update shape (codebase convention: POST + verb, no REST PATCH).

Request:

```
POST /api/auth/me/update
Content-Type: application/json
X-CSRF-Token: <token>
```

```
{
  "type": "update_auth_me_request",
  "payload": {
    "default_language": "pl"
  }
}
```

default_language is validated against the union of supported client codes (iOS-canonical + frontend-canonical forms accepted — e.g. "zh-Hans", "zh", "pt-BR", "pt", "nb", "no"). null clears the preference.

Response (200):

Returns the same UserResponse envelope shape as GET /api/auth/me, with the updated default_language echoed in payload.

Errors:

- 404 — caller's user row was not found (rare; only surfaces if an admin concurrently deactivates the user between auth-dep resolution and the update commit).
- 422 — default_language not in the supported-language allowlist.

GET /api/auth/users

List users. Requires manage:users.

Query parameters:

Parameter	Type	Description
include_inactive	bool	Include deactivated users (default false)
as_of	datetime	Optional point-in-time query timestamp

Returns UserListResponse with payload and count.

POST /api/auth/users

Create a user. Requires manage:users and CSRF for cookie auth. Body is CreateUserRequest wrapping username, password, optional email, role, and is_active.

POST /api/auth/users/{user_id}/update

Update email, role, and active state for an existing user. Requires manage:users and CSRF for cookie auth. Returns UserResponse.

POST /api/auth/users/{user_id}/deactivate

Deactivate a user through the canonical kill-switch flow. Requires manage:users and CSRF for cookie auth. The path segment is resolved as the target username; self-deactivation returns 400, unknown active users return 404. The service also revokes active sessions and emits admin.user_deactivated for verify-cache eviction.

POST /api/auth/users/{user_id}/change-password

Change a password. Users may change their own password; admins may change any password. Requires CSRF for cookie auth and is account-rate limited. Body is ChangePasswordRequest with current and new password.

POST /api/auth/users/{user_id}/admin-reset-password

Admin password reset without the current password. Requires manage:users, CSRF for cookie auth, and is account-rate limited. Body is AdminResetPasswordRequest.

CSRF Protection

The csrf_token cookie is readable by JavaScript (not HttpOnly). For cookie-authenticated mutating requests outside /api/auth/login, /api/auth/refresh, and /api/auth/logout, include its value as a header:

`X-CSRF-Token: <value from csrf_token cookie>`

New CSRF tokens are issued on login and refresh. There is no separate endpoint for obtaining CSRF tokens.

Bearer-authenticated requests do not require X-CSRF-Token. This is intentional: the request is authorized by the explicit Authorization: Bearer ... header rather than an ambient browser cookie.

REST Endpoints

REST endpoints return the same Data schemas used by WebSocket messages (OrderData, SignalData, ExecutionData, PositionData, CandleData from messaging.schemas.data). This means the wire format is identical whether data arrives via REST or the WebSocket feed.

REST envelopes — PayloadResponse / PayloadListResponse

REST responses are JSON objects with a typed envelope plus a payload. Two shapes ship from src/snapper/api/schemas/base.py:

- `PayloadResponse[T, P]` — singleton response, fields: type (Literal discriminator), envelope provenance (`public_id`, `session_id`, `sequence_id`, `timestamp`, `topic`) and payload: `P`.
- `PayloadListResponse[T, P]` — list response, same envelope plus payload: `list[P]` and `count: int` (always equal to `len(payload)`).

Per-item provenance is preserved on items that originate from a bitemporal DB projection (the item carries its own `public_id/session_id/sequence_id`); minted results share the envelope's provenance. Each item's data-type shape is identical to what the WebSocket feed delivers for the same domain object — only the wrapping differs.

OpenAPI request-body schemas for routes that use the `json_body` dependency are patched into FastAPI's generated OpenAPI from `openapi_extra` metadata. Optional request bodies, such as `refresh` and `delegate deactivate`, stay optional in the generated schema. The mounted `/api/mcp` Streamable HTTP sub-app is intentionally outside FastAPI's OpenAPI route list; its tool contract is documented in [ai-integration.md](#).

Provenance on Reads vs. Mutations

- **GET reads** — No client provenance is expected. When telemetry recording is enabled (disabled by default), the server records the request in the telemetry table for observability; otherwise no provenance is recorded for reads. Either way, it does not stamp provenance onto the response items beyond what was stored at write time.

- **Mutations (POST)** — All mutations use command-style POST with a `PayloadRequest` envelope carrying provenance (`public_id`, `session_id`, `sequence_id`, `timestamp`) and domain intent in payload. The server-side `ClientProvenanceMiddleware` extracts these fields, emits a structured info log, and runs a per-session `GapDetector` to warn on sequence gaps. Gaps are logged as warnings but never reject requests.

ClientProvenanceMiddleware

`ClientProvenanceMiddleware` is an ASGI middleware that processes every mutation request:

1. Intercepts the request body transparently (replays chunks to downstream handlers).
2. Extracts `session_id`, `sequence_id`, and `public_id` from the JSON body when present.
3. Emits a structured log with `client_session_id`, `client_sequence_id`, `client_public_id`, and the request path.
4. Runs a per-session `GapDetector` to detect sequence gaps from the client.
5. Records the mutation in the control table via a finally block, so the row is persisted whether the request succeeded or failed.

The control recording is non-blocking: any DB failure is logged and swallowed so the response already sent to the client is never invalidated. The control row includes the redacted request payload, HTTP method, path, outcome (ok, error, or exception), and server-side provenance (`session_id` and `sequence_id` from the middleware's own `SequenceTracker`).

All data endpoints support bitemporal querying via the optional `as_of` parameter (UTC datetime). When provided, the query returns data as it was known at that point in time. When omitted, the current time is used.

GET /api/health

Public health check endpoint. No authentication required.

status reflects the health of enabled long-running CORE processes: "healthy" when all are running, "error" if any are missing. In `SERVER_API_ONLY` mode the status is always "healthy" (no processes are started by design).

Response (200):

```
{
  "type": "health_check_response",
  "sequence_id": 1,
  "public_id": "<uuid7>",
  "timestamp": "2026-01-18T12:00:00Z",
  "session_id": "<server-session>",
  "topic": null,
  "payload": {
    "type": "health_check",
```

```

    "status": "healthy",
    "version": "0.1.0",
    "connections": {
      "type": "connection_stats",
      "active_connections": 5,
      "zmq_subscribers": 12,
      "subscriber_tasks": 12,
      "active_topics": 8,
      "active_clients": 3
    },
    "topics": {
      "type": "health_topics",
      "available": 7,
      "active": 3
    },
    "gap_detection": {
      "type": "gap_detection_stats",
      "bridge": {
        "type": "gap_stats",
        "gaps_detected": 0,
        "session_resets": 0,
        "duplicates": 0,
        "mid_stream_joins": 0,
        "rejected_unstamped": 0
      },
      "rest_clients": {}
    }
  }
}

```

The outer `health_check_response` envelope carries server-side provenance; the inner `health_check` payload holds the runtime snapshot. `gap_detection` provides observability into sequence gap detection across the ZMQ bridge and per-session REST client detectors.

GET /api/candles

Fetch OHLCV candlestick data. Requires `read:market_data` permission.

Request:

```
GET /api/candles?instrument=BTC-USD&exchange=kraken&timeframe=1h&limit=100
```

Parameters:

Parameter	Type	Required	Description
instrument	string	yes	Instrument symbol (e.g., BTC-USD)
exchange	string	yes	Exchange name (kraken, kraken_futures, kraken_equities, walumat, polygon)
timeframe	string	yes	Candle timeframe (e.g., 1m, 5m, 15m, 1h, 4h, 1d)

Parameter	Type	Required	Description
limit	int	no	Number of candles, max 1000 (default 100)
as_of	datetime	no	Point-in-time query, UTC (default: current time)

Returns 200 OK with an empty payload array when no candles match the query (unknown instrument, no warm-cache rows, etc.) — the CandleListResponse envelope is the canonical shape for empty results too.

Response (200):

```
{
  "type": "candle_list",
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "payload": [
    {
      "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
      "type": "candle",
      "timestamp": "2026-01-18T12:00:00Z",
      "instrument": "BTC-USD",
      "exchange": "kraken",
      "timeframe": "1h",
      "open_at": "2026-01-18T11:00:00Z",
      "open": 42000.0,
      "high": 42500.0,
      "low": 41800.0,
      "close": 42300.0,
      "volume": 1234.56,
      "vwap": 42150.0,
      "trades": 5678
    }
  ],
  "count": 1
}
```

GET /api/candles/db

Explicit DB-only candle read for operators. Parameters and response shape match GET /api/candles, but the route bypasses the in-process market cache unconditionally so incident response can verify persisted candles rows directly.

GET /api/candles/cache

Explicit cache-only candle diagnostic read. Parameters match GET /api/candles except as_of is not accepted. For cache-eligible timeframes (1m, 5m, 15m, 30m) the route returns cache-shaped payload data with diagnostic fields such as is_warm, source, and

sample_count; cold cache returns an empty payload with is_warm=false instead of silently falling back to DB. Long frames (1h, 4h, 1d) fall through to persisted rows because the cache does not serve them.

GET /api/orders

Fetch orders with optional filtering. Requires read:orders permission.

Request:

```
GET /api/orders?symbol=BTC-USD&exchange=kraken&limit=100&offset=0
```

Parameters:

Parameter	Type	Required	Description
symbol	string	no	Filter by instrument symbol
exchange	string	no	Filter by exchange (paper, kraken, kraken_futures, walutomat)
limit	int	no	Number of orders, 1-1000 (default 100)
offset	int	no	Number of orders to skip (default 0)
as_of	datetime	no	Point-in-time query, UTC (default: current time)
operator_public_id	string	no	Scope to a single operator (403 if foreign)
wallet_public_id	string	no	Scope to a single wallet (403 if inaccessible)

Response (200):

```
{
  "type": "order_list",
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "payload": [
    {
      "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
      "type": "order",
      "timestamp": "2026-01-18T12:00:00Z",
      "instrument": "BTC-USD",
      "exchange": "kraken",
      "client_order_id": "signal-a1b2c3d4",
      "exchange_order_id": "KRAKEN-456",
      "created_at": "2026-01-18T12:00:00Z",
      "updated_at": "2026-01-18T12:01:00Z",
      "side": "buy",
      "order_type": "limit",

```

```

        "price": 42000.0,
        "size": 0.1,
        "filled_size": 0.1,
        "average_price": 42000.0,
        "status": "filled",
        "time_in_force": "GTC",
        "mode": "live",
        "error": null
    }
],
"count": 1
}

```

POST /api/orders

Create a manual order via a manual_once execution plan. Requires create:orders permission.

Request:

```

POST /api/orders
Content-Type: application/json
X-CSRF-Token: <csrf_token>

{
  "type": "create_order_command",
  "session_id": "ui",
  "sequence_id": 0,
  "public_id": "<uuid7>",
  "timestamp": "2026-04-10T12:00:00Z",
  "payload": {
    "instrument": "BTC-USD",
    "instrument_public_id": "<uuid>",
    "exchange": "kraken",
    "mode": "live",
    "side": "buy",
    "order_type": "limit",
    "quantity": 0.5,
    "price": 50000.0,
    "wallet_public_id": "<uuid>",
    "idempotency_key": "<uuid7>"
  }
}

```

Payload fields:

Field	Type	Required	Description
instrument	string	yes	Native symbol (e.g., BTC-USD)
instrument_public_id	string	yes	Instrument UUID
exchange	string	yes	Exchange name
mode	string	no	live (default) or paper

Field	Type	Required	Description
side	string	yes	buy or sell
order_type	string	yes	market, limit, stop, stop_limit
quantity	float	yes	Order size (must be > 0)
price	float	cond	Required for limit and stop_limit
stop_price	float	cond	Required for stop and stop_limit
wallet_public_id	string	yes	Target wallet UUID
operator_public_id	string	no	Operator identity
idempotency_key	string	no	Dedup key (409 on duplicate)

Response (200): ExecutionPlanResponse envelope with plan details.

Errors: 403 (disabled/forbidden), 409 (idempotency conflict), 422 (validation).

POST /api/orders/{plan_public_id}/cancel

Cancel an active execution plan by its plan public id. Requires cancel:orders permission and caller access to the plan's wallet.

The route transitions the plan to cancel_requested, hydrates the venue-assigned exchange_order_id from the active orders row if available, and inserts a cancel TradeCommand so the outbox dispatcher can publish OrderCancelData on the orders.commands.{ex}.{instr}.cancel topic. If the cancel TradeCommand insert fails the plan is transitioned to failed with last_error, and HTTP 500 is returned — PlanExecutorService will re-emit the cancel on the next startup (idempotent via has_pending_cancel_command).

Request:

```
POST /api/orders/<plan_public_id>/cancel
Content-Type: application/json
X-CSRF-Token: <csrf_token>

{
  "type": "cancel_order_command",
  "session_id": "ui",
  "sequence_id": 0,
  "public_id": "<uuid7>",
  "timestamp": "2026-04-10T12:01:00Z",
  "payload": {
    "reason": "changed mind"
  }
}
```

Response (200): ExecutionPlanResponse with status cancel_requested.

Errors: 403 (wallet not accessible), 404 (not found), 409 (already terminal or concurrent change), 500 (cancel command insert failed).

POST /api/orders/by-client-order-id/{client_order_id}/cancel

UI convenience route that cancels by the child order's client_order_id (the value displayed on the Orders table). Resolves the owning plan via the trade_commands table and delegates to the shared cancel flow described above.

Requires cancel:orders permission. Same response shape and error codes as the plan-id route, plus 404 when no plan-linked trade_commands row is found for the given client_order_id.

Request:

```
POST /api/orders/by-client-order-id/<client_order_id>/cancel
Content-Type: application/json
X-CSRF-Token: <csrf_token>

{
  "type": "cancel_order_command",
  "session_id": "ui",
  "sequence_id": 0,
  "public_id": "<uuid7>",
  "timestamp": "2026-04-11T12:00:00Z",
  "payload": {
    "reason": "cancelled from Orders table"
  }
}
```

Response (200): ExecutionPlanResponse with status cancel_requested.

Errors: 403 (wallet not accessible), 404 (no linked plan), 409 (already terminal), 500 (cancel command insert failed).

GET /api/instrument-capabilities

Fetch instrument order capability matrix. Requires read:market_data.

Parameters:

Parameter	Type	Required	Description
exchange	string	no	Filter by exchange
instrument_public_id	string	no	Filter by instrument
as_of	datetime	no	Point-in-time query

Response (200): InstrumentCapabilityListResponse with capability flags.

GET /api/venue-fee-schedules

Fetch venue fee schedules. Requires read:market_data.

Parameters:

Parameter	Type	Required	Description
exchange	string	no	Filter by exchange
as_of	datetime	no	Point-in-time query

Response (200): VenueFeeScheduleListResponse with fee tiers.

GET /api/signals

Fetch trading signals with optional filtering. Requires read:market_data permission.

Request:

```
GET /api/signals?instrument=BTC-USD&strategy=rsi_btc_1h&exchange=paper&hours=24&limit=100
```

Parameters:

Parameter	Type	Required	Description
instrument	string	no	Filter by instrument symbol
strategy	string	no	Filter by strategy name
exchange	string	no	Filter by exchange (paper, kraken, kraken_futures, walutomat)
hours	int	no	Hours of history, max 168 (default 24)
limit	int	no	Number of signals, max 1000 (default 100)
as_of	datetime	no	Point-in-time query, UTC (default: current time)
operator_public_id	string	no	Scope to a single operator (403 if foreign)
wallet_public_id	string	no	Scope to a single wallet (403 if inaccessible)

Response (200):

```
{
  "type": "signal_list",
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
```

```
"topic": null,
"payload": [
  {
    "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
    "type": "signal",
    "timestamp": "2026-01-18T12:00:00Z",
    "instrument": "BTC-USD",
    "exchange": "paper",
    "side": "buy",
    "strength": 0.85,
    "reason": "RSI 28.5 <= 30",
    "strategy_name": "rsi_btc_1h",
    "price": 42000.0,
    "fired_at": "2026-01-18T12:00:00Z"
  }
],
"count": 1
}
```

GET /api/executions

Fetch order fills/executions. Requires read:orders permission.

Request:

```
GET /api/executions?limit=100
```

Parameters:

Parameter	Type	Required	Description
limit	int	no	Number of executions, max 1000 (default 100)
as_of	datetime	no	Point-in-time query, UTC (default: current time)
operator_public_id	string	no	Scope to a single operator (403 if foreign)
wallet_public_id	string	no	Scope to a single wallet (403 if inaccessible)

Response (200):

```
{
  "type": "execution_list",
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:01:00Z",
  "topic": null,
  "payload": [
    {
      "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
      "type": "execution",
      "timestamp": "2026-01-18T12:01:00Z",
      "trade_id": "TTRAD-456",

```

```
    "exchange_order_id": "KRAKEN-456",
    "client_order_id": "signal-a1b2c3d4",
    "instrument": "BTC-USD",
    "exchange": "kraken",
    "side": "buy",
    "size": 0.1,
    "price": 42000.0,
    "last_size": 0.1,
    "last_price": 42000.0,
    "fee": 0.001,
    "fee_asset": "USD",
    "status": "filled",
    "executed_at": "2026-01-18T12:00:59Z"
  },
  "count": 1
}
```

GET /api/positions

Fetch current portfolio positions. Requires read:positions permission.

Request:

```
GET /api/positions
```

Parameters:

Parameter	Type	Required	Description
as_of	datetime	no	Point-in-time query, UTC (default: current time)
operator_public_id	string	no	Scope to a single operator (403 if foreign)
wallet_public_id	string	no	Scope to a single wallet (403 if inaccessible)

Response (200):

```
{
  "type": "position_list",
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "payload": [
    {
      "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
      "type": "position",
      "timestamp": "2026-01-18T12:00:00Z",
      "instrument": "BTC-USD",
      "exchange": "kraken",
      "quantity": 0.5,
      "average_price": 41500.0,

```

```

        "unrealized_pnl": 250.0,
        "realized_pnl": 100.0,
        "mode": "live",
        "position_cycle_public_id": "019e1a2b-4d5e-7f6a-8b9c-0d1e2f3a4b5c"
    }
],
"count": 1
}

```

The position_cycle_public_id field is null when no open position cycle exists for the position (e.g. flat positions or positions without cycle tracking).

POST /api/execution-plans

Create a bracket (SL/TP) execution plan on an open position cycle. Requires create:orders permission.

Request:

```

POST /api/execution-plans
Content-Type: application/json
X-CSRF-Token: <csrf_token>

```

Body (BracketCreateCommand envelope):

```

{
  "type": "create_bracket_command",
  "public_id": "019e1a2b-0000-7000-8000-000000000001",
  "session_id": "ui-session-1",
  "sequence_id": 1,
  "timestamp": "2026-04-12T18:00:00Z",
  "payload": {
    "position_cycle_public_id": "019e1a2b-4d5e-7f6a-8b9c-0d1e2f3a4b5c",
    "sl_price": 48000.0,
    "tp_price": 55000.0
  }
}

```

At least one of sl_price or tp_price is required.

Response (200): ExecutionPlanResponse wrapping the new armed bracket plan.

Errors: 422 (invalid params, missing capability), 409 (cycle not open, duplicate), 403 (wallet inaccessible), 503 (executor unavailable).

POST /api/execution-plans/{plan_public_id}/cancel

Cancel a bracket execution plan. Armed brackets transition to cancelled directly. Active brackets transition to cancel_requested with cancel TradeCommands emitted. Requires cancel:orders permission.

Request:

```
POST /api/execution-plans/{plan_public_id}/cancel
Content-Type: application/json
X-CSRF-Token: <csrf_token>
```

Response (200): ExecutionPlanResponse wrapping the updated plan.

Errors: 404 (not found), 409 (already terminal or concurrent cancel).

GET /api/execution-plans/{plan_public_id}

Retrieve a single execution plan by public_id. Requires read:orders permission.

Response (200): ExecutionPlanResponse wrapping the plan.

GET /api/execution-plans/{plan_public_id}/decisions

List decision audit rows for an execution plan. Requires read:orders permission.

Response (200): List of decision rows with action, reason, and timestamp.

POST /api/trailing-stops

Create a trailing stop execution plan on an open position cycle. The trailing stop ratchets the stop price as the market moves favorably and triggers a reduce_only market close on breach.

Request body:

- position_cycle_public_id (string, required): Target position cycle.
- trailing_pct (float, required): Trailing distance as percentage ($0 < x < 100$).
- min_lock_pct (float, optional, default 0): Minimum profit % before trailing activates.
- idempotency_key (string, optional): Dedup key for retries.

Permission: create:orders. CSRF token required.

Response (200): ExecutionPlanResponse with plan_type: "trailing_stop", status: "armed".

Response (409): Cycle not open or duplicate trailing stop on same cycle.

Response (422): Invalid params, missing capability, or no open position.

POST /api/trailing-stops/{plan_public_id}/cancel

Cancel a trailing stop. Armed stops transition directly to cancelled. Active stops (with in-flight child orders) transition to cancel_requested.

Permission: cancel:orders. CSRF token required.

GET /api/trailing-stops/{plan_public_id}

Retrieve a trailing stop plan by public_id. Only returns plans with plan_type: "trailing_stop".

Permission: read:orders.

GET /api/trailing-stops/{plan_public_id}/decisions

List decision audit rows for a trailing stop plan.

Permission: read:orders.

GET /api/trailing-stops/by-cycle/{cycle_public_id}

Get live trailing stop state for a position cycle. Returns peak_price and current_stop from the evaluator's in-memory state.

Permission: read:orders.

Response (200): TrailingStopStateResponse with live state, or { "type": "message", "payload": "none" } if no active trailing stop.

Response (404): Position cycle not found.

GET /api/position-cycles/open

List all open position cycles with age information. Admin-only endpoint for diagnosing orphaned cycles (open cycles without a matching trading engine).

Query parameters:

- min_age_hours (float, optional, default 0): Only return cycles open longer than this.

Permission: manage:users (admin only).

Response (200): PositionCycleListResponse envelope whose payload items carry cycle_public_id, shard_key, instrument_public_id, exchange, mode, wallet_public_id, direction, max_qty (per-cycle peak, not lifetime), opened_at, and age_hours.

POST /api/position-cycles/close-orphan

Close a specific orphaned position cycle. The operator should verify via GET /open that the cycle is genuinely orphaned before calling this.

Query parameters:

- cycle_public_id (string, required): Public ID of the cycle to close.

Permission: manage:users (admin only). CSRF token required.

Response (200): OrphanSweepResponse envelope with payload { "closed_count": 1, "closed_cycle_ids": ["<id>"] }.

Response (404): No active open cycle found for the given public_id.

POST /api/position-cycles/sweep-orphans

Bulk-close all open position cycles older than min_age_hours. Default threshold is 72 hours (3 days). Review via GET /open?min_age_hours=72 before running.

Query parameters:

- min_age_hours (float, optional, default 72, minimum 1): Minimum age threshold.

Permission: manage:users (admin only). CSRF token required.

Response (200): OrphanSweepResponse envelope with payload { "closed_count": N, "closed_cycle_ids": [...] }.

Response (400): min_age_hours must be at least 1.

GET /api/exchanges

List distinct exchange names from active symbol aliases. Requires read:market_data permission.

Request:

```
GET /api/exchanges
```

Response (200):

```
{
  "type": "exchange_list",
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "payload": ["kraken", "polygon", "walutomat"],
  "count": 3
}
```

GET /api/exchanges/{exchange}/instruments

List distinct native symbols available on a given exchange. Requires read:market_data permission.

Request:

```
GET /api/exchanges/kraken/instruments
```

Response (200):

```
{
  "type": "instrument_list",
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "payload": ["BTC-USD", "ETH-USD", "SOL-USD"],
  "count": 3
}
```

GET /api/exchanges/{exchange}/instruments/detail

List capability-aware instrument rows for an exchange. Requires `read:market_data`. Each row includes the native symbol plus trade and market-data capability flags, instrument kind, and expiry metadata so clients can show market-data-only instruments without a second request.

GET /api/instruments/{exchange}/{native_symbol}/related

Return related instruments for a concrete exchange/native-symbol pair. Requires `read:market_data`. Used by cross-asset and continuous-contract UI flows to navigate from an instrument to its configured underlying, front month, and sibling contracts.

GET /api/status

System-wide status including trader process, backtests, and active strategies. Requires `read:system_status` permission.

Request:

```
GET /api/status
```

Response payload excerpt (200):

The actual response is a `SystemStatusResponse` envelope with provenance fields and this object under payload.

```
{
  "trader": {
    "status": "running",
    "pid": null,
    "started_at": null,
    "command": null,
  }
}
```

```

        "exit_code": null,
        "error": null
    },
    "backtests": {},
    "strategies": [
        {
            "strategy_name": "rsi_btc_1h",
            "status": "running",
            "details": {},
            "signals_generated": 42,
            "trades_executed": 5,
            "last_signal": "buy",
            "last_signal_time": "2026-01-18T11:45:00Z",
            "pnl": 150.25,
            "pid": 12345,
            "uptime": "2h 15m"
        }
    ]
}

```

GET /api/ws/stats

WebSocket and ZMQ bridge statistics. Requires read:system_status permission.

Request:

```
GET /api/ws/stats
```

Response payload excerpt (200):

The actual response is a WsStatsResponse envelope with provenance fields and this object under payload.

```

{
  "websocket": {
    "active_connections": 5,
    "topic_subscribers": {
      "market.kraken.BTC-USD.candles.1h": 3,
      "signals.paper.BTC-USD.rsi_btc_1h": 2
    },
    "client_count": 5
  },
  "zmq_bridge": {
    "active_topics": 12,
    "subscriber_tasks": 12,
    "available_topics": ["market.", "signals.", "system.heartbeats.",
"admin.", "orders.commands.", "orders.events.", "accruals.", "backtest.",
"alerts.", "plans.decisions.", "ai_reviews.", "processes.events.summary.",
"processes.events.configured.", "processes.events.runs.",
"strategies.events.list."]
  },
  "connections": {
    "active_connections": 5,
    "zmq_subscribers": 12,
    "subscriber_tasks": 12,
    "active_topics": 8,

```

```

    "active_clients": 5
  },
  "topics": {
    "market.kraken.BTC-USD.candles.1h": {
      "active_subscribers": 3,
      "received": 1200,
      "forwarded": 1180,
      "throttled": 20,
      "dropped": 0,
      "timeout": 0,
      "errors": 0,
      "invalid_messages": 0,
      "last_message_ts": 1737208800.0,
      "throttle_ms": 100,
      "pattern": "market."
    }
  },
  "subscriptions": {
    "per_topic": {
      "market.kraken.BTC-USD.candles.1h": 3
    },
    "per_client": {
      "140234567890": ["market.kraken.BTC-USD.candles.1h"]
    }
  },
  "config": {
    "broker_xpub": "tcp://127.0.0.1:7501",
    "heartbeat_interval_ms": 15000
  }
}

```

GET /api/zmq/health

ZMQ bridge health check. Requires read:system_status permission.

Request:

```
GET /api/zmq/health
```

Response payload excerpt (200):

The actual response is a ZmqHealthResponse envelope with provenance fields and this object under payload.

```

{
  "status": "healthy",
  "timestamp": "2026-01-18T12:00:00Z",
  "components": {
    "zmq_context": "ok",
    "websocket_manager": "ok",
    "active_connections": 5
  },
  "config": {
    "available_topics": ["market.", "signals.", "system.heartbeats.",
      "admin.", "orders.commands.", "orders.events.", "accruals.", "backtest.",
      "alerts.", "plans.decisions.", "ai_reviews.", "processes.events.summary."],

```

```

"processes.events.configured.", "processes.events.runs.",
"strategies.events.list."]
  },
  "connections": {
    "active_connections": 5,
    "zmq_subscribers": 12,
    "subscriber_tasks": 12,
    "active_topics": 8,
    "active_clients": 3
  },
  "message_stats": {},
  "errors": []
}

```

GET /api/market/cache/health

Diagnostic snapshot of the in-process market cache, configured pair stats, and persist-policy universe. Requires read:market_data. Returns counts for cached instruments, cached pairs, and persisted instrument universe size.

GET /api/market/cache/stats/configured

Return cached Pearson and cointegration stats for every configured market-stats pair. Requires read:market_data. Configured-but-cold pairs return placeholders with is_warm=false so dashboards can render stable "computing" rows.

GET /api/market/cache/stats/{exchange_a}/{symbol_a}/{exchange_b}/{symbol_b}

Return cached stats for one configured pair. Requires read:market_data. Unknown exchanges return 400; unconfigured pairs return 404; configured-but-not-yet-computed pairs return 200 with is_warm=false.

GET /api/market/coverage

Per-exchange market-data coverage over active instruments. Requires read:system_status.

Query parameters:

Parameter	Type	Description
tick_window_seconds	int	Freshness window for ticks (default 600)
candle_window_seconds	int	Freshness window for candles (default 1800)

The payload includes one row per exchange with instrument count, fresh_ticks, fresh_candles, gated_off, and dark counts.

GET /api/market/feed-health

Current per-symbol feed-health rows. Requires read:system_status.

Query parameters:

Parameter	Type	Description
exchange	string	Optional lowercase exchange filter
fresh_within_seconds	int	Optional staleness filter; older snapshots are dropped

Process Management

Process endpoints manage background services (feeds, strategies, executors, brokers). Most require manage:processes permission (operator/admin). The summary endpoint requires only read:system_status (viewer+).

GET /api/processes/available

List registered process templates that can be instantiated. Requires manage:processes permission.

Response (200):

```
{
  "type": "available_processes",
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "payload": [
    {
      "name": "zmq_broker",
      "class_path":
"snapper.messaging.infrastructure.broker.ZmqBrokerProcess",
      "method": "start",
      "description": "ZeroMQ XPUB/XSUB message broker",
      "lifecycle": "long_running",
      "role": "core",
      "tags": ["infrastructure"],
      "parameters_schema": null
    }
  ],
  "count": 1
}
```

GET /api/processes/configured

List configured process instances with runtime state. Requires manage:processes permission.

Response (200):

```
{
  "type": "configured_processes",
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "payload": [
    {
      "type": "configured_process",
      "name": "zmq_broker",
      "enabled": true,
      "running": true,
      "mode": "thread",
      "class_path":
"snapper.messaging.infrastructure.broker.ZmqBrokerProcess",
      "method": "start",
      "parameters": {},
      "note": null,
      "lifecycle": "long_running",
      "role": "core",
      "tags": ["infrastructure"],
      "parameters_schema": null,
      "is_one_shot": false,
      "active_public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
      "kind": "instance",
      "wallet_public_id": null,
      "parent_template": null,
      "coordinator": "coord-0",
      "managed_remotely": false
    }
  ],
  "count": 1
}
```

GET /api/processes/summary

Lightweight process category counts for the overview dashboard. Requires read:system_status permission.

Response (200):

```
{
  "type": "process_summary_response",
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "session_id": "<server-session>",
  "sequence_id": 1,
```

```

    "timestamp": "2026-01-18T12:00:00Z",
    "topic": null,
    "payload": {
      "type": "process_summary",
      "coordinator": "coord-0",
      "feeds": { "running": 2, "total": 3 },
      "strategies": { "running": 1, "total": 2 },
      "executors": { "running": 1, "total": 1 },
      "brokers": { "running": 1, "total": 1 },
      "processes": []
    }
  }
}

```

POST /api/processes

Create a new process configuration from a registered template. Requires manage:processes permission. Returns 201 on success.

Request:

```

POST /api/processes
Content-Type: application/json
X-CSRF-Token: <csrf_token>

{
  "type": "process_create_request",
  "public_id": "<uuid7>",
  "session_id": "<client-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "payload": {
    "name": "kraken_feed_btc",
    "template": "kraken_feed_publisher",
    "enabled": true,
    "mode": "thread",
    "parameters": { "symbols": ["BTC-USD"] },
    "note": "Kraken BTC feed"
  }
}

```

Response (201):

```

{
  "type": "process_create_response",
  "public_id": "<uuid7>",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "payload": {
    "type": "process_create",
    "public_id": "<uuid7>",
    "session_id": "<server-session>",
    "sequence_id": 1,
    "timestamp": "2026-01-18T12:00:00Z",
    "topic": null,

```

```

    "status": "created",
    "process": {
      "name": "kraken_feed_btc",
      "template": "kraken_feed_publisher"
    }
  }
}

```

GET /api/processes/schema/{name}

Get the configuration schema and defaults for a registered process template. Requires manage:processes permission.

Response (200):

```

{
  "type": "process_schema_response",
  "public_id": "<uuid7>",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "payload": {
    "type": "process_schema",
    "public_id": "<uuid7>",
    "session_id": "<server-session>",
    "sequence_id": 1,
    "timestamp": "2026-01-18T12:00:00Z",
    "topic": null,
    "name": "kraken_feed_publisher",
    "description": "Kraken market data feed publisher",
    "class_path":
"snapper.messaging.publishers.kraken.KrakenMarketDataPublisher",
    "method": "start",
    "default_enabled": true,
    "default_mode": "thread",
    "default_parameters": {},
    "lifecycle": "long_running"
  }
}

```

POST /api/processes/{name}/start

Start a configured process. Requires manage:processes permission.

Request:

```

POST /api/processes/zmq_broker/start
Content-Type: application/json
X-CSRF-Token: <csrf_token>

{
  "type": "process_start_request",
  "public_id": "<uuid7>",

```

```
"session_id": "<client-session>",
"sequence_id": 1,
"timestamp": "2026-01-18T12:00:00Z",
"payload": {
  "mode": "process"
}
}
```

All payload fields are optional. mode overrides the stored execution mode for this run only — it does not persist to Settings. Omitting it uses the stored value.

Response (200):

```
{
  "type": "process_start_response",
  "public_id": "<uuid7>",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "payload": {
    "type": "process_start",
    "public_id": "<uuid7>",
    "session_id": "<server-session>",
    "sequence_id": 1,
    "timestamp": "2026-01-18T12:00:00Z",
    "topic": null,
    "status": "success",
    "name": "zmq_broker",
    "process_public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
    "message": "Process started"
  }
}
```

POST /api/processes/{name}/stop

Stop a running process. Requires manage:processes permission.

Request:

```
POST /api/processes/zmq_broker/stop
X-CSRF-Token: <csrf_token>
```

Response (200):

```
{
  "type": "process_stop_response",
  "public_id": "<uuid7>",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "payload": {
    "type": "process_stop",
    "public_id": "<uuid7>",
```

```

    "session_id": "<server-session>",
    "sequence_id": 1,
    "timestamp": "2026-01-18T12:00:00Z",
    "topic": null,
    "status": "success",
    "name": "zmq_broker",
    "message": "Process stopped"
  }
}

```

GET /api/processes/runs

List historical process runs. Requires manage:processes permission.

Parameters:

Parameter	Type	Required	Description
limit	int	no	Number of runs to return (default 50)
name	string	no	Filter by process name

Response (200):

```

{
  "type": "process_runs",
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T18:00:00Z",
  "topic": null,
  "payload": [
    {
      "type": "process_run",
      "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
      "process_name": "zmq_broker",
      "status": "succeeded",
      "role": "core",
      "lifecycle": "long_running",
      "parameters": {},
      "result": null,
      "error": null,
      "tags": ["infrastructure"],
      "started_at": "2026-01-18T10:00:00Z",
      "completed_at": "2026-01-18T18:00:00Z"
    }
  ],
  "count": 1
}

```

Strategies

GET /api/strategies

List configured strategy processes with lightweight status. Requires read:strategies permission (viewer+).

Response (200):

```
{
  "type": "strategy_list",
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "payload": [
    {
      "type": "strategy_process",
      "name": "strategy_rsi_btc_1h",
      "running": true,
      "enabled": true,
      "mode": "process"
    }
  ],
  "count": 1
}
```

Settings

Settings-management endpoints require configure:system permission (admin only). GET /api/settings/features is the public exception.

GET /api/settings

List all settings, optionally filtered by category.

Parameters:

Parameter	Type	Required	Description
category	string	no	Filter by setting category

Response (200):

```
{
  "type": "setting_list",
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "payload": [
```

```

{
  "type": "setting_read",
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5c",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "key": "polygon_api_key",
  "value": "IvLG...",
  "category": "api",
  "description": "Polygon.io market-data API key",
  "updated_at": "2026-01-15T10:00:00Z",
  "updated_by": "admin"
},
"count": 1
}

```

Note: per-wallet trading credentials (kraken, walumat, kraken_futures) are NOT exposed through the settings endpoints. They live in the wallet_credentials table and are managed via the dedicated /api/wallets/{wallet_public_id}/credentials* routes (src/snapper/server/credential_routes.py — GET for the active summaries, POST for create, and POST rotate for replacing an existing credential row). Seed files (dev.toml / prod.toml — see [Configuration / Wallet Credentials](#)) remain the bootstrap source for dev/prod parity.

GET /api/settings/features

Public feature-flag projection used by the frontend before auth-gated navigation renders. No authentication required. Currently exposes ai_integration_enabled; disabled AI integration also makes /api/mcp and /api/ai-delegates/* return the shared feature_disabled envelope.

GET /api/settings/categories

List distinct setting category names.

Response (200):

```

{
  "type": "setting_categories",
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "payload": ["exchanges", "strategy", "system"],
  "count": 3
}

```

GET /api/settings/push-beta/users

Return the active push-beta gate configuration. Requires configure:system. If the underlying push_beta_config setting is absent or malformed, the endpoint returns the default disabled gate with an empty allowlist so admins can see the effective routing state.

POST /api/settings/push-beta/users

Replace the push-beta gate configuration in one call. Requires configure:system and CSRF for cookie auth. The request body is UpdatePushBetaUsersCommand; user_public_ids becomes the complete allowlist after the write, so callers should read, edit locally, then submit the full desired set.

POST /api/settings/{key}/set

Set (create or update) a setting value.

Request:

```
POST /api/settings/polygon_api_key/set
Content-Type: application/json
X-CSRF-Token: <csrf_token>

{
  "public_id": "<uuid7>",
  "session_id": "<client-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "payload": {
    "value": "new-api-key-value",
    "category": "api",
    "description": "Polygon.io market-data API key"
  }
}
```

Payload fields:

Field	Type	Required	Description
value	string	yes	Setting value
category	string	no	Setting category (default: system)
description	string	no	Human-readable description

Response (200):

```
{
  "type": "setting_response",
  "public_id": "<uuid7>",
  "session_id": "<server-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "payload": {
    "type": "setting_read",
    "key": "polygon_api_key",
    "value": "new-api-key-value",
    "category": "api",
    "description": "Polygon.io market-data API key",
    "updated_at": "2026-01-18T12:00:00Z",
    "updated_by": "admin"
  }
}
```

POST /api/settings/{key}/remove

Soft-delete a setting by key (sets known_to to current time).

Request:

```
POST /api/settings/polygon_api_key/remove
Content-Type: application/json
X-CSRF-Token: <csrf_token>

{
  "public_id": "<uuid7>",
  "session_id": "<client-session>",
  "sequence_id": 2,
  "timestamp": "2026-01-18T12:00:01Z",
  "payload": {}
}
```

Response (200):

```
{
  "type": "message",
  "public_id": "<uuid7>",
  "session_id": "<server-session>",
  "sequence_id": 2,
  "timestamp": "2026-01-18T12:00:01Z",
  "payload": "Setting 'polygon_api_key' deleted successfully"
}
```

Underlying Assets

GET /api/underlyings

List all underlying assets with instrument counts.

```
GET /api/underlyings?as_of=2026-06-20T12:00:00Z
```

Returns PayloadListResponse with UnderlyingAssetData items (ticker, name, asset_class, sector, instrument_count).

GET /api/underlyings/{ticker}/instruments

List instruments mapped to an underlying asset.

```
GET /api/underlyings/SPX/instruments?relationship_type=derivative
```

Query parameters:

- relationship_type (optional): Filter by exact, derivative, or proxy
- as_of (optional): Point-in-time query timestamp

GET /api/underlyings/{ticker}/front-month

Return the front-month (nearest non-expired) futures contract for an underlying.

```
GET /api/underlyings/SPX/front-month?exchange=kraken_equities&contract_family=ES
```

Query parameters:

- exchange (optional): Filter by exchange
- contract_family (optional): Filter by product root (e.g., ES vs MES)
- as_of (optional): Point-in-time query timestamp

Returns PayloadResponse with FrontMonthData (instrument_public_id, native_symbol, exchange, expiry_at, relationship_type, contract_family). Returns 404 if no active futures contracts exist.

GET /api/underlyings/{ticker}/contracts

List all futures contracts for an underlying asset.

```
GET /api/underlyings/SPX/contracts?include_expired=true&contract_family=ES
```

Query parameters:

- exchange (optional): Filter by exchange
- contract_family (optional): Filter by product root

- `include_expired` (optional, default false): Include expired contracts
- `as_of` (optional): Point-in-time query timestamp

Returns `PayloadListResponse` with `ContractData` items. Each item includes `is_front_month` (true for nearest non-expired within same contract family).

GET /api/underlyings/{ticker}/continuous

Build a continuous futures series for an underlying from active contract metadata and candle rows.

```
GET /api/underlyings/SPX/continuous?
exchange=kraken_equities&contract_family=ES&timeframe=1h&limit=500
```

Query parameters:

- `exchange` (optional): Filter by exchange
- `contract_family` (optional): Filter by product root
- `timeframe` (optional): Candle timeframe, default 1h
- `limit` (optional): Maximum points, route-bounded
- `as_of` (optional): Point-in-time query timestamp

Returns a continuous-series response with the selected contract windows and candle points. Returns 400 for invalid parameters and 404 when the underlying cannot be resolved.

Multi-Tenant (Wallets, Operators, Scope Grants, Credentials)

ADMIN principals see the full catalogue; VIEWER and OPERATOR principals see only the subset covered by their operator memberships and active scope grants.

GET /api/wallets

List wallets accessible to the current principal. ADMIN sees all; OPERATOR/VIEWER sees only wallets covered by at least one active scope grant from their operator set.

GET /api/operators

List operators accessible to the current principal. ADMIN sees all; OPERATOR/VIEWER sees only operators in `principal.operator_public_ids`.

GET /api/scope-grants

List active scope grants on a given wallet. Non-ADMIN callers must have visibility into the target wallet; otherwise 403. Required query parameter: wallet_public_id.

POST /api/scope-grants

Create a new scope grant. Requires manage:scope_grants (ADMIN only). Returns 409 on overlap conflict. Body fields: operator_public_id, wallet_public_id, scope_kind (underlying or instrument), underlying_public_id or instrument_public_id, optional note.

POST /api/scope-grants/handover

Atomic SCD2 close + insert transfer of a scope grant to a different operator. Requires manage:scope_grants. Body: from_grant_public_id, to_operator_public_id, optional reason. Returns 404 if source grant missing, 400 on self-handover, 409 on cross-scope overlap.

POST /api/scope-grants/{grant_public_id}/revoke

Atomic SCD2 close (no replacement row) of an active scope grant + emits admin.scope_revoked on the bus for live AI_DELEGATE subscription revalidation. Requires manage:scope_grants (ADMIN only). Body: optional reason (flows to event payload, NOT to the closed row). Returns 404 if the grant does not exist or is already closed (double-revoke).

POST /api/wallets

Create a new wallet. Requires manage:wallet_credentials (ADMIN only). Returns 409 if (label, is_paper) active-unique index is violated. Body: label, optional description, is_paper (default false).

GET /api/wallets/{wallet_public_id}/credentials

List active credentials on a wallet as summaries (no encrypted payload on the wire). Requires read:wallet_credentials (ADMIN only).

POST /api/wallets/{wallet_public_id}/credentials

Create a new wallet credential. Requires manage:wallet_credentials. Plaintext credential_payload is Fernet-encrypted server-side before DB insert. Body: exchange, credential_type (api_key_secret, rsa_pem, oauth, paper), credential_payload (dict), optional label. Returns 409 if (wallet, exchange) already exists.

POST /api/wallets/{wallet_public_id}/credentials/{credential_public_id}/rotate

SCD2 close + insert rotation. Old credential row closed, new row inserted with updated encrypted payload. Requires manage:wallet_credentials. Returns 404 if credential not found. Validates credential_payload against the existing credential type before encrypting.

WebSocket

Connection and Authentication

The WebSocket endpoint is at /api/ws. Authentication is message-based, not query-parameter-based, but the connection must already carry the active auth session cookie. Missing or invalid cookies are rejected with close code 4401 before the auth_required challenge.

Connection flow:

1. Client connects to ws://host:port/api/ws (or wss:// over TLS) with the active auth session cookie
2. Server validates the origin header and session cookie
3. Server sends auth_required message
4. Client sends authenticate message with a WebSocket token
5. Server validates the token and sends auth_ok
6. Server sends auth_complete with available topics for the user's role
7. Client can now subscribe to topics

Obtaining a WebSocket token:

Call POST /api/auth/ws_token with an access token (Bearer header or cookie auth). The route returns a short-lived, one-shot ws_token without rotating the refresh-token pair. POST /api/auth/refresh can also return ws_token during session rotation, but long-running clients should prefer /api/auth/ws_token.

Client-originated control frames (authenticate, reauth, subscribe, unsubscribe, get_subscriptions, ping) inherit StrictDataSchema; clients must stamp public_id, session_id, sequence_id, and timestamp before sending. topic defaults to null and can be omitted on client frames.

Authentication Messages

Server sends after connection:

```
{
  "type": "auth_required",
  "public_id": "019e1a2b-0000-7000-8000-0000000000901",
  "session_id": "<server-ws-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "timeout": 10
}
```

Client sends to authenticate:

```
{
  "type": "authenticate",
  "public_id": "019e1a2b-0000-7000-8000-0000000000a01",
  "session_id": "<client-ws-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "ws_token": "eyJhbGciOi..."
}
```

Server sends on success:

```
{
  "type": "auth_ok",
  "public_id": "019e1a2b-0000-7000-8000-0000000000902",
  "session_id": "<server-ws-session>",
  "sequence_id": 2,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "exp": "2026-01-18T12:30:00Z"
}
```

Server sends session info:

```
{
  "type": "auth_complete",
  "public_id": "019e1a2b-0000-7000-8000-0000000000903",
  "session_id": "<server-ws-session>",
  "sequence_id": 3,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "available_topics": [
    "market.",
  ]
}
```

```

    "signals.",
    "system.heartbeats.",
    "admin.",
    "orders.commands.",
    "orders.events.",
    "accruals.",
    "backtest.",
    "alerts.",
    "plans.decisions.",
    "ai_reviews.",
    "processes.events.summary.",
    "processes.events.configured.",
    "processes.events.runs.",
    "strategies.events.list."
  ],
  "user_role": "operator",
  "session_expires_at": "2026-01-18T12:15:00Z",
  "ws_token_exp": "2026-01-18T12:30:00Z"
}

```

available_topics contains allowed registry roots for the user's role. Subscriptions may use a registry root such as market. or a full concrete topic such as market.kraken.BTC-USD.candles.1h. Intermediate prefixes such as market.kraken. are rejected. Backtests also allow scoped prefixes: backtest., backtest.{wallet_public_id}., and backtest.{wallet_public_id}.{run_public_id}., subject to RBAC and wallet scope.

Server sends on failure:

```

{
  "type": "auth_failed",
  "public_id": "019e1a2b-0000-7000-8000-0000000000904",
  "session_id": "<server-ws-session>",
  "sequence_id": 4,
  "timestamp": "2026-01-18T12:00:00Z",
  "topic": null,
  "reason": "Invalid token"
}

```

Reauthentication

Before the WebSocket token expires, the server sends a reauth_required message. Clients may also refresh proactively before that deadline. In both cases they obtain a new ws_token via POST /api/auth/refresh and send it as a reauth message.

Server warning:

```

{
  "type": "reauth_required",
  "public_id": "019e1a2b-0000-7000-8000-0000000000905",
  "session_id": "<server-ws-session>",
  "sequence_id": 5,
  "timestamp": "2026-01-18T12:28:00Z",
}

```

```
"topic": null,  
"deadline": "2026-01-18T12:29:00Z"  
}
```

Client sends new token:

```
{  
  "type": "reauth",  
  "public_id": "019e1a2b-0000-7000-8000-0000000000a02",  
  "session_id": "<client-ws-session>",  
  "sequence_id": 2,  
  "timestamp": "2026-01-18T12:28:30Z",  
  "ws_token": "eyJhbGciOi...(new token)..."  
}
```

Server confirms:

```
{  
  "type": "reauth_ok",  
  "public_id": "019e1a2b-0000-7000-8000-0000000000906",  
  "session_id": "<server-ws-session>",  
  "sequence_id": 6,  
  "timestamp": "2026-01-18T12:28:30Z",  
  "topic": null,  
  "exp": "2026-01-18T13:00:00Z"  
}
```

If the client does not reauthenticate in time:

```
{  
  "type": "auth_expired",  
  "public_id": "019e1a2b-0000-7000-8000-0000000000907",  
  "session_id": "<server-ws-session>",  
  "sequence_id": 7,  
  "timestamp": "2026-01-18T12:30:00Z",  
  "topic": null  
}
```

Subscription Management

Subscribe to topics:

```
{  
  "type": "subscribe",  
  "public_id": "019e1a2b-0000-7000-8000-0000000000a03",  
  "session_id": "<client-ws-session>",  
  "sequence_id": 3,  
  "timestamp": "2026-01-18T12:00:01Z",  
  "topics": ["market.kraken.BTC-USD.candles.1h", "signals.paper.BTC-  
USD.rsi_btc_1h"]  
}
```

Server confirms subscription:

```
{
  "type": "subscription_success",
  "public_id": "019e1a2b-0000-7000-8000-000000000908",
  "session_id": "<server-ws-session>",
  "sequence_id": 8,
  "timestamp": "2026-01-18T12:00:01Z",
  "topic": null,
  "action": "subscribe",
  "status": "subscribed",
  "topics": ["market.kraken.BTC-USD.candles.1h", "signals.paper.BTC-USD.rsi_btc_1h"],
  "denied_topics": [],
  "active_subscriptions": ["market.kraken.BTC-USD.candles.1h", "signals.paper.BTC-USD.rsi_btc_1h"],
  "message": null
}
```

Unsubscribe:

```
{
  "type": "unsubscribe",
  "public_id": "019e1a2b-0000-7000-8000-000000000a04",
  "session_id": "<client-ws-session>",
  "sequence_id": 4,
  "timestamp": "2026-01-18T12:00:02Z",
  "topics": ["market.kraken.BTC-USD.candles.1h"]
}
```

List active subscriptions:

```
{
  "type": "get_subscriptions",
  "public_id": "019e1a2b-0000-7000-8000-000000000a05",
  "session_id": "<client-ws-session>",
  "sequence_id": 5,
  "timestamp": "2026-01-18T12:00:02Z"
}
```

Response:

```
{
  "type": "subscriptions_list",
  "public_id": "019e1a2b-0000-7000-8000-000000000909",
  "session_id": "<server-ws-session>",
  "sequence_id": 9,
  "timestamp": "2026-01-18T12:00:02Z",
  "topic": null,
  "subscriptions": ["signals.paper.BTC-USD.rsi_btc_1h"],
  "available_topics": ["market.", "signals.", "system.heartbeats.", "orders.events.", "alerts.", "backtest."],
  "total_available": 6
}
```

Ping/Pong

Client sends:

```
{
  "type": "ping",
  "public_id": "019e1a2b-0000-7000-8000-000000000a06",
  "session_id": "<client-ws-session>",
  "sequence_id": 6,
  "timestamp": "2026-01-18T12:00:30Z"
}
```

Server responds:

```
{
  "type": "pong",
  "public_id": "019e1a2b-0000-7000-8000-00000000090a",
  "session_id": "<server-ws-session>",
  "sequence_id": 10,
  "timestamp": "2026-01-18T12:00:30Z",
  "topic": null,
  "active_connections": 5
}
```

Server Data Messages

Data messages are flat JSON objects forwarded directly from the ZMQ bus. There is no "data" wrapper. Messages with malformed JSON are dropped by the bridge before forwarding and are counted in the `invalid_messages` metric for the topic.

All data messages carry `session_id` and `sequence_id` provenance fields (see REST section above). All WebSocket control messages (auth, subscribe, ping/pong, errors) inherit from `StrictDataSchema`, so they also carry the same provenance envelope. Clients can use these fields to detect gaps without server-side replay support.

Candle

```
{
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "type": "candle",
  "timestamp": "2026-01-18T11:00:00Z",
  "session_id": "019e0000-0000-7000-0000-000000000001",
  "sequence_id": 42,
  "instrument": "BTC-USD",
  "exchange": "kraken",
  "timeframe": "1h",
  "open_at": "2026-01-18T11:00:00Z",
  "open": 42000.0,
  "high": 42500.0,
  "low": 41800.0,
  "close": 42300.0,
  "volume": 1234.56,
}
```

```
"vwap": 42150.0,
"trades": 5678
}
```

Tick

```
{
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "type": "tick",
  "timestamp": "2026-01-18T12:00:00Z",
  "instrument": "BTC-USD",
  "exchange": "kraken",
  "volume": 1234.5,
  "bid": 42000.0,
  "ask": 42001.0,
  "last": 42000.5
}
```

Trade

```
{
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "type": "trade",
  "timestamp": "2026-01-18T12:00:00Z",
  "instrument": "BTC-USD",
  "exchange": "kraken",
  "executed_at": "2026-01-18T12:00:00Z",
  "price": 42000.5,
  "volume": 0.25,
  "side": "buy"
}
```

Signal

```
{
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "type": "signal",
  "timestamp": "2026-01-18T12:00:00Z",
  "session_id": "019e0000-0000-7000-0000-000000000001",
  "sequence_id": 7,
  "instrument": "BTC-USD",
  "exchange": "paper",
  "side": "buy",
  "strength": 0.85,
  "reason": "RSI 28.5 <= 30",
  "price": 42000.0,
  "strategy_name": "rsi_btc_1h",
  "fired_at": "2026-01-18T12:00:00Z"
}
```

Execution

```
{
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "type": "execution",
  "timestamp": "2026-01-18T12:01:00Z",
  "trade_id": "TTRAD-456",
  "exchange_order_id": "KRAKEN-456",
  "client_order_id": "signal-a1b2c3d4",
  "instrument": "BTC-USD",
  "exchange": "kraken",
  "side": "buy",
  "size": 0.1,
  "price": 42000.0,
  "fee": 0.001,
  "fee_asset": "USD",
  "status": "filled",
  "executed_at": "2026-01-18T12:00:59Z"
}
```

Order

```
{
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "type": "order",
  "timestamp": "2026-01-18T12:00:00Z",
  "exchange_order_id": "KRAKEN-456",
  "client_order_id": "signal-a1b2c3d4",
  "instrument": "BTC-USD",
  "exchange": "kraken",
  "side": "buy",
  "status": "submitted",
  "order_type": "limit",
  "size": 0.1,
  "filled_size": 0.0,
  "price": 42000.0,
  "average_price": null,
  "reason": null,
  "time_in_force": "GTC",
  "error": null,
  "created_at": "2026-01-18T12:00:00Z",
  "updated_at": null
}
```

Heartbeat

```
{
  "public_id": "019e1a2b-3c4d-7e5f-8a9b-0c1d2e3f4a5b",
  "type": "heartbeat",
  "timestamp": "2026-01-18T12:00:00Z",
  "session_id": "019e1a2b-0000-7000-8000-000000000001",
  "sequence_id": 42,
  "component": "zmq_broker",
  "sequence": 42,
  "status": "healthy",
}
```

```
"lag_ms": 5,  
"meta": {}  
}
```

Error

```
{  
  "type": "error",  
  "message": "Topic 'invalid.topic' does not exist",  
  "timestamp": "2026-01-18T12:00:00Z"  
}
```

Available Topic Patterns

Topics use dot-separated hierarchical names. Subscribe using a full topic string or one of the registry roots returned in `available_topics`. Intermediate prefixes are rejected; use `market.`, not `market.kraken.`. Backtest subscriptions are the exception and support the scoped prefixes documented in the WebSocket auth section above.

Market Data

- `market.{exchange}.{instrument}.candles.{timeframe}` -- OHLCV candles
- `market.{exchange}.{instrument}.ticks` -- Price ticks
- `market.{exchange}.{instrument}.trades` -- Individual trades
- `market.paper.{source_exchange}.{instrument}.candles.{timeframe}` -- Paper candles sourced from a real exchange
- `market.paper.{source_exchange}.{instrument}.ticks` -- Paper ticks sourced from a real exchange

Signals

- `signals.{exchange}.{instrument}.live` -- Live trading signals
- `signals.paper.{instrument}.{strategy_name}` -- Paper trading signals

Order Events

- `orders.events.{exchange}.{instrument}.submitted` -- Order submitted
- `orders.events.{exchange}.{instrument}.accepted` -- Order accepted
- `orders.events.{exchange}.{instrument}.rejected` -- Order rejected
- `orders.events.{exchange}.{instrument}.executed` -- Order fill
- `orders.events.{exchange}.{instrument}.cancelled` -- Order cancelled
- `orders.events.{exchange}.{instrument}.expired` -- Order expired
- `orders.events.{exchange}.{instrument}.replaced` -- Order replaced

Order Commands

- `orders.commands.{exchange}.{instrument}.submit -- Submit order`
- `orders.commands.{exchange}.{instrument}.cancel -- Cancel order`
- `orders.commands.{exchange}.{instrument}.replace -- Replace order`

System

- `system.heartbeats -- Global heartbeat`
- `system.heartbeats.strategy.{name} -- Strategy heartbeat`
- `system.heartbeats.executor.{exchange} -- Single-wallet executor heartbeat`
- `system.heartbeats.executor.{exchange}.{wallet_short} -- Per-wallet executor heartbeat; wallet_short is exactly 12 lowercase hex characters`
- `system.heartbeats.feed.{exchange} -- Live feed heartbeat`
- `system.heartbeats.feed.paper.{source} -- Paper replay feed heartbeat`
- `admin.{resource} -- Administrative events (admin only)`
- `processes.events.summary.{coord_slug} -- Process summary snapshots`
- `processes.events.configured.{coord_slug} -- Configured process-name snapshots`
- `processes.events.runs.{process_name} -- Per-process lifecycle transitions`
- `strategies.events.list.{coord_slug} -- Strategy class-path snapshots`

Alerts and Reviews

- `alerts.{user_public_id}.{alert_type} -- Notification stream`
- `plans.decisions.{plan_public_id} -- Execution-plan decision events`
- `ai_reviews.{user_public_id}.{strategy_public_id}.{suffix} -- AI delegate review frames`
- `accruals.{exchange}.{instrument}.{accrual_type} -- Funding, rollover, and borrow accruals`
- `backtest.{wallet_public_id}.{run_public_id}.{event} -- Backtest lifecycle and progress frames`

Error Handling

HTTP Status	Description
401	Missing or invalid authentication
403	Insufficient permissions or invalid CSRF token
404	Resource not found
422	Request validation error

HTTP Status	Description
429	Rate limit exceeded (Retry-After header included)
500	Internal server error

Usage Example (Python)

```
import httpx

BASE_URL = "http://localhost:8000/api"

async def main():
    async with httpx.AsyncClient() as client:
        # Login: LoginRequest envelope around LoginBody payload
        login_resp = await client.post(
            f"{BASE_URL}/auth/login",
            json={
                "type": "login_request",
                "sequence_id": 1,
                "public_id": "<client-uuid7>",
                "timestamp": "2026-01-18T12:00:00Z",
                "session_id": "<client-session>",
                "topic": None,
                "payload": {
                    "username": "admin",
                    "password": "password123",
                    "remember_me": False,
                },
            },
        )

        refresh_resp = await client.post(f"{BASE_URL}/auth/refresh")
        # RefreshResponse envelope: ws_token lives under .payload
        ws_token = refresh_resp.json()["payload"]["ws_token"]

        candles_resp = await client.get(
            f"{BASE_URL}/candles",
            params={
                "instrument": "BTC-USD",
                "exchange": "kraken",
                "timeframe": "1h",
            },
        )
        candles = candles_resp.json()
        print(candles)
```

Usage Example (JavaScript)

This example is intentionally minimal. The shipped frontend client reuses cached WS tickets, schedules proactive reauthentication before expiry, and sends heartbeat pings every 5 seconds.

```

async function connect() {
  const refreshResp = await fetch("/api/auth/refresh", {
    method: "POST",
    credentials: "include",
  });
  // RefreshResponse envelope wraps the data in `.payload`.
  const { payload } = await refreshResp.json();
  const { ws_token } = payload;

  const protocol = location.protocol === "https:" ? "wss:" : "ws:";
  const ws = new WebSocket(`${protocol}://${location.host}/api/ws`);
  const sessionId = crypto.randomUUID();
  let sequenceId = 0;

  const controlFrame = (type, body = {}) => ({
    type,
    public_id: crypto.randomUUID(),
    session_id: sessionId,
    sequence_id: ++sequenceId,
    timestamp: new Date().toISOString(),
    ...body,
  });

  ws.onopen = () => {
    console.log("Connected, waiting for auth_required...");
  };

  ws.onmessage = (event) => {
    const msg = JSON.parse(event.data);

    if (msg.type === "auth_required") {
      ws.send(JSON.stringify(controlFrame("authenticate", {
        ws_token,
      })));
    }

    if (msg.type === "auth_complete") {
      console.log("Authenticated. Topics:", msg.available_topics);
      ws.send(JSON.stringify(controlFrame("subscribe", {
        topics: ["market.kraken.BTC-USD.candles.1h"],
      })));
    }

    if (msg.type === "candle" || msg.type === "tick") {
      console.log("Data:", msg);
    }

    if (msg.type === "reauth_required") {
      refreshAndReauth(ws);
    }
  };

  setInterval(() => {
    if (ws.readyState === WebSocket.OPEN) {
      ws.send(JSON.stringify(controlFrame("ping")));
    }
  }, 30000);
}

```

Backtests

Backtest endpoints manage strategy backtesting runs. Read endpoints require `read:backtests` permission (viewer+). Mutation endpoints require `manage:backtests` (operator+). Reads are wallet-scoped when the caller has an active wallet selected; admins without a wallet selection see all runs.

POST /api/backtests

Create and launch a new backtest run. Requires an active wallet selection.

Request body:

```
{
  "strategy_class": "sma_cross",
  "instrument_public_id": "BTC-USD",
  "exchange": "kraken",
  "timeframe": "1h",
  "start_date": "2026-01-01T00:00:00Z",
  "end_date": "2026-06-01T00:00:00Z",
  "initial_cash": 10000.0,
  "strategy_params": {"fast": 10, "slow": 30}
}
```

Response: BacktestRunResponse with the created run details.

GET /api/backtests

List backtest runs with optional filters.

Query parameters:

Parameter	Type	Description
strategy	string	Filter by strategy name
status	string	Filter by status (pending, running, completed, failed, cancelled)
config_hash	string	Pairing-stable SHA-256 config hash used by comparison auto-pairing
limit	int	Page size (1-100, default 20)
offset	int	Page offset (default 0)
as_of	datetime	Temporal query timestamp

GET /api/backtests/strategy-classes

List registered strategy-class identifiers accepted by BacktestCreateBody.strategy_class. Used by the UI to populate the create-backtest strategy selector.

POST /api/backtests/compare

Create or return an idempotent existing comparison row for two terminal runs. Manual mode supplies run_a_public_id and run_b_public_id; auto mode supplies config_hash and optionally anchor_run_public_id. Requires an active wallet and read:backtests. See [backtesting.md](#) for pairing semantics.

GET /api/backtests/compare

List recent comparison rows for the caller's active wallet.

Query parameters: limit, offset, and as_of.

GET /api/backtests/compare/{comparison_public_id}

Fetch comparison metadata plus recomputed metrics, equity, trades, and signals diffs from the current artifact rows.

GET /api/backtests/{run_id}

Get backtest run detail by public ID.

POST /api/backtests/{run_id}/cancel

Cancel a pending or running backtest. Sets status to cancel_requested via SCD2 close-and-insert. Returns 409 if the run is already in a terminal state.

POST /api/backtests/{run_id}/rerun

Create a new backtest run with the same configuration as the original.

GET /api/backtests/{run_id}/trades

Get paginated trades for a completed backtest run.

GET /api/backtests/{run_id}/signals

Get paginated signals for a completed backtest run.

GET /api/backtests/{run_id}/events

Get lifecycle events for a backtest run (started, failed, etc.).

GET /api/backtests/{run_id}/equity

Get paginated equity curve points for a run.

Query parameters: limit (1-20000, default 5000), after, and as_of.

AI Reviews

Endpoints for the human-in-the-loop review queue. Strategies emit `ai_reviews.*` requests via the `create_ai_review_and_await()` primitive (see [strategies.md](#)); the selected, registered AI delegate replies via these routes.

POST /api/ai-reviews/{review_public_id}/decision

REST mirror of the `submit_ai_review_decision` MCP tool. Body is `AiReviewDecisionCommand`, an envelope wrapping an inner `AiReviewDecisionRequest`:

```
{
  "type": "ai_review_decision_command",
  "sequence_id": 1,
  "public_id": "<uuid7>",
  "timestamp": "2026-01-18T12:00:00Z",
  "session_id": "<client-session>",
  "topic": null,
  "payload": {
    "decision": "approve",
    "rationale": "Risk profile within bounds."
  }
}
```

decision is "approve" or "reject" (validated against `AiReviewDecisionEnum`; unknown values yield 422 with `error_code='invalid_decision'`). rationale is optional free text, ≤ 4096 chars per the `ai_reviews.rationale` column constraint. Requires

`Permission.CREATE_ORDERS`, but that permission is not sufficient by itself: the caller must resolve to a registered `ai_delegates` row and have a live scope grant for the review wallet and instrument. The dispatch fans out on `bus.ai_review_decision` and emits `ai_reviews.{user}.{strategy}.decision_ack` on the WS surface.

Responses use `AiReviewDecisionResponse` — the canonical envelope shared with the MCP `CallToolResult` payload:

```
{
  "success": true,
  "error_code": null,
  "message": "Decision recorded.",
  "details": { "...": "..." }
}
```

Error status codes: 404 (review not found), 403 (caller not authorized for this review), 409 (already resolved by peer), 410 (deadline elapsed), 422 (invalid decision).

GET /api/ai-reviews/pending

List pending CONSULT reviews where the caller is the selected AI delegate. Used by the bridge for catch-up after WS reconnect and by operator dashboards. Snapshot is bounded by limit and ordered by fanout_after ASC (oldest first). The route first requires Permission.READ_SIGNALS (callers without it receive 403); callers that pass that gate but are not registered as an AI delegate principal receive 422.

Query parameters:

Parameter	Type	Description
limit	int	Page size, 1..500 (default 100)
wallet_public_id	string	Optional wallet filter

Response: PendingReviewListResponse (items + count) where each PendingReviewSummaryItem carries the resolved instrument ticker plus the raw signal_envelope payload (thesis, side, news anchors) so the AI delegate inbox can render a meaningful row without a follow-up read.

Devices

APNs device registration for the iOS app (push notifications for order / position / system events) plus per-device alert preferences. All rows are bound to user_public_id and follow the SCD2 close-and-insert convention.

POST /api/devices

Register or refresh the caller's APNs device token. Body is RegisterDeviceCommand. Same-token re-registrations collapse onto the existing SCD2 row via upsert_notification_device and return the stable public_id reused across versions. Returns NotificationDeviceResponse.

GET /api/devices

List the caller's active devices, newest registered first. Returns NotificationDeviceListResponse (payload + count — PayloadListResponse base); rows whose active version has token_status='user_unregistered' (tombstones) are excluded.

DELETE /api/devices/{device_public_id}

Soft-delete a device via SCD2 close + tombstone successor: the active row is closed and a successor with token_status='user_unregistered' is inserted so an as_of query after the close sees an inactive row rather than a gap. Returns MessageResponse. Marking a nonexistent or not-owned device inactive returns 404.

PATCH /api/devices/{device_public_id}/prefs

Upsert a per-(device, alert_type, scope) preference for the caller. Body is UpdateDevicePrefCommand. The composite scope key is (device_public_id, alert_type, operator_public_id, wallet_public_id) — three null-permutations map to the three partial unique indexes on device_alert_prefs. Returns DeviceAlertPrefResponse.

POST /api/devices/{device_public_id}/prefs/{pref_public_id}/revoke

Close one device-scoped alert preference (SCD2 in place). Backend convention is POST + envelope (not DELETE) so every write carries client-side provenance for the gap detector — same as cancel_order / revoke_scope_grant. The pref row is closed by stamping known_to at the revoke timestamp; no successor is inserted, which frees the slot in the partial unique index. Body is RevokeDevicePrefCommand; returns RevokeDevicePrefResponse.

GET /api/devices/{device_public_id}/prefs

List active per-(alert_type, scope) prefs for a caller-owned device. Returns DeviceAlertPrefListResponse. Tombstoned device prefs are filtered out server-side via the join to notification_devices on token_status='active'.

Alerts

Real-time alert feed for the dashboard + iOS notification panel. Alert events are minted by various subsystems and surfaced via WS on alerts.* (see [messaging.md](#) for the canonical AlertType enumeration) plus this REST catch-up tail.

GET /api/alerts/history

Keyset-paginated page of the caller's active alert_events. Order: (timestamp DESC, public_id DESC).

Query parameters:

Parameter	Type	Description
limit	int	Page size, 1..200 (default 50)
before	string	Opaque cursor token from a previous page's next_cursor; None or malformed values map to page 1 (max 160 chars)

The before cursor is decoded server-side back into the internal AlertListCursor and applied as a keyset predicate against Repository.list_recent_alerts_for_user. Returns AlertHistoryResponse.

GET /api/alerts/{alert_public_id}

Fetch a single active alert_events row if owned by the caller. Returns AlertEventResponse. Ownership is enforced server-side via user_public_id equality; 404 is returned for unknown IDs **and** for IDs owned by other users, so the endpoint never leaks the existence of foreign alert events.

Alert Defaults

Per-user fallback preferences that apply when no per-device device_alert_prefs row matches.

GET /api/alert_defaults

List the caller's active user-level fallback prefs. Returns UserAlertDefaultListResponse (payload + count — PayloadListResponse base). An empty list is the legitimate "no overrides" state — clients should fall through to the in-app default UI; the route does **not** auto-create rows on first read.

PATCH /api/alert_defaults

Upsert the caller's user-level fallback pref for a given alert_type via SCD2 close-and-insert. Body is UpdateUserAlertDefaultCommand.

Metrics

Observability endpoints surfaced for the operations dashboard and the iOS health widget. All routes require `Permission.READ_SYSTEM_STATUS`. See [observability.md](#) for the underlying retention / system-metrics pipeline and snapshot schemas.

GET /api/metrics/notifications

Current notify-sidecar outbox counters (per-status row counts — *not* a rolling window). Returns `NotificationMetricsResponse`.

GET /api/metrics/system

Most recent `SystemMetricsSnapshot` from the in-memory ring buffer (CPU, memory, GC, file descriptors, plus the process connection count under `process.num_connections`). Returns `SystemMetricsResponse`.

GET /api/metrics/system/history

Bounded slice of the in-memory system-metrics history (no cursor / page tokens — the caller filters with `since` / `until` + `limit`). Returns `SystemMetricsHistoryResponse`.

Query parameters:

Parameter	Type	Description
<code>since</code>	<code>datetime</code>	Lower bound (inclusive)
<code>until</code>	<code>datetime</code>	Upper bound (inclusive)
<code>limit</code>	<code>int</code>	Page size, 1..100000 (default 720 — roughly the last hour at the default 5-second cadence)

Ring-buffer depth is capped by `SYSTEM_METRICS_HISTORY_CAP` (default 17280, ~24 h at 5 s cadence — see [configuration.md](#)).

POST /api/metrics/system/tracemalloc/start

Arm Python tracemalloc with an auto-stop deadline. Body-less; `duration_s` is supplied as a query parameter. Returns `TracemallocStateResponse`.

POST /api/metrics/system/tracemalloc/stop

Disarm tracemalloc and cancel any pending auto-stop deadline. Returns `TracemallocStateResponse`.

GET /api/metrics/retention

Most recent retention-scheduler run summary. Returns RetentionRunResponse whose payload is a RetentionRunData with top-level fields run_started_at, run_completed_at, dry_run, and results: list[RetentionPolicyResult]. Each per-policy result inside results carries archived_rows, purged_rows, files_written, and the window boundaries (day_start, day_end).

GET /api/metrics/db/tables

Most recent per-table row-count snapshot. Refreshed on the cadence configured by DB_METRICS_INTERVAL_SECONDS. Returns DbStatsResponse.

GET /api/metrics/rest-rate

Per-exchange REST call rates + venue rate-limit utilization (rps_1s / rps_10s / rps_60s rolling windows, plus limit_rps + utilization when the venue's documented cap is known). Read from the in-process RestCallTracker snapshot. Returns RestRateResponse.

AI Delegates

AI delegate management is feature-gated by ai_integration_enabled; when disabled, these routes return a 503 feature_disabled envelope matching /api/mcp.

Route	Description
POST /api/ai-delegates	Create a delegate and return its 90-day access JWT once
GET /api/ai-delegates	List active delegates owned by the caller
GET /api/ai-delegates/{delegate_public_id}	Fetch one owned delegate; 404 also covers foreign IDs
PATCH /api/ai-delegates/{delegate_public_id}	SCD2 close+insert new caps for an owned delegate
POST /api/ai-delegates/{delegate_public_id}/deactivate	Deactivate via the shared kill-switch flow and revoke active tokens

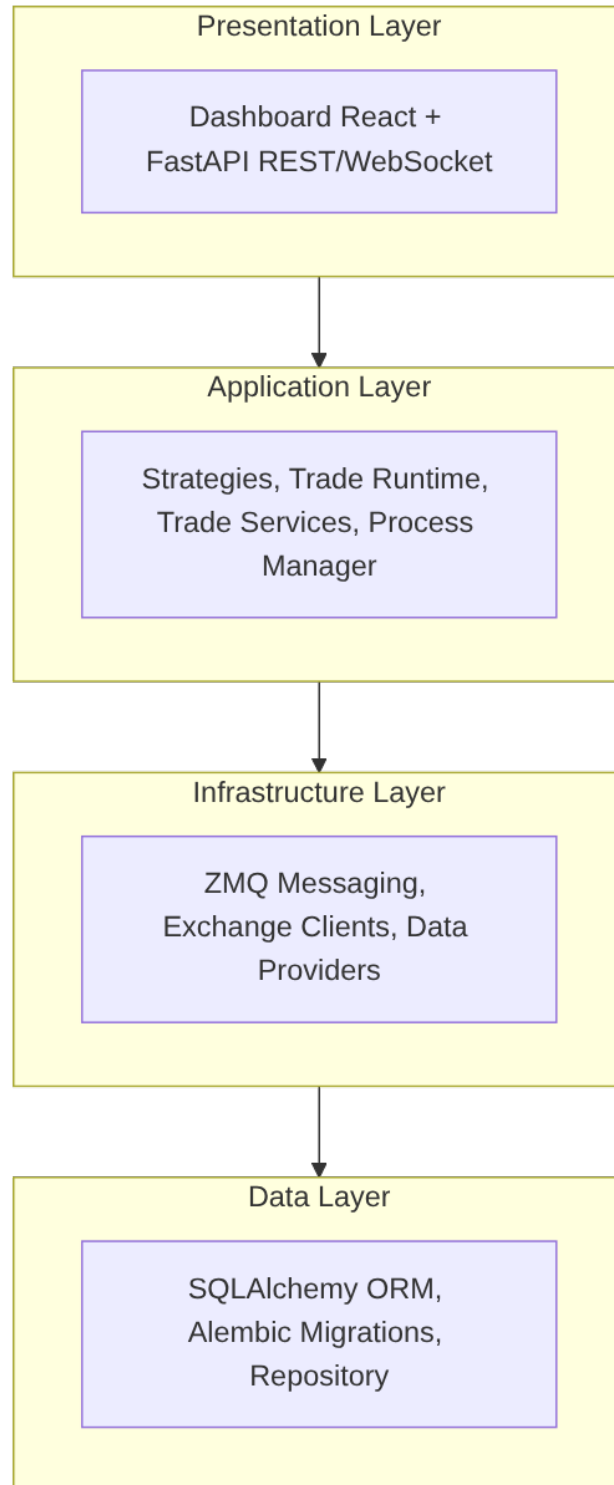
The routes are documented end-to-end in [ai-integration.md](#) alongside the Claude Code / Claude Desktop / Cursor / Windsurf wire-up instructions. They follow the same PayloadResponse envelope convention used by the auth routes above.

Architecture

Snapper is a trading platform built with a layered architecture using asynchronous processing and ZeroMQ messaging.

The trading path is facts-canonical: Order and Execution are the canonical business facts, while Position, Balance, and equity are derived projections that can be rebuilt from facts plus checkpoints.

System Layers



Components

Core (src/snapper/core/)

Fundamental types and aliases used throughout the application:

- TradeSide — Trade direction (buy, sell)
- OrderType — Order type (market, limit, stop, stop_limit)
- OrderStatus — Order status in lifecycle
- ExecutionMode — Execution mode (live, paper)
- OrderExchange — Order-capable exchanges (paper, kraken, kraken_futures, walutomat)

Config (src/snapper/config/)

Two-layer configuration:

1. Bootstrap Settings (bootstrap.py)

Settings from environment variables required before database connection:

- Database URL
- Master password for encryption
- HTTP and ZMQ server endpoints

2. App Settings (app.py)

Settings from database (encrypted):

- Exchange API keys
- Trading parameters
- Authentication configuration

Data (src/snapper/data/)

Persistence layer with SQLAlchemy:

• ORM Models (models.py):

- Instrument — Financial instruments (natural key: symbol_public_id + exchange). Instrument.public_id is the stable identity used by fact tables (orders, executions, positions, signals, candles). The versioned business attributes symbol_public_id and exchange are resolved via ensure_instrument()
- Candle — OHLCV candles
- Trade — Transactions
- Order — Canonical order lifecycle facts (includes mode: "live" or "paper")
- Execution — Canonical confirmed fills
- Position — Derived position projection (includes mode: "live" or "paper")
- TradeCommand — Durable trade intent written by the engine before execution

- VenueEvent — Durable venue observations and acknowledgements persisted by executors
- TradeProjectionCheckpoint — Materialized position/balance snapshot for fast recovery
- Signal — Signal events
- User — System users
- Setting — Settings (encrypted)
- Symbol — Symbol identity with versioned attributes (native_symbol, base, quote, asset_type) via SCD Type 2. Archive symbols for filesystem paths are derived from the anchor row (first version per public_id)
- SymbolAlias — Exchange-specific symbol aliases (one row per (symbol_public_id, exchange, channel))
- SymbolExchangeCapability — Exchange-specific symbol capabilities
- ProcessRun — Background process execution records
- InstrumentSpec — Instrument trading specifications (tick_size, margins, expiry_at, instrument_kind)
- UnderlyingAsset — Canonical asset identity (e.g., S&P 500, Gold) linking instruments across exchanges
- InstrumentUnderlyingMapping — Temporal link from instrument to underlying (exact/derivative/proxy)
- MarketSnapshot — Real-time market data snapshots
- InstrumentFeedHealth — Current per-symbol feed-health snapshot keyed by coordinator, exchange, native symbol, stream kind, and timeframe
- UserLoginEvent — Authentication event log
- Control — Always-on audit for commands, auth events, subscribe/unsubscribe, REST mutations. Includes redacted payload, outcome, and client causation linkage (client_session_id, client_public_id)
- Telemetry — Toggleable high-volume table for pings, heartbeats, pongs, and GET read requests. Recording gated by TELEMETRY_RECORDING_ENABLED setting

Most ORM models carry provenance via TemporalMixin (the few exceptions — UserActiveToken, AiDelegate, AiReview, AiReviewEvent, InstrumentFeedHealth — opt out because they manage their own lifecycle / revocation semantics):

- session_id (str, required) — producer session identity
- sequence_id (int, required) — per-table monotonic counter for gap detection

StrictDataSchema (Pydantic base) requires these fields at construction. Producers must obtain values from a SequenceTracker before creating the event, ensuring every payload is complete and identifiable from birth.

All ORM models use a dual-key identity pattern:

- `id` (INTEGER, internal PK) — row/version identifier for SCD2 close operations: SELECT → lock → close by id → INSERT new version. Never exposed outside the repository layer.
- `public_id` (UUID7 string) — logical entity identifier for domain joins and external references. Stable across SCD2 versions.
- `timestamp` (DateTime, system time / `known_from`)
- `known_to` (DateTime NOT NULL, default `KNOWN_TO_MAX` = 9999-12-31T23:59:59 UTC, active rows have `known_to` == `KNOWN_TO_MAX`; query pattern: WHERE `timestamp` <= :t AND `known_to` > :t)

No hard foreign keys between temporal entities. All domain joins use `child.instrument_public_id == Instrument.public_id` (or `Execution.order_public_id == Order.public_id`) with temporal filters. This enables future archival of closed SCD2 rows without breaking referential integrity.

- **SQLAlchemyRepository** (repository.py) — Async CRUD for SQLite/PostgreSQL
- **DatabaseRepository** (repository.py) — Sync access for scripts, archiver, and background updaters
- **CandleCacheArchiver** (archiver.py) — Exports active candle data to polygon-compatible CSV cache files per exchange/archive_symbol
- **EventArchiver** (archiver.py) — Exports append-only event rows (Tick, Trade, Signal, Execution, Telemetry, Control) to per-day CSV archive files with full temporal metadata, merge/dedup, and optional purge
- **CandleAuditArchiver** (archiver.py) — Exports all candle SCD2 versions (closed + active) with full temporal metadata, grouped by open_at date
- **StateArchiver** (archiver.py) — Generic archiver for state-SCD2 tables (Order, Position, Instrument, Setting, Symbol, etc.) with closed_only filtering and Symbol anchor row protection
- **Archive symbol resolution** (archive_symbols.py) — Stable filesystem-safe symbol naming derived from Symbol anchor rows, with seniority-based collision handling

Strategies (src/snapper/strategies/)

Trading strategies framework:

- **BaseStrategy** (base.py) — Base class for all strategies
- **StrategySignal** — Trading signal structure
- **StrategyConfig** — Strategy configuration

Built-in strategies:

- **RSIReversion** (rsi.py) — Mean reversion on RSI
- **MACDCrossover** (macd.py) — MACD crossover

- CointegrationPairs (cointegration.py) — Pairs trading

Indicators (src/snapper/indicators/)

Technical indicators:

- rsi — Relative Strength Index (Python implementation)
- macd — Moving Average Convergence Divergence
- ta_lib_adapter — TA-Lib library adapter

Messaging (src/snapper/messaging/)

ZeroMQ pub/sub architecture:



Components:

- **Broker** (infrastructure/broker.py) — XPUB/XSUB proxy
- **Publishers** (publishers/) — Market data publication
- **Executors** (executors/) — Per-wallet order execution on exchanges. One executor process per (exchange, wallet) pair is spawned at boot from active wallet_credentials rows; each instance filters incoming commands by wallet_public_id and loads its credentials via CredentialResolver during startup. Persisted executor_<exchange> rows are templates only; runtime instances are named executor_<exchange>_w<wallet_short> and bare templates are not started as live executors.
- **Schemas** (schemas/) — Pydantic message models
- **Topics** (topics/) — ZMQ topic definitions

Message types (in messaging.schemas.data):

- TickData — Price tick
- CandleData — OHLCV candle
- SignalData — Trading signal
- TradeData — Trade execution
- OrderData — Order status
- ExecutionData — Order execution
- OrderRequestData — Order request
- HeartbeatData — Component heartbeat

Every Data class carries public_id: str (UUID7), type: Literal[...], and timestamp: datetime.

Venue reconciliation (executors)

Executors can periodically poll exchange REST APIs to reconcile local pending-order state against the exchange's authoritative view. This detects two failure modes that WebSocket streams alone miss: **fill gaps** (exchange has more filled quantity than the executor observed) and **disappeared orders** (pending orders absent from the open-orders snapshot).

Always-on — runs alongside the executor's order handler + heartbeat tasks (no feature flag; the defensive polling is cheap enough that gating it provided no value).

_reconcile_fill_gap (in messaging/executors/base.py) emits a corrective ExecutionUpdate covering the observed cum-qty delta, stamping the synthetic execution with `exec_id=recon-<exchange_oid>-<monotonic_ns>` so downstream de-dup stays correct.

_reconcile_disappeared_order (in messaging/executors/base.py) calls `get_order()` on orders missing from the open-orders snapshot to determine their actual terminal status (CLOSED / CANCELED / EXPIRED), emits any residual fill gap first, then publishes the terminal ExecutionUpdate.

Known limitation — market orders without price

If a market order's exchange snapshot reports `price=None`, the fill-gap path logs an ERROR ("no price on market order, skipping corrective fill") and skips the corrective emission. The executor does not subscribe to ticks and cannot approximate the fill price without degrading VWAP accuracy silently. Recovery is lifecycle-dependent: if the order stays open and a later snapshot populates price, the next reconciliation iteration emits the corrective fill; if the order reaches a terminal state first, the skipped gap persists until reconciled manually against the exchange's fill history. Operators seeing this ERROR log should not patch Snapper to approximate; the skip is an intentional trade-off (predictable behaviour + observable gap) over silent approximation drift.

Application (src/snapper/application/)

Business logic:

- **Engine** (engine/) — TraderCoordinator is the multi-symbol trade runtime coordinator; TradingEngineService is the per-instrument engine that turns signals into TradeCommand rows and order requests
- **Trade** (trade/) — Trade-domain services: `trade_service.py` (in-memory command and position read model, fill deduplication, funding accrual application, and reconciliation halt feedback), `balance_service.py` (cash/equity/exposure projection), `outbox.py` (outbox-driven publishing with wake-up + polling fallback), `reconciler.py` (stale-command scan and reconciliation failure feedback)

- **Process Manager** (process_manager/) — Process management
- **Services** (services/) — Application services, including market_cache.py (in-process 1m candle cache and pair-stat snapshots) and market_persist_policy.py (runtime policy deciding which market-data streams are persisted to DB versus cache-only)
- **Updaters** (updaters/) — Data updates (symbols, historical)
- **Risk** (risk/) — Risk defaults and sizing models
- **Portfolio** (portfolio/) — Signed long/short portfolio accounting

Infrastructure (src/snapper/infrastructure/)

External integrations:

- **Exchanges** (exchanges/) — Exchange clients (Kraken, Walutomat)
- **Market Data** (market_data/) — WebSocket feeds
- **Symbols** (symbols/) — Symbol mapping
- **Security** (security/) — Settings encryption
- **Logging** (logging/) — Logging configuration

Server (src/snapper/server/)

FastAPI application:

- **App Factory** (app.py) — create_app() with lifespan management
- **REST API** — HTTP endpoints
- **WebSocket** — Real-time streaming via ZMQ bridge
- **Static Files** — Frontend dashboard
- **ClientProvenanceMiddleware** (provenance_middleware.py) — Extracts client provenance from mutation requests, runs per-session gap detection, records mutations to control table and GET reads to telemetry table

API (src/snapper/api/)

Shared API schemas and WebSocket auth helpers (not route definitions):

- **Schemas** (schemas/) — Pydantic request/response models (health, process, settings).

Schema class hierarchy:

```

text
BaseModel
├── PartialBody          - extra="ignore", strict=True (GapEnvelope,
WsTokenPayload)
├── StrictBody           - extra="forbid", strict=True (request bodies,
nested DTOs)
└── StrictDataSchema     - + provenance (all event payloads, REST/WS/ZMQ)

```

```

├── PayloadRequest[T]
├── PayloadResponse[T]
├── PayloadListResponse[T]
├── ExchangeResponse      – extra="allow", strict=True (exchange API
parsing)
├── ExchangeRequest      – extra="forbid", strict=True (outgoing exchange
requests)
├── BaseSettings          – BootstrapSettingsLoader (env var config)

```

- **Auth** (auth/) — WebSocket token service and schemas

Route modules live closer to their domains: server/app.py (assembly and data endpoints), server/process_routes.py, server/strategy_routes.py, config/settings_routes.py, and auth/routes.py.

REST endpoints return the same Data schemas used by WebSocket messaging (messaging.schemas.data): OrderData, SignalData, ExecutionData, PositionData, CandleData. Old REST-specific schema modules were removed.

Auth (src/snapper/auth/)

Authentication system:

- JWT tokens with access/refresh via HTTP-only cookies
- CSRF protection
- Role-based access (viewer, operator, admin, ai_delegate). The ai_delegate role is the principal type minted by the AI Delegates flow (see [ai-integration.md](#)) and is gated by its own permission matrix separate from human operators.
- WebSocket authentication

MCP (src/snapper/mcp/)

Model Context Protocol server mounted at /api/mcp as a Starlette sub-application. Exposes read/query tools (tools.py) consumed by AI delegates over Streamable HTTP. The mount re-applies its own middleware stack: feature-flag gating, bearer auth (auth.py), and per-principal rate limiting (rate_limiting.py); each tool handler runs output sanitization (output_sanitizer.py) and returns a structured error envelope (error_envelope.py). See [ai-integration.md](#) for the delegate flow and the @mateusz-klatt/snapper-mcp client plugin.

CLI (src/snapper/cli/)

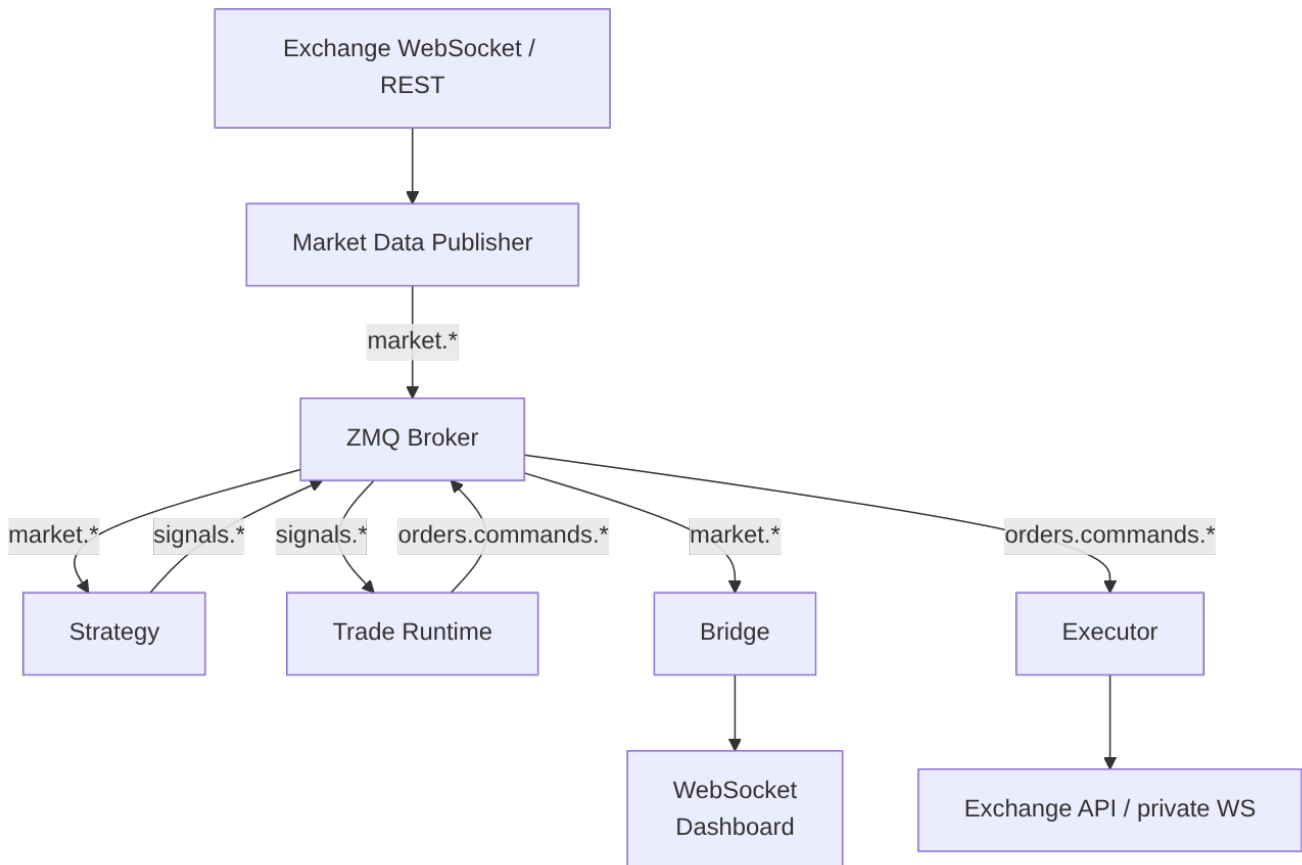
Command-line interface (Typer):

- Server and infrastructure management
- Database operations
- User management

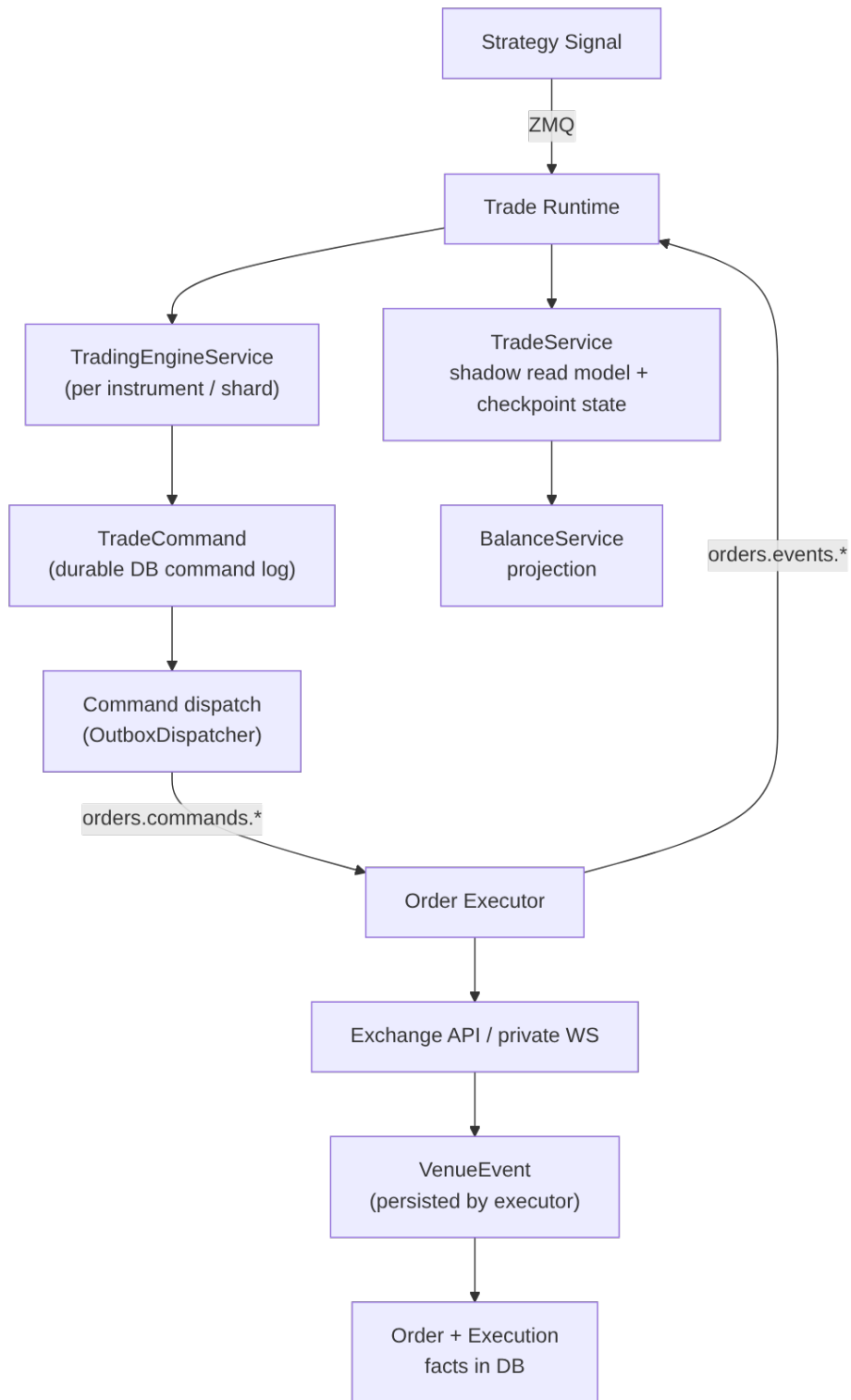
- Market data updates

Data Flow

Market Data and Signal Flow



Order Flow



Database

SQLite for development, PostgreSQL for production.

Schema

```
-- Market data (joined to instruments via instrument_public_id)
instruments          -- Financial instruments (logical key: symbol_public_id +
exchange)
candles              -- OHLCV data
ticks                -- Real-time price snapshots
trades               -- Transaction history
market_snapshots     -- Real-time market data (SCD2 per instrument, one active
row each)
instrument_feed_health -- Current per-symbol feed health snapshots

-- Trading (joined to instruments via instrument_public_id)
orders               -- Order history (mode: live/paper)
executions           -- Order executions (joined to orders via order_public_id)
positions            -- Portfolio positions (mode: live/paper)
position_cycles       -- Open/closed cycles tracking per-cycle peak qty
signals              -- Strategy signals
execution_plans       -- Plan envelopes (brackets, trailing stops, manual)
execution_plan_checkpoints -- Materialized plan state snapshots
execution_plan_decisions -- Per-tick plan decisions (audit trail)

-- Trade runtime (durable command path)
trade_commands        -- Durable trade intent (engine writes before
execution)
venue_events          -- Durable venue observations and acknowledgements
from executors
trade_projection_checkpoints -- Materialized position/balance snapshots

-- Multi-tenant
wallets               -- Trading accounts (label, is_paper)
wallet_credentials    -- Per-wallet exchange credentials (Fernet-
encrypted)
wallet_operator_scope_grants -- Per-wallet operator scope grants
operators             -- Operator identities (multi-tenant trade actors)
user_operator_memberships -- Operator memberships per user
user_trading_caps     -- Per-user trading-caps overrides

-- Symbol management
symbols               -- Symbol identity + versioned attributes (SCD2)
symbol_aliases        -- Exchange-specific symbol mappings
symbol_exchange_capabilities -- Exchange-specific capabilities
underlying_assets     -- Underlying-asset definitions (YAML-sourced)
instrument_underlying_mappings -- Active instrument → underlying mappings
continuous_contract_configs -- Continuous-contract series config

-- Instrument specifications
instrument_specs       -- Trading specifications (tick_size, lot_size,
limits)
instrument_order_capabilities -- Per-instrument order-side capability flags
venue_fee_schedules   -- Maker/taker fee schedules per venue

-- Funding + accruals
funding_rates         -- Per-instrument funding rate history (perps)
accrual_ledger        -- Funding / rollover / borrow accrual entries

-- Backtest
backtest_runs         -- One row per backtest run
backtest_events       -- Lifecycle events (started/failed/completed/...)
```

```

backtest_signals      -- Strategy signals emitted during a run
backtest_trades       -- Trades minted by the backtest engine
backtest_equity_points -- Equity curve samples
backtest_results      -- Final per-run metrics
backtest_comparisons  -- Pairing-grouping fingerprint anchors

-- AI Reviews + Delegates
ai_delegates          -- AI delegate principals (PAT-style auth)
ai_reviews            -- Human-in-the-loop review queue rows
ai_review_events      -- Per-review lifecycle audit trail

-- Notifications
alert_events          -- APNs alert payloads
alert_deliveries      -- Per-device delivery attempts
device_alert_prefs    -- Per-(device, alert_type, scope) preferences
notification_devices  -- Registered APNs device tokens
user_alert_defaults   -- Per-user fallback alert preferences

-- System
users                 -- User accounts
user_active_tokens    -- Active JWT inventory (revocation)
settings              -- Settings (encrypted)
user_login_events     -- Authentication event log
process_runs          -- Background process execution records
control               -- Audit log (commands, auth events, mutations)
telemetry              -- High-volume operational events (toggleable)

```

Migrations

Alembic manages migrations:

```

snapper db-upgrade    # Apply migrations
snapper db-downgrade  # Rollback

```

Trade Runtime

The trade runtime always uses the durable (outbox-driven) dispatch path.

TradingEngineService writes TradeCommand rows to the database; OutboxDispatcher polls the table and publishes undispatched commands to ZMQ. Executor VenueEvent writes are fail-closed — a failed persist raises before the executor acknowledges the venue event.

The TraderCoordinator class acts as the trade runtime coordinator and integrates TradeService (command lifecycle) and BalanceService (balance tracking). It spawns the outbox dispatcher (when a SQLAlchemyRepository is wired) plus per-exchange reconciliation loops that feed the circuit breaker. Canonical Order and Execution rows are persisted on the exchange/executor path.

Multiple TraderCoordinator processes can run against the same DB + broker with deterministic SHA-256 shard partitioning via `snapper.core.partitioning.ShardOwnership`. Each coordinator owns $\sim 1/N$ of the `shard_key` set; signals, venue events, recovery, outbox dispatch, and reconciliation all filter by ownership so no two instances process the same shard. Default `--instance-count 1` is byte-identical to single-coordinator behavior. See [docs/operations.md](#) for the systemd template recipe and scale-up / scale-down / crash-recovery procedures.

Bitemporal Model

Most entity tables inherit `TemporalMixin` which provides `id`, `public_id`, `timestamp` (bus-time / `known_from`), and `known_to` columns. A small number of tables that manage their own lifecycle (notably `user_active_tokens`, `ai_delegates`, `ai_reviews`, `ai_review_events`, and `instrument_feed_health`) opt out.

Key concepts:

- **SCD Type 2** — No in-place UPDATE or DELETE. Mutations use a close+insert pattern: the current row is closed (`known_to` set to now) and a new row is inserted with the updated values and `known_to = KNOWN_TO_MAX`.
- **KNOWN_TO_MAX** — Sentinel value (9999-12-31T23:59:59 UTC) marking the active version of a row.
- **Point-in-time queries** — REST endpoints accept an `as_of` query parameter to retrieve the state of data at a specific moment.
- **Helpers:**
 - `where_active(model, at)` — SQLAlchemy filter for temporal queries. `at`: datetime is required — no silent default-to-now
 - `where_active_now(model)` — convenience for operations that genuinely mean "current state" (auth login, heartbeat). Not a shortcut for skipping `as_of` threading
 - `close_and_insert()` / `close_and_insert_sync()` — Repository helpers that atomically close the current row and insert a new version

Batch atomicity guarantees

True SCD2 batch writes (`upsert_candles`, `upsert_market_snapshots`) and single-row close+insert methods (`revise_*`, `close_and_insert`) commit as all-or-nothing transactions. The pattern is:

```
async with self.session() as s:           # opens a transaction
    for row in rows:
        # SELECT ... FOR UPDATE the active version
        # UPDATE close (set known_to = bus_time)
        # s.add(new row)
    await s.commit()                       # single commit
```

If any row in the loop raises (`IntegrityError`, serialization failure, connection drop, await cancellation), the transaction aborts and **no** close/insert pair from this batch lands in the DB. The batch must be retried in full; there is no partial-progress mode.

`upsert_candles` batch-loads existing versions for unique (`instrument_public_id`, `timeframe`, `open_at`) keys before closing and inserting rows. If the incoming batch contains duplicate candle natural keys, those rows keep the sequential close+insert path because an earlier row in that group can create the version a later row must close. Lookup batches are chunked to stay below SQLite parameter limits.

Append-only batch writes do NOT use the SCD2 close-insert pattern and split into two shapes:

- `upsert_trades` delegates to the dialect-aware `_upsert_batch`. On native dialects (SQLite / Postgres) it uses `INSERT ... ON CONFLICT DO NOTHING` in a single `s.execute` — atomic per-batch, and duplicates are silently skipped. On the fallback row-by-row path it wraps each row in a `begin_nested()` `SAVEPOINT`; successful rows stay in the outer transaction and are committed at the final `s.commit()`, while rows that hit `IntegrityError` are skipped. This trades all-or-nothing for conflict tolerance.
- `upsert_ticks` issues a Core bulk insert via `session.execute(insert(Tick), list(rows))`. The function has two branches: when the caller passes a session, commit is external (caller-managed); otherwise it opens its own session and commits internally. Atomic per-batch on every dialect — it is not routed through `_upsert_batch` and has no duplicate-skipping behaviour.

Serialization on the same natural key: on **PostgreSQL**, row-level `SELECT ... FOR UPDATE` serializes concurrent transactions via MVCC row locks — a second session waiting on a locked active row observes the first session's close+insert only after the first commits; both batches complete without error and produce a clean SCD2 chain.

On **SQLite** (`aiosqlite`), `SELECT ... FOR UPDATE` is a no-op and the default `BEGIN DEFERRED` isolation does not acquire a write lock at `SELECT` time — two concurrent tasks can both read the same active row before either writes. The first writer to `COMMIT` wins; the second writer's `INSERT` then collides on the `public_id` `UNIQUE` index and raises `sqlalchemy.exc.IntegrityError`. The final DB state is still a valid SCD2 chain (exactly one active row, carried-forward `public_id` preserved, no orphan inserts) — one writer is simply reported the conflict via exception rather than implicitly serialized. Callers on SQLite that race on the same natural key must either retry the failed batch or rely on the single-publisher-per-exchange invariant to avoid the contention entirely.

At-scale implication: under heavy multi-publisher contention on the same natural key, batches serialize strictly. If a long-running batch holds the lock for many milliseconds per row, throughput falls linearly in contention. The current single-publisher-per-exchange shape makes this a non-issue; revisit if multi-publisher contention on the same natural key becomes a real workload.

Non-atomic fallback (_upsert_batch row-by-row path): the dialect-agnostic fallback (_upsert_batch in data/repository.py) wraps each row in a begin_nested() SAVEPOINT and continues on IntegrityError. Successful rows are NOT committed immediately — they are held in the outer transaction and finalised only by the final await s.commit(). This trades all-or-nothing for conflict tolerance — the caller sees inserted = N_successful, not all-or-nothing. Used only for append-only paths where a duplicate key is a genuine no-op. Never used for SCD2 close-and-insert.

Processes

The system manages processes through Process Manager:

- **Core** — Broker, Feed, Bridge (required — startup is aborted if any enabled long-running CORE process fails to start)
- **Strategy** — Trading strategies
- **Task** — One-time tasks
- **Backtest** — Backtesting processes

Processes can be:

- long_running — Run continuously (services)
- one_shot — Execute once (tasks)

Deployment Modes

By default, the FastAPI lifespan starts all enabled processes (all-in-one dev mode). Set SERVER_API_ONLY=true to skip process autostart -- the ZMQ-WebSocket bridge still starts, so the frontend receives live data from a separately-running engine. Useful when broker/strategies/executors run on different hosts.

Dedicated feed container (NYSE event-loop split): the market-data publishers (Kraken spot/equities/futures, Walutomat, paper) can be moved out of the FastAPI backend into a separate container so the API event loop stops sharing one CPU core with five WS publishers + their parse/persist work. Under NYSE open burst the shared loop saturated one core and starved the trade writer (drops); isolating each publisher onto its own process/core removes the contention.

PROCESS_AUTOSTART_PROFILE selects which registered processes a node autostarts (see snapper.core.types.ProcessAutostartProfileEnum):

- all (default) — every enabled process (single-container / dev).
- api — everything EXCEPT market-data publishers (the backend container: broker, executors, strategies, API).

- feed — ONLY market-data publishers (the feed container). Set by the feed-engine CLI command, which launches each publisher as its own OS subprocess (mode=PROCESS, uvloop) via ProcessLauncherService.start_feed_publishers. A market-data publisher is any registered process tagged both market-data and publisher.

A market-data publisher is filtered out of the backend's start_all_processes under the api profile, and get_core_health honours the same filter so a publisher that intentionally runs in the feed container is not reported as a missing CORE process on the backend.

The ZMQ broker stays in the backend (its tags are infrastructure, not publisher, so the api profile keeps it). Cross-container wiring uses two endpoint pairs because ZMQ bind rejects hostnames while connect resolves them:

- ZMQ_BROKER_BIND_XSUB / ZMQ_BROKER_BIND_XPUB — the routable interface the broker binds (tcp://0.0.0.0:7500 / :7501). Empty (default) falls back to the connect endpoints, so single-container behaviour is unchanged.
- ZMQ_BROKER_XSUB / ZMQ_BROKER_XPUB — the connect endpoints every process (backend's own components AND the feed container's publishers) dials. Both containers set these to the backend service name (tcp://snapper:7500 / :7501).

DB_POOL_SIZE / DB_MAX_OVERFLOW clamp each publisher subprocess's SQLAlchemy pool (PostgreSQL only). get_repository caches one engine per process, so splitting publishers across processes otherwise multiplies the default pool (5 + 10) per process toward Postgres max_connections. Keep count(publishers) * (DB_POOL_SIZE + DB_MAX_OVERFLOW) + backend pool under the server's max_connections.

Deploy notes (see docker-compose.yml snapper-feed service):

- Bring up the backend (api) and feed (feed) together. A partial docker compose up snapper with the api profile but no feed container stops all market-data ingest, so deploy both atomically.
- The broker's bound endpoint is seeded into its DB process-config parameters by registry sync on a fresh database. An existing deployment whose broker config already carries tcp://127.0.0.1:7500 must have that row re-synced (or the parameters cleared) so the broker picks up tcp://0.0.0.0:*; otherwise the feed container cannot reach it.
- Verify outbound Kraken WS from the feed container routes through snapper-egress (same egress_pool setting + network as the backend).

Multi-instance deployment (AI-review fanout dedup): multi-worker uvicorn is a supported production topology. The shared ZMQ XPUB-XSUB broker delivers every internal bus event to every FastAPI worker's `AiReviewService.start_bus_listener`; each subscribed topic has its own dedup story so connected clients see exactly one external WS frame per source event regardless of worker count:

- `bus.delegate_offline` is idempotent at the DB layer — the handler runs a per-row `atomic_dispatch_fanout_with_audit CAS UPDATE`, so only one worker wins per affected review row even when every worker receives the bus event.
- `bus.ai_review_decision` is process-local — the handler resolves an `asyncio.Future` from the per-worker registry populated by the worker that ran `create_review` for the strategy await loop. Other workers stash the event in a bounded pending-resolution cache (1024-entry hard cap, 30s TTL); cache memory across the cluster is bounded by $N \times 1024$ entries even at peak.
- `bus.caps_violation_after_ai_approve` is gated by deterministic partitioning. `handle_caps_violation_bus_message` short-circuits on `ShardOwnership.owns(msg.review_public_id)`; exactly one worker (the SHA-256 owner of the review row) re-publishes the external `ai_reviews.*.caps_violation` frame. Other workers observe the bus event, log a debug skip, and return `False`. `instance_count == 1` (default deployment) makes `ShardOwnership.owns` return `True` unconditionally so behaviour is byte-identical to single-worker.

The same partitioning primitive backs the Trade Runtime shard ownership; AI-review reuses it via the `set_shard_ownership(ShardOwnership(coordinator_instance_id, coordinator_instance_count))` injection seam wired by the FastAPI lifespan after the publisher seam and before the bus listener starts.

Operational scaling: raise `SNAPPER_COORDINATOR_INSTANCE_COUNT` to `N` and start `N` FastAPI workers each with a distinct `SNAPPER_COORDINATOR_INSTANCE_ID` in `[0, N)` (env vars only — snapper server does not expose per-worker CLI flags; the `--instance-id / --instance-count` CLI flags exist on `snapper trade-zmq` for the Trade Runtime coordinator partitioning, not on the FastAPI server). Each worker subscribes to every bus topic; only the SHA-256 owner runs the caps-violation external fanout.

Execution Plans

The Execution Plans framework provides a unified control plane above the existing TradeCommand spine. All manual and algorithmic trading actions become `ExecutionPlan` instances evaluated by pluggable `PlanEvaluator` classes running inside `PlanExecutorService`.

Tables: execution_plans (plan state + lifecycle), execution_plan_checkpoints (high-churn evaluator state), execution_plan_decisions (tiered decision log), instrument_order_capabilities (capability matrix), venue_fee_schedules (fee tiers), position_cycles (one open→close lifetime per shard, brackets attach by position_cycle_public_id).

Plan types: manual_once (shipped), bracket (shipped — attaches to a position_cycles row), trailing_stop (shipped — stateful ratcheting stop with checkpoint persistence, separate /api/trailing-stops route module), peg, scheduler.

Manual order create flow: POST /api/orders creates a manual_once plan (pending), stamps child_client_order_id, native_instrument, and venue_order_type into plan.params, then inserts a create TradeCommand for the outbox dispatcher and transitions the plan to active. The OutboxDispatcher inside TraderCoordinator publishes the command as OrderRequestData on the orders.commands.{ex}.{instr}.submit topic.

Fill propagation: PlanExecutorService subscribes to orders.events. and market. on the broker XPUB and consumes every fill. Cumulative ExecutionData.size is compared against plan.filled_quantity with a small epsilon so duplicate or stale frames are dropped. Partial fills update filled_quantity and keep the plan active; full fills transition the plan to completed with completed_at stamped via SCD2.

Cancel flow: POST /api/orders/{plan_public_id}/cancel (plan id path) or POST /api/orders/by-client-order-id/{client_order_id}/cancel (UI convenience route) resolves the plan, verifies wallet scope, transitions the plan to cancel_requested, hydrates exchange_order_id from the active orders row, and inserts a cancel TradeCommand. On insert failure the plan is rolled back to failed and the route returns HTTP 500 — this is terminal (no retry). PlanExecutorService._recover_plans on the next service restart re-emits cancel commands only for plans still in cancel_requested state (deduplicated via has_pending_cancel_command); failed plans are not retried. The outbox dispatcher branches on command_type and re-hydrates exchange_order_id one more time at publish time so a late venue ACK is picked up, then publishes OrderCancelData on the orders.commands.{ex}.{instr}.cancel topic.

Listen loop resilience: per-message exceptions (parse failures, handler errors) are caught inside the loop so a single bad frame cannot silently stop the service. Transient recv_multipart errors retry with a 100 ms backoff. asyncio.CancelledError still unwinds the loop for graceful shutdown. After _setup_subscriber() the service sleeps 500 ms for ZMQ XPUB/XSUB slow-joiner stabilization before recovery to avoid losing terminal events triggered by stranded-cancel re-emits.

Position cycles: the position_cycles table represents one flat→non-flat→flat lifetime per shard via the standard TemporalMixin SCD2 envelope. A partial unique index uq_pc_shard_open_active enforces at most one open cycle per shard_key at any instant. Five repository methods cover the lifecycle: insert_position_cycle, close_position_cycle (SCD2 close-and-insert), get_open_position_cycle (lookup by shard_key because paper

shards embed `strategy_tag`), `flip_position_cycle` (TOCTOU-hardened atomic close+open with `shard_key` match assertion), and `update_position_cycle_max_qty` (monotonic — silent no-op when the new peak does not strictly exceed the existing one). The repository methods are called from the trader (not from `TradeService`, which stays sync and pure); `ShardState` carries trader-owned cache fields `active_cycle_public_id` and `active_cycle_max_qty`.

The trader's `_sync_fill_to_trade_service` samples `old_qty` before `apply_venue_event` and `new_qty` after, then delegates to `_sync_position_cycle_on_fill`. That helper classifies the transition via `TradeService._detect_cycle_transition` (returns `Literal["open", "close", "flip", "scale_up"] | None` with a `1e-12` epsilon to match the existing zero-snap) and issues the matching repository call. Four guards apply: degraded-identity fail-closed when `engine.wallet_public_id` is falsy, idempotency on open via a `get_open_position_cycle` pre-check, async `instrument_public_id` resolution via `get_instrument_public_id_by_symbol`, and a symmetric DB fallback on close/flip/scale_up paths so a degraded restart cannot reuse a stale cycle row from a prior life of the shard. When a flip's instrument resolution fails, the trader degrades to close-only — the old cycle is closed cleanly, the new leg is left uncovered with an audit-gap warning, and the cache is cleared so subsequent fail-soft paths handle themselves correctly.

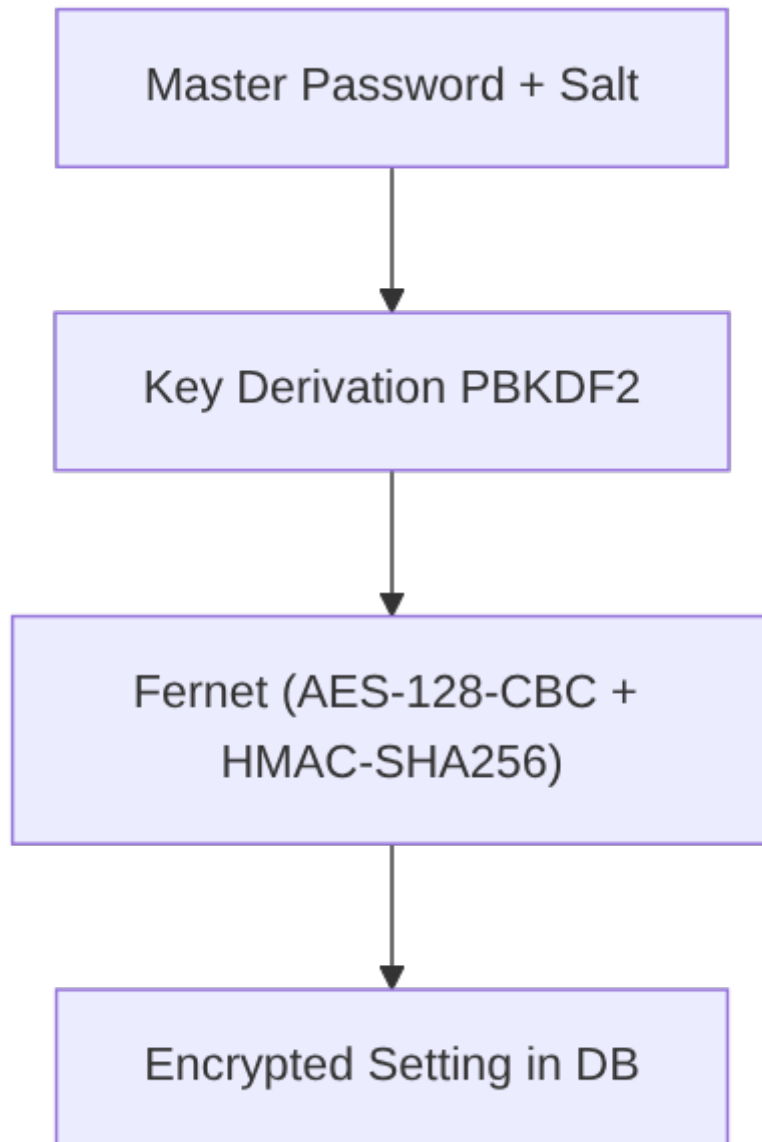
`_recover_engine_state` runs `_reconcile_position_cycles` as a fourth recovery phase, after engine state is rebuilt and before the trading loop starts. It iterates `self.engines` (engines, not `TradeService`, because full replay rebuilds `engine.position_qty` but not the projection) and handles four cases: recovered flat with a stale open row → close the row; recovered non-flat with a matching-direction open row → hydrate the cache and bump `max_qty` if downtime scaled-up beyond the stored peak; recovered non-flat with an opposite-direction row → atomic flip via `flip_position_cycle` (or degrade to close-only on unresolved instrument); recovered non-flat with no row → bootstrap a synthetic cycle. Brackets attach to `position_cycle_public_id`, not to an order, so a flat reopen does not inherit stale stop levels. Orphan cycles (open rows without a matching engine) can be detected via `GET /api/position-cycles/open` and closed individually with `POST /api/position-cycles/close-orphan` or in bulk with `POST /api/position-cycles/sweep-orphans`.

max_qty is per-cycle, not lifetime. Each `position_cycles` row tracks the peak absolute quantity reached within its own `opened_at` → `closed_at` window. A new cycle on the same shard starts a fresh accumulator at `abs(opening_qty)`; it does not inherit or max against prior cycles' peaks. This matches the bracket-evaluator contract: a bracket attaching to cycle C1 is sized against C1's peak, not against all-time peak. A future UI surfacing `max_qty` must scope to the individual cycle row; summing or max-ing across cycles would conflate independent trading episodes.

Security

Settings Encryption

Sensitive data (API keys) are encrypted in the database:



Backtesting

The backtesting subsystem runs strategy simulations against historical candle data. It supports both direct repository-backed candle streaming and ZMQ replay through the same strategy processing contract.

Architecture:

- **BacktestConfig** (Pydantic): Strategy, instruments, date range, fill model
- **DirectDbEngine**: Candle-driven simulation loop with time-batched processing
- **ZmqReplayEngine**: Replay mode that publishes historical candles through ReplayPublisher on the same topic shape used by live feeds

- **ResultCollector:** In-memory artifact buffer (signals, trades, equity)
- **BacktestMetrics:** Pure-math metrics computation (Sharpe, Sortino, CAGR, etc.)
- **BacktestRunnerProcess:** ONE_SHOT RegisterableProcess orchestrating lifecycle
- **BacktestRepository:** Separate repo sharing session_factory, SCD2 status transitions

Lifecycle: pending -> running -> completed | failed | cancelled

Boot reconciliation: Orphaned runs (pending/running/cancel_requested) are swept to failed at server startup via reconcile_stale_runs(). This runs only on the normal startup path (skipped in SERVER_API_ONLY mode).

Storage: 7 bitemporal tables (BacktestRun, BacktestEvent, BacktestSignal, BacktestTrade, BacktestEquityPoint, BacktestResult, BacktestComparison) all with TemporalMixin.

Authentication

- JWT tokens via HTTP-only cookies (access: 15min, refresh: 7 days)
- CSRF tokens for mutating requests
- WebSocket tokens (one-time)
- Bcrypt password hashing

Kraken FCM market data (TradFi)

Kraken's FCM platform hosts index-futures, energy, metals, and yield contracts (collective name: "TradFi" in this project). Unlike crypto spot + perpetuals, FCM has **no public order API** — Snapper treats these instruments as observation-only.

Data sources

- REST instrument metadata: iapi.kraken.com/api/internal/markets/all/futures-contracts. Reverse-engineered from pro.kraken.com (requires Origin: <https://pro.kraken.com> + Referer: <https://pro.kraken.com/> headers). Powers the `kraken_equities_symbol_updater` process.
- REST historical candles: iapi.kraken.com/api/internal/markets/{ws_symbol}/ticker/history. Same Origin/Referer requirement. Drives `KrakenEquitiesExchangeClient.get_ohlcv()` and the `kraken_equities_aggregates_backfill` process. Intervals {1, 5, 15, 30, 60, 1440} minutes accepted.
- WebSocket ticks + trades: <wss://ws-equities.kraken.com> (Kraken Spot WS v2 protocol with `asset_class: futures_contract`). Feed is delayed ~10 minutes per FCM policy. The outer envelope's `delayed: true` flag propagates onto every `TickData.is_delayed=True` so ZMQ subscribers can gate accordingly.

Capability model

Each TradFi symbol is seeded with a `SymbolExchangeCapability(exchange="kraken_equities", can_trade=False, can_market_data=True)` row. Every order-entry REST route (`/api/orders`, `/api/execution-plans`, `/api/trailing-stops`) calls `snapper.server._capability_guard.require_tradable` before emitting a `TradeCommand` — market-data-only submits fail with HTTP 422 `error_code=instrument_market_data_only`. The AST meta-test at `tests/meta/test_order_entry_capability_guard.py` enforces that every enumerated submit handler imports and calls the guard.

Frontend UX

`GET /api/exchanges/{exchange}/instruments/detail` exposes capability-aware rows (`InstrumentDetailData` — `symbol` + `can_trade` + `can_market_data` + `instrument_kind` + `expiry_at`). `MarketData.tsx` renders a "Market-data only" badge next to the page title when the currently-selected instrument is observation-only and marks every `can_trade=false` row in the instrument dropdown with the same pill. `NewOrderModal.tsx` consumes the same endpoint, shows an amber inline notice, suffixes option labels with — market-data only, and disables the submit button. Submit failures branch on `APIError.details.error_code` via `errorMessages.ts`.

Cross-asset strategy pattern (informational)

Strategies may subscribe TradFi candles via the normal `market.kraken_equities.{symbol}.candles.{timeframe}` topic and emit signals whose instrument targets a `can_trade=True` execution instrument (crypto spot, xStocks). See `src/snapper/strategies/examples/tradfi_observe_crypto_execute.py` for an illustrative EMA-crossover implementation + activation instructions in the module docstring. The backtest engines process multi-feed batches and route simulated fills to the signalled execution instrument. `BacktestConfig.target_execution_exchange` selects the fill venue when it differs from the source feed, and both direct-DB and ZMQ replay modes build topics for every configured exchange/instrument feed.

Operational runbook

See `docs/operations.md` "Kraken Equities (TradFi) market data" for enable-the-feed steps (SQL DB Setting toggles), backfill invocation, quarterly rotation cadence, and troubleshooting (iapi reachability checks, WS disconnect loops).

Backtesting

Snapper ships two backtest execution engines, both producing the same artifacts (signals, trades, equity points) so a strategy author can pick either path with confidence.

Execution modes

The `execution_mode` field on `BacktestConfig` selects which engine the runner instantiates:

- `direct_db` — `DirectDbEngine` reads candles directly from the repository and feeds them into the strategy in-process. Lowest overhead; recommended for fast iteration.
- `zmq_replay` — `ZmqReplayEngine` spins up a private XPUB/XSUB broker on OS-assigned local ports, attaches the strategy as a SUB, and publishes candles via a `ReplayPublisher`. Exercises the full live message-bus path, so any production wiring bug surfaces in the backtest.

Both engines call the same `process_time_batch` helper for fill simulation, signal recording, and equity sampling, so artifact parity is structural rather than aspirational.

Choosing between modes

Concern	<code>direct_db</code>	<code>zmq_replay</code>
Speed	fastest (no IPC)	slower (per-candle ZMQ hop)
Fidelity	bypasses message bus	mirrors live wire path
Concurrency	at most one active run (global <code>uq_bt_single_running</code> invariant — see below)	at most one active run (same DB-enforced invariant)
Cancel SLA	bounded by <code>cancel_poll_ms</code>	bounded by <code>cancel_poll_ms</code>

At-most-one-running invariant

Migration 0001_init includes a partial unique index `uq_bt_single_running` on `backtest_runs(status)` filtered to `status='running'` AND `known_to=KNOWN_TO_MAX`. This is a declarative DB guarantee — a second runner trying to transition `pending` → `running` receives `IntegrityError` at COMMIT, the runner catches it via `is_single_running_conflict` and transitions to `failed` with a clear error message. No engine, no broker, no ports consumed for the loser.

The constraint is dialect-aware (PostgreSQL via `asyncpg` + SQLite via `aiosqlite`); see `snapper.data.backtest_conflict` for detection details.

Cancellation

CancelProbe polls `backtest_runs.status` between time batches (Direct-DB) or per processed candle (ZMQ replay). Throttled by `config.cancel_poll_ms` (default 500 ms), each DB read bounded by `probe_timeout_s` (default 1.0 s). A stuck DB logs a warning and the next probe re-tries — cancel detection degrades gracefully under SQLite lock contention rather than hanging behind the driver's 30 s lock timeout.

End-to-end cancel SLA target: ≤ 5 s from the API call `POST /api/backtests/{run_id}/cancel` (which sets status to `cancel_requested`) to terminal status `cancelled`.

start_date warmup gating

Candles before `config.start_date` feed strategy indicators (so SMA/MACD warm up properly) but do **not** generate fills, signals, or equity points. The `process_time_batch` helper enforces this with a single `if batch[0].open_at < config.start_date: return guard` before fill simulation.

Both engines honour the gate identically; the parity test `tests/integration/test_backtest_hardening.py::test_warmup_gating_parity_between_engines` asserts field-for-field equality.

ZMQ replay implementation notes

The replay path adds a few primitives the live system does not need:

- **Echo-ack handshake:** the publisher sends a sentinel `WARMUP_PUBLIC_ID` candle on every expected topic each round; the strategy mixin's `_listen_loop` ACKs each topic into `state.acked_topics` and only sets `state.subscriber_ready` when the set equals `state.expected_topics`. Up to `WARMUP_MAX_RETRIES=5` rounds of `WARMUP_READY_TIMEOUT_S=1.0` s each; failure raises `BacktestReadinessTimeoutError`.
- **Drain coordinator:** counts published vs processed candles. After the publisher streams the last candle it calls `mark_done_publishing()` and waits on `drained.wait()` bounded by `DRAIN_TIMEOUT_S=10.0` s. Failure raises `BacktestDrainTimeoutError` with both counters in the message so an operator can tell publisher- stall from strategy-death.
- **Bounded cleanup:** the engine cancels publisher and listen tasks then `asyncio.wait(timeout=2.0)` — tasks that refuse to terminate are logged as leaks rather than silently kept alive.

- **Shielded DB writes:** the runner wraps every terminal-status DB write (cancelled, failed, fail-event insert) in `asyncio.shield` so a second cancel arriving mid-write does not interrupt the write coroutine.

Advanced metrics, live progress, comparison

The backtest surface ships three interlocking capabilities:

Advanced metrics

Eight additional metric columns on `backtest_results` — five promoted from the `extra_metrics` JSON blob (`sortino_ratio`, `cagr`, `calmar_ratio`, `expectancy`, `avg_trade_pnl`) plus three net-new metrics computed in `snapper.application.backtest.metrics`:

- **max_drawdown_duration_seconds** — longest peak-to-recovery window; unrecovered runs use the final suffix from last peak to end-of-curve. Degenerate input (fewer than two points, flat curve, strictly monotone) returns `None` + a `metric_warning` event.
- **exposure_ratio** — fraction of total run duration during which the portfolio held a non-zero position; leading-edge interval attribution. Zero-trade runs with non-zero duration return `0.0` (not degenerate).
- **turnover_ratio** — total notional traded divided by mean equity. Zero-trade returns `0.0`; non-positive mean equity returns `None` + warning.

Read-side fallback at `GET /api/backtests/{id}` collapses legacy `extra_metrics` JSON into the typed slots using explicit `is not None` coalescing (never Python truthiness — preserves legitimate `0.0` values). The response strips the five promoted names from the emitted `extra_metrics` so mixed-vintage rows never surface the same metric twice.

Live progress (4-segment WS topic family)

Runner emits `BacktestProgressData` events on `backtest.{wallet_public_id}. {run_public_id}. {event}` where `event` $\in$ `{started, progress, milestone, completed, failed, cancelled}`. The existing ZMQ→WS bridge forwards the envelope to every subscribed client.

- `BacktestProgressEvent` is the canonical Literal source of truth in `snapper.messaging.schemas.data`; topic validator, emitter, and frontend handlers all derive from it.
- `BacktestProgressEmitter` throttles progress at 250 ms, emits milestone events exactly once per 25/50/75 pct bucket (bypassing the throttle so milestones + progress at the same step do not collide), and fires started / terminal events once. Degrades gracefully when `total_candles=None` by disabling milestones and pinning `progress_pct=0.0`.

- Per-subscription RBAC: admins subscribe to any wallet prefix; non-admin roles subscribe only to prefixes matching their `active_wallet_public_id`. Foreign-wallet or bare backtest. subscriptions are denied at the handler before the bridge is touched.
- `BacktestProgressData.milestone` carries a `@model_validator` enforcing the cross-field invariant milestone is not None iff `event == 'milestone'` so a malformed payload cannot reach the frontend chip logic.

Config hash for auto-pair

Every new run persists a `config_hash` via `compute_fingerprint(config, for_pairing=True)` — SHA-256 over the 10 pairing-stable fields (`strategy_class`, `instruments`, `start_date`, `end_date`, `initial_balance`, `strategy_params`, `timeframe`, `fill_model`, `slippage_bps`, `commission_bps`), plus `target_execution_exchange` when it is set on the config (see "Cross-asset execution" → "Fingerprint + pairing" below — legacy default-None runs keep their exact prior hash). `execution_mode`, `snapshot_as_of`, `warmup_bars`, and `buffer_size` are explicitly excluded so Direct-DB and ZMQ replay runs on the same config share a hash.

GET `/api/backtests?config_hash={hash}` returns sibling runs for the auto-pair UI; the repository query is wallet-scoped and indexed on (`wallet_public_id`, `config_hash`, `timestamp`).

Comparison

POST `/api/backtests/compare` creates (or returns existing) a comparison row; GET `/api/backtests/compare/{id}` returns the comparison metadata plus the diff **recomputed from current artifact rows** so metric-schema changes never stale a persisted diff. The diff surfaces four shapes:

- `metrics_diff` — per-name `{run_a, run_b, delta, pct}` with explicit is not None precedence.
- `equity_overlay` — full-outer-joined equity samples on `point_time` with nullable per-leg fields.
- `trades_diff` — multiset matching on (`instrument`, `executed_at`, `side`, `quantized_qty`, `quantized_price`) at 1e-8 precision so IEEE drift collapses but real ticks stay distinct. Common entries carry `pnl_a/pnl_b/pnl_delta`.
- `signals_diff` — multiset on (`instrument`, `signal_time`, `signal_type`).

Route ordering: compare endpoints are registered **before** `/api/backtests/{run_id}` so `/compare` is not captured as `id="compare"`. Pair is normalised to lexical (min, max) before insert.

Duplicate submit is idempotent via SELECT-then-INSERT with an IntegrityError rollback+re-SELECT race recovery path.

No-active-wallet fail-closed

Every backtest read endpoint (list, detail, trades, signals, events, equity, compare) returns **400 no active wallet selected** when `principal.active_wallet_public_id` is `None`. WS subscribes respond with a `subscription_success` frame whose status is denied (or partial when some requested topics succeeded), with the offending topics surfaced in `denied_topics`. The picker triggers a `selectWalletAndRefresh` on change that mints a new JWT carrying the chosen wallet claim before swapping the client scope, so REST and WS both authorise against the same wallet after the picker moves.

Cross-asset execution

Cross-asset strategies observe market data on one venue / instrument and emit signals that execute on a *different* venue / instrument (for example `TradFiObserveCryptoExecute` observes MNQM6-CME candles on `kraken_equities` and emits BUY/SELL signals on BTC-USD / `kraken`). The backtest engine supports this through a single config field.

`TradFiObserveCryptoExecute` is an illustrative, non-runnable example: it is **not** auto-registered, so `BacktestConfig.validate_strategy_class` rejects the config below as written. To run it, first copy the example module to the strategies package root and decorate it with `@register_strategy`, or substitute a registered strategy class (`RSIReversion`, `MACDCrossover`, or `CointegrationPairs`):

```
BacktestConfig(
    strategy_class="TradFiObserveCryptoExecute",
    instruments={
        "kraken_equities": ["<MNQM6-CME-public-id>"],
        "kraken": ["<BTC-USD-public-id>"],
    },
    target_execution_exchange="kraken",
    start_date=..., end_date=...,
    wallet_public_id=...,
    initial_balance=10_000.0,
    strategy_params={"fast_period": 12, "slow_period": 26, "min_candles": 30},
)
```

Attribution semantics

When `target_execution_exchange` is not `None`, the batch processor substitutes at simulated-fill time:

- `target_exchange = str(config.target_execution_exchange)` — carries the *venue label* only. It is NOT validated against the feed populating `latest_closes[target_instrument]`; the caller is responsible for configuring instruments so the target symbol's feed matches the target venue.

- `target_instrument = signal.instrument` — comes directly from the strategy's `StrategySignal`, unchanged since pre-cross-asset.
- `target_close = latest_closes[target_instrument]` — the *price* at which the simulated fill executes. The source feed's close price is no longer used on cross-asset runs.

When `target_execution_exchange` is `None` (default), the fill is attributed to `event.exchange + signal.instrument`. In legacy single-feed strategies `signal.instrument == event.instrument` by convention (verified across every in-tree emitter at `rsi.py`, `macd.py`, `cointegration.py`), so the recorded exchange / instrument / price are byte-identical with the pre-v1.2 path — every existing single-feed backtest keeps its exact fingerprint and result rows.

Missing-target-close policy

If `signal.instrument` has no entry in `latest_closes` (typically during the warmup overlap window where the strategy fires before the target feed has primed), the engine:

1. Skips `simulate_market_fill` + the trade row (no fill attempted).
2. Increments `collector.cross_asset_blocked_fills` by one, logged at `DEBUG` with `reason='missing_target_close'`.
3. Still records the signal row with `price = float(signal.price)` (source-close fallback) so downstream analytics see the strategy's intent.

At run end the counter surfaces through

`BacktestResultInsertRow.extra_metrics["cross_asset_blocked_fills"]` *only when positive*.

Single-feed runs keep `extra_metrics == {}` byte-identical with pre-cross-asset behaviour.

Scope limits

Two explicit non-goals in the current implementation:

- **Same-symbol multi-venue** — `Portfolio.positions` is keyed by instrument only, so running BTC-USD on Kraken Spot + Kraken Futures simultaneously in one backtest is NOT supported. Cross-asset strategies in scope use distinct symbols across feeds.
- **Multi-target cross-asset** — a single `target_execution_exchange` per run means one strategy cannot emit signals for different target venues inside the same backtest.

Supported: cross-asset via REST

`POST /api/backtests` accepts an optional `target_execution_exchange` field on `BacktestCreateBody`. When set, simulated fills are attributed to that order-capable venue while candles still feed from exchange. When unset, the run stays single-exchange and

byte-identical to legacy behaviour. The same field round-trips through DB persistence + the rerun endpoint, and surfaces on BacktestRunData for frontend display. Allowed target values: paper / kraken / kraken_futures / walutomat.

Fingerprint + pairing

compute_fingerprint includes target_execution_exchange in the payload **only when non-None**, so legacy (default-None) runs keep their exact pre-cross-asset hash in both the default and for_pairing=True paths. This preserves dedup cache validity and the auto-pair UI grouping logic in the _resolve_auto_pair resolver in backtest_routes.py. Explicit cross-asset runs (field set to a concrete venue like "kraken") generate distinct fingerprints so they never collide with single-venue baselines.

CLI

Snapper provides a command-line interface built on the Typer framework. All commands are available through the `snapper` command.

Usage

```
snapper [COMMAND] [OPTIONS]
```

Help for a specific command:

```
snapper [COMMAND] --help
```

Server and Infrastructure

server

Starts the FastAPI server with the web dashboard.

```
snapper server [OPTIONS]
```

Options:

Option	Type	Default	Description
<code>--host</code>	string	from env	Listen address
<code>--port</code>	int	from env	Server port
<code>--reload / --no-reload</code>	bool	from env (<code>SERVER_RELOAD</code>)	Auto-reload for development; explicit flag overrides the env-var setting

Environment:

Set `SERVER_API_ONLY=true` to skip process autostart while keeping the ZMQ-WebSocket bridge alive (useful when the engine runs as a separate process). Multi-worker uvicorn (multiple FastAPI processes sharing a broker) is supported — see the AI-review fanout dedup contract in `docs/architecture.md` (Deployment Modes); set `SNAPPER_COORDINATOR_INSTANCE_ID` + `SNAPPER_COORDINATOR_INSTANCE_COUNT` per worker (env vars only — `snapper server` does not expose per-worker CLI flags; the --

instance-id / --instance-count flags in the table below are on snapper trade-zmq) to enable the shared partitioning that prevents duplicate caps-violation WS frames under $N > 1$.

Examples:

```
# Start with default settings (all-in-one dev mode)
snapper server

# Start on all interfaces
snapper server --host 0.0.0.0 --port 8000

# Development mode with auto-reload
snapper server --reload

# API-only mode (engine runs separately)
SERVER_API_ONLY=true snapper server
```

broker

Starts the ZeroMQ XPUB/XSUB broker for message routing.

```
snapper broker [OPTIONS]
```

Options:

Option	Type	Default	Description
--xsub	string	from env	XSUB endpoint (publishers)
--xpub	string	from env	XPUB endpoint (subscribers)

Example:

```
snapper broker --xsub tcp://127.0.0.1:7500 --xpub tcp://127.0.0.1:7501
```

trade-zmq

Starts the central trade runtime coordinator.

The process still runs as the trade-zmq command, but after the trade runtime redesign it also hosts TradeService, BalanceService, the outbox dispatcher, and the reconciliation loops. It writes TradeCommand rows, consumes orders.events.* to keep projections in sync, and persists checkpoints for recovery.

```
snapper trade-zmq [OPTIONS]
```

Options:

Option	Type	Default	Description
--signal-topics	string	signals.	Signal topics to subscribe
--instance-id	int	0 (or \$SNAPPER_COORDINATOR_INSTANCE_ID)	Multi-instance partitioning: zero-based id for this coordinator in a multi-instance deployment. CLI flag > env var > default.
--instance-count	int	1 (or \$SNAPPER_COORDINATOR_INSTANCE_COUNT)	Multi-instance partitioning: total coordinator instance count across the deployment. Must match across all coordinators.

Example:

```
snapper trade-zmq --signal-topics "signals.paper.BTC-USD.rsi_btc_1h,signals.kraken.BTC-USD.live"
```

Multi-instance deployment:

```
# Coordinator 0 of 2
snapper trade-zmq --instance-id 0 --instance-count 2

# Coordinator 1 of 2 (separate process, same DB + broker)
snapper trade-zmq --instance-id 1 --instance-count 2
```

Each instance owns $\sim 1/N$ of the shard_keys via deterministic SHA-256 hashing. Default --instance-count 1 is byte-identical to single-coordinator behavior (every shard owned by the single coordinator). See [docs/operations.md](#) for systemd template unit + scale-up/down/crash-recovery runbooks.

executor

Starts the order execution service for a selected exchange.

```
snapper executor [OPTIONS]
```

Options:

Option	Type	Default	Description
-e, --exchange	string	kraken	Exchange (kraken, walumat)

Examples:

```
# Kraken executor
snapper executor -e kraken

# Walumat executor
snapper executor --exchange walumat
```

feed

Starts the market data publisher via WebSocket.

```
snapper feed [OPTIONS]
```

Options:

Option	Type	Default	Description
--symbols	string	BTC/ USD	Comma-separated symbols passed to the Kraken publisher. Symbols must be in native dash format (e.g. BTC-USD); the literal code default BTC/USD is the WebSocket output format, and only native symbols actually subscribe (slash-format inputs are skipped as unknown).

Example:

```
snapper feed --symbols "BTC-USD,ETH-USD"
```

feed-engine

Runs the dedicated feed-container entrypoint. It syncs the process registry, starts enabled market-data publishers as OS subprocesses, and exits non-zero if any supervised publisher crashes so the orchestrator restarts the feed container.

```
snapper feed-engine
```

zmq-logger

Monitors and logs ZMQ message traffic.

```
snapper zmq-logger [OPTIONS]
```

Options:

Option	Type	Default	Description
--payload/--no-payload	bool	false	Log full payload
--file/--no-file	bool	true	Write to audit file
--audit-file	string	data/zmq_audit.jsonl	Audit file path
--max-length	int	200	Max payload preview length

Example:

```
snapper zmq-logger --payload --audit-file logs/zmq.jsonl
```

egress

Runs the snapper-egress sidecar from the unified Snapper image. Additional arguments are forwarded to the egress endpoint.

```
snapper egress [--instance-id snapper-egress-prod]
```

See [snapper-egress.md](#) for tunnel settings and verification steps.

Database

db-init

Initializes the database schema.

```
snapper db-init
```

db-upgrade

Runs Alembic migrations to a specified revision.

```
snapper db-upgrade [OPTIONS]
```

Options:

Option	Type	Default	Description
--revision	string	head	Target revision

Examples:

```
# Migrate to latest version
snapper db-upgrade

# Migrate to specific revision
snapper db-upgrade --revision abc123
```

db-downgrade

Rolls back Alembic migrations.

```
snapper db-downgrade [OPTIONS]
```

Options:

Option	Type	Default	Description
--revision	string	-1	Steps to roll back

Examples:

```
# Roll back one migration
snapper db-downgrade

# Roll back to specific revision
snapper db-downgrade --revision base
```

db-seed

Seeds the database with environment-specific data from TOML profiles. Automatically run by make migrate-dev and make migrate-prod.

```
snapper db-seed [OPTIONS]
```

Options:

Option	Type	Default	Description
--profile	string	dev	Seed profile name (e.g. dev, prod)

Users

init-admin

Creates the initial administrator user.

```
snapper init-admin [OPTIONS]
```

Options:

Option	Type	Default	Description
--username	string	admin	Username
--password	string	<i>(prompted, hidden, confirmed)</i>	Password — no default; the command prompts with hidden input and confirmation when the flag is omitted, so the value never lands in shell history.

Example:

```
# Prompts for password with hidden input + confirmation; no inline value
# means nothing lands in shell history
snapper init-admin --username admin
```

list-users

Displays a list of all system users.

```
snapper list-users
```

reset-password

Resets a user's password.

```
snapper reset-password USERNAME [OPTIONS]
```

Arguments:

- USERNAME — Username (required)

Options:

Option	Type	Default	Description
--new-password	string	(prompt)	New password

Example:

```
# Prompts for the new password with hidden input + confirmation; the secret
# never lands in shell history when --new-password is omitted
snapper reset-password john
```

Symbol Synchronization

update-kraken-symbols

Synchronizes symbol mappings from the Kraken API.

```
snapper update-kraken-symbols [OPTIONS]
```

Options:

Option	Type	Default	Description
-f, --force	bool	false	Force update

update-kraken-futures-symbols

Syncs Kraken Futures (crypto perpetuals) symbol mappings from the exchange API.

```
snapper update-kraken-futures-symbols [OPTIONS]
```

Options:

Option	Type	Default	Description
-f, --force	bool	false	Force update

update-kraken-equities-symbols

Syncs Kraken Equities (FCM Futures) symbol mappings from the exchange API.

```
snapper update-kraken-equities-symbols [OPTIONS]
```

Options:

Option	Type	Default	Description
-f, --force	bool	false	Force update

update-walutomat-symbols

Synchronizes symbol mappings from the Walutomat API.

```
snapper update-walutomat-symbols [OPTIONS]
```

Options:

Option	Type	Default	Description
-f, --force	bool	false	Force update

update-polygon-symbols

Synchronizes symbol mappings from the Polygon.io API.

```
snapper update-polygon-symbols [OPTIONS]
```

Options:

Option	Type	Default	Description
-f, --force	bool	false	Force update
--insert-new	bool	false	Insert new symbols

Note: This operation may take 10-15 minutes (44k+ symbols).

update-underlyings

Syncs underlying asset definitions from YAML to the database. Matches active instruments to underlyings via pattern rules, upserts mappings, and applies YAML-fallback instrument_type/expiry_override to InstrumentSpec when API-sourced values are NULL.

```
snapper update-underlyings [OPTIONS]
```

Option	Type	Default	Description
-f, --force	bool	false	Bypass safety guard for stale cleanup

reconcile-symbol-aliases

Closes active symbol_aliases rows whose owning capability row is no longer tradeable and no longer market-data-enabled. The command is idempotent and can be scoped to one exchange.

```
snapper reconcile-symbol-aliases [OPTIONS]
```

Option	Type	Default	Description
-e, --exchange	string	all	Exchange to reconcile (kraken, kraken_futures, kraken_equities, walutomat, polygon, or all)

Market Snapshots

update-kraken-market-snapshot

Updates Kraken market snapshots with current prices.

```
snapper update-kraken-market-snapshot
```

update-kraken-futures-market-snapshot

Updates Kraken Futures market snapshots with current prices.

```
snapper update-kraken-futures-market-snapshot
```

update-kraken-equities-market-snapshot

Updates Kraken Equities market snapshots with current prices.

```
snapper update-kraken-equities-market-snapshot
```

update-kraken-futures-funding-rates

Backfills historical funding rates for Kraken Futures perpetuals into the funding_rates table. Duplicates are silently skipped via the partial unique index.

```
snapper update-kraken-futures-funding-rates [OPTIONS]
```

Options:

Option	Type	Default	Description
--symbol / -s	list[str]	None	One or more native perpetual symbols (-s BTC-USD-PERP -s ETH-USD-PERP)
--all	bool	false	Backfill every mapped Kraken Futures perpetual

update-walutomat-market-snapshot

Updates Walutomat market snapshots with current prices.

```
snapper update-walutomat-market-snapshot
```

Data Backfill

polygon-backfill-aggregates

Downloads historical aggregated data from the Polygon.io API to the on-disk CSV cache. This command is download-only: it writes CSV files and never touches the database. Populating the candles table is the separate second step, handled by [polygon-load-csv](#).

The full workflow is two steps:

- 1. polygon-backfill-aggregates downloads candles into the CSV cache.
- 2. polygon-load-csv reads that cache and upserts the candles into the database (no Polygon API calls).

```
snapper polygon-backfill-aggregates [OPTIONS]
```

Options:

Option	Type	Default	Description
-s, --symbol	string[]	from settings	Symbols to download
--all	bool	false	All mapped symbols
-m, --multiplier	int	1	Timeframe multiplier
-t, --timespan	string	minute	Timespan (minute, hour, day)
-d, --days	int	None (effective default 30)	Days back
--resume/--no-resume	bool	true	Skip days already cached on disk
--csv/--no-csv	bool	true	Save to CSV files

Examples:

```
# Download specific symbols to the CSV cache
snapper polygon-backfill-aggregates -s AAPL -s MSFT --days 365

# Download all mapped symbols
```

```
snapper polygon-backfill-aggregates --all --timespan day

# Hourly download, skipping already-cached days
snapper polygon-backfill-aggregates -s SPY -t hour -m 1 --resume

# Then load the downloaded cache into the database
snapper polygon-load-csv --all --timespan day
```

polygon-load-csv

Loads candles from the on-disk Polygon CSV cache into the database. This is the cache-only counterpart to `polygon-backfill-aggregates`: it reads the CSV files produced by that command and upserts the candles into the candles table. It never contacts the Polygon API.

Run this after `polygon-backfill-aggregates` has populated the cache. Archive-symbol cache directories with no current Symbol identity are skipped with a warning, so a stale cache directory never aborts the load.

When neither `--symbol` nor `--all` is given, the symbols default to the settings-configured Polygon instruments (the `instruments` setting). The wildcard sentinel `["*"]` in that setting is treated like `--all` and loads every cached archive symbol; an explicit settings list loads only those symbols.

```
snapper polygon-load-csv [OPTIONS]
```

Options:

Option	Type	Default	Description
<code>-s, --symbol</code>	string[]	from settings	Symbols to load (native or Polygon format)
<code>--all</code>	bool	false	Load every archive symbol present in the cache
<code>-t, --timespan</code>	string	day	Timespan subtree to read (minute, hour, day)
<code>--since</code>	string	None	Earliest day to load, inclusive (YYYY-MM-DD)
<code>--until</code>	string	None	Latest day to load, inclusive (YYYY-MM-DD)

Examples:

```
# Load every cached minute symbol into the database
snapper polygon-load-csv --all --timespan minute

# Load a single symbol within a date window
snapper polygon-load-csv -s X:BTCUSD --since 2024-01-01 --until 2024-01-31
```

kraken-futures-backfill-candles

Backfills historical OHLCV candles for Kraken Futures perpetuals.

```
snapper kraken-futures-backfill-candles [OPTIONS]
```

Options:

Option	Type	Default	Description
-s, --symbol	str[]	None	Native symbols to backfill (e.g. BTC-USD-PERP)
--all	bool	false	Backfill all mapped Kraken Futures symbols
-t, --timeframe	str	1h	Candle interval (1m, 1h, 4h, 1d)
-d, --days	int	90	Days back to fetch
--resume / --no-resume	bool	--resume	Resume from latest stored candle

kraken-equities-backfill-candles

Backfills historical OHLCV candles for Kraken Equities (TradFi FCM) contracts via the internal iapi.kraken.com ticker-history endpoint. Response is ~10-minute delayed per FCM policy. Per-chunk upstream failures raise RuntimeError so outages are distinguishable from legitimately-empty candle windows.

```
snapper kraken-equities-backfill-candles [OPTIONS]
```

Options:

Option	Type	Default	Description
-s, --symbol	str[]	None	Native symbols to backfill (e.g. MNQM6-CME)
--all	bool	false	Backfill all mapped Kraken Equities symbols
-t, --timeframe	str	1h	Candle interval (1m, 5m, 15m, 30m, 1h, 1d)
-d, --days	int	30	Days back to fetch
--resume / --no-resume	bool	--resume	Resume from latest stored candle

build-continuous

Builds and displays a continuous-contract series for an underlying.

```
snapper build-continuous TICKER EXCHANGE CONTRACT_FAMILY [OPTIONS]
```

Arguments:

Argument	Description
TICKER	Underlying asset ticker (e.g. SPX, GOLD)
EXCHANGE	Exchange to source contracts from
CONTRACT_FAMILY	Product root (e.g. ES, GC)

Options:

Option	Type	Default	Description
-t, --timeframe	str	1d	Candle timeframe
-m, --method	str	panama	Adjustment method (unadjusted, ratio, panama)
--start	str	365 days ago	Series start time (ISO)
--end	str	now	Series end time (ISO)
--rollover-days	int	0	Days before expiry to roll (0..365)

polygon-backfill-grouped

Downloads grouped daily data from the Polygon.io API to the CSV cache. Like polygon-backfill-aggregates, this is download-only and writes CSV files without touching the database.

```
snapper polygon-backfill-grouped [OPTIONS]
```

Options:

Option	Type	Default	Description
--market / -m	str	crypto	Market type (crypto, stocks, fx)
--days / -d	int	3	Number of recent days to fetch
--locale / -l	str	global	Market locale (global, us)
--csv / --no-csv	flag	--csv	Save data to CSV.gz files
--adjusted / --unadjusted	flag	--adjusted	Use adjusted prices

Data Archive

archive

Export data to CSV archive files. Supports candle cache projection, append-only event tables (ticks, trades, signals, executions, telemetry, control), and SCD2 state tables (orders, positions, instruments, instrument_specs, settings, symbols, symbol_aliases, symbol_exchange_capabilities, users, user_login_events, market_snapshots, process_runs, underlying_assets, instrument_underlying_mappings, continuous_contract_configs). Merges with existing files and deduplicates rows.

```
snapper archive [OPTIONS]
```

Options:

Option	Type	Default	Description
--table	TEXT	candles	Table to archive. Accepts candles, candles-audit, any event table (ticks, trades, signals, executions, telemetry, control), or any state table (see the section intro for the full list).
--exchange	TEXT	None	Exchange filter (e.g. polygon, kraken)
--symbol	TEXT	None	Native symbol filter (e.g. BTC-USD), resolved to stable archive_symbol
--timeframe	TEXT	1m	Candle timeframe (1m, 5m, 1h, 1d) — candles only
--day	str	None	Single day to archive (YYYY-MM-DD; parsed in the handler)
--from	str	None	Start of date range (YYYY-MM-DD; parsed in the handler)
--to	str	None	End of date range (YYYY-MM-DD; parsed in the handler)
--dry-run	FLAG	False	Report counts without writing files
--purge	FLAG	False	Delete exported rows from DB after writing (event tables, plus candles-audit/state tables when combined with --closed-only)
--closed-only	FLAG	False	Only closed SCD2 versions (candles-audit and state tables)
--output-dir	str	data	Base output directory

Output structure:

```
Candles: {output_dir}/{exchange}/cache/{timespan}/{archive_symbol}/{year}/{date}.csv
Events:  {output_dir}/archive/{table}/{exchange}/{archive_symbol}/{year}/{date}.csv
Flat:    {output_dir}/archive/{table}/{year}/{date}.csv  (telemetry, control)
```

Examples:

```
snapper archive --day 2024-03-15 --exchange polygon --symbol BTC-USD
snapper archive --from 2024-01-01 --to 2024-03-31 --dry-run
snapper archive --day 2024-03-15 --timeframe 1d
snapper archive --table ticks --day 2024-03-15 --exchange polygon
snapper archive --table trades --from 2024-01-01 --to 2024-03-31 --purge
snapper archive --table control --day 2024-03-15
snapper archive --table candles-audit --day 2024-03-15 --timeframe 1m
snapper archive --table candles-audit --day 2024-03-15 --closed-only --purge
snapper archive --table orders --day 2024-03-15
snapper archive --table symbols --day 2024-03-15 --closed-only --purge
snapper archive --table instruments --day 2024-03-15
```

Notes:

- --symbol resolves the current native_symbol to the stable archive_symbol via the Symbol anchor row. After a symbol rename, the CLI argument and the output directory may differ.
- --day or --from/--to is required.
- Daily timespan (1d) produces monthly CSV files to match Polygon layout.
- --purge is supported for event tables, plus candles-audit and state tables when combined with --closed-only (which protects active rows); it is **not** supported for the candle cache export.
- Event archive CSV files include full temporal metadata (public_id, timestamp, known_to, session_id, sequence_id).
- Merge/dedup for events uses (public_id, timestamp, known_to) as key.
- Executions are partitioned by exchange/archive_symbol via order -> instrument.
- candles-audit exports all SCD2 versions (closed + active) grouped by open_at date. --purge requires --closed-only to protect active rows.

restore

Restore archived CSV data back into the database. Reads CSV files exported by snapper archive and inserts rows, deduplicating against existing data by (public_id, timestamp, known_to).

```
snapper restore [OPTIONS]
```

Options:

Option	Type	Default	Description
--table	TEXT	(required)	Table name to restore into
--file	str	None	Single CSV file to restore

Option	Type	Default	Description
--dir	str	None	Directory to scan recursively for CSV files
--source	TEXT	audit	Restore mode (audit for full history)

Examples:

```
snapper restore --table ticks --file data/archive/ticks/polygon/BTC-USD/2024/2024-01-01.csv
snapper restore --table settings --dir data/archive/settings/
snapper restore --table candles --dir data/archive/candles/polygon/BTC-USD/
```

Notes:

- At least one of --file or --dir is required.
- Rows already present in DB (matching public_id + timestamp + known_to) are skipped.
- Audit restore inserts all temporal columns exactly as exported.

One-time cache directory migration

Renames legacy Polygon cache directories from API ticker format (X_BTCUSD) to archive_symbol format (BTC-USD).

```
python -m scripts.migrate_polygon_cache_dirs [--dry-run] [--cache-root PATH]
```

Encryption Management

settings-rotate-encryption

Rotates encryption keys for settings in the database.

```
snapper settings-rotate-encryption [OPTIONS]
```

Options:

Option	Type	Default	Description
--new-password	string	<i>(prompted, hidden, confirmed)</i>	New master password — no default; the command prompts with hidden input and confirmation when the flag is omitted, so the value never lands in shell history. Inline --new-password X is still accepted for non-interactive automation but the same shell-history caveats apply.

Option	Type	Default	Description
--old-password	string	from env	Current master password (optional; reads from env/bootstrap when omitted). Avoid inline use — passing a secret on the command line leaks it to shell history.
--dry-run	bool	false	Show changes without applying

Example:

```
# Simulate rotation – prompts hidden for the new master password
snapper settings-rotate-encryption --dry-run

# Actual rotation – same prompt flow, then re-encrypts every encrypted setting
snapper settings-rotate-encryption
```

The CLI prompts for --new-password with hidden input and confirmation when the flag is omitted (preferred), so the master password never lands in shell history, scrollback, or CI logs. After a successful rotation the command prints only a reminder to update MASTER_PASSWORD in .env — it does NOT echo the value you entered. Inline --new-password X is still accepted for non-interactive automation but the same shell-history caveats apply.

Example Workflows

New Installation Initialization

```
# 1. Initialize database
snapper db-init

# 2. Create administrator (prompts for password – never put it on the cmdline)
snapper init-admin --username admin

# 3. Synchronize symbols
snapper update-kraken-symbols
snapper update-polygon-symbols

# 4. Start server
snapper server
```

Full Trading System Startup

```
# Terminal 1: ZMQ Broker
snapper broker

# Terminal 2: Data feed
snapper feed --symbols "BTC-USD,ETH-USD"
```

```
# Terminal 3: Executor
snapper executor -e kraken

# Terminal 4: Trade runtime
snapper trade-zmq

# Terminal 5: Server with dashboard
SERVER_API_ONLY=true snapper server
```

Historical Data Backfill

Backfilling Polygon candles is a two-step workflow: download to the CSV cache, then load the cache into the database.

```
# Step 1 - download last 30 days for SPY to the CSV cache (no DB write)
snapper polygon-backfill-aggregates -s SPY --days 30 --timespan minute

# Step 1 - download daily data for all symbols to the CSV cache
snapper polygon-backfill-aggregates --all --timespan day --days 365

# Step 2 - load the downloaded CSV cache into the database (no API calls)
snapper polygon-load-csv --all --timespan day
```

Notifications

notify

Long-running iOS Push Foundation sidecar (ZMQ alerts.* → APNs HTTP/2). Subscribes to the alerts. ZMQ prefix, fans out each received AlertEventData to the target user's active devices via the outbox-backed NotifySidecar, and retries server/throttled failures on its own 30-second scheduler. Configuration is read from the apns_* settings (provisioned through the seed profile).

```
snapper notify
```

Run under systemd / K8s with Restart=always — the sidecar exits only on unhandled errors or SIGINT; the outbox drain on the next start recovers any queued deliveries left behind.

AI Integration

dev-mint-pat

Mints a long-lived AI delegate JWT into data/dev-pat.json (mode 0600) for the local Snapper MCP bridge. See [ai-integration.md](#) for the wider Claude Code / Cursor / Windsurf wire-up.

```
snapper dev-mint-pat [OPTIONS]
```

Options:

Option	Env var	Default	Description
--base-url URL	SNAPPER_DEV_BASE_URL	http://localhost:8000	Base URL of the running backend
--admin-username USER	SNAPPER_DEV_ADMIN_USERNAME	None (resolved from seed profiles, mcp then dev)	Admin login
--admin-password PASS	SNAPPER_DEV_ADMIN_PASSWORD	None (resolved from seed profiles, mcp then dev)	Admin password
--output PATH	SNAPPER_DEV_PAT_OUTPUT	data/dev-pat.json	Output file path
--label LABEL	—	Local Dev MCP	Delegate label

The script drives the production REST endpoints (POST /api/auth/login + POST /api/ai-delegates) so the seeded delegate exercises the same code path the AI Integration UI uses. Tokens never appear in stderr — HTTP error bodies are passed through a JWT-shape redaction helper before printing.

Backtesting

backtest-run

Run a backtest synchronously. Creates a DB record, executes the engine in-process, and persists results (signals, trades, equity, metrics).

```
snapper backtest-run \  
  --strategy sma_cross \  
  --instrument BTC-USD \  
  --exchange kraken \  
  --start 2026-01-01 \  
  --end 2026-06-01 \
```

```
--timeframe 1h \  
--initial-cash 10000 \  
--params '{"fast": 10, "slow": 30}'
```

Options:

Option	Type	Default	Description
--strategy	str	required	Registered strategy class name
--instrument	str	required	Instrument public ID
--exchange	str	required	Exchange name
--start	str	required	Start date (ISO format)
--end	str	required	End date (ISO format)
--timeframe	str	1h	Candle timeframe
--initial-cash	float	10000	Starting cash balance
--wallet	str	cli	Wallet public ID
--params	str	{}	Strategy params as JSON
--execution-mode	str	direct_db	Engine type (direct_db or zmq_replay)
--fill-model	str	market	Fill simulation model
--slippage-bps	float	0.0	Per-fill slippage in basis points
--commission-bps	float	0.0	Per-fill commission in basis points

backtest-list

List backtest runs with optional filters.

```
snapper backtest-list --status completed --limit 10
```

Options:

Option	Type	Default	Description
--strategy	str	None	Filter by strategy name
--status	str	None	Filter by status (pending, running, completed, failed, cancel_requested, cancelled)
--wallet	str	None	Filter by wallet public ID

Option	Type	Default	Description
--limit	int	20	Maximum number of rows to return

backtest-cancel

Cancel a pending or running backtest run.

```
snapper backtest-cancel <run-public-id>
```

backtest-rerun

Re-run a backtest with the same configuration (including strategy params).

```
snapper backtest-rerun <run-public-id>
```

Configuration

Snapper's configuration system is based on two layers: environment variables (bootstrap) and database settings.

Environment Variables

Settings loaded from .env file or environment variables.

Database

Variable	Default	Description
DB_URL	sqlite+aiosqlite:///./data/snapper.db	SQLAlchemy connection URL

Database URL examples:

```
# SQLite (development default)
DB_URL=sqlite+aiosqlite:///./data/snapper.db

# PostgreSQL (production)
DB_URL=postgresql+asyncpg://user:password@localhost:5432/snapper
```

Encryption

Variable	Default	Description
MASTER_PASSWORD	snapper_default_master_password_v1	Password for encrypting settings in database (salt derived automatically)

Important: Change default values in production environment.

HTTP Server

Variable	Default	Description
SERVER_HOST	127.0.0.1	Listen address
SERVER_PORT	8000	Server port
SERVER_RELOAD	false	Auto-reload for development

Variable	Default	Description
SERVER_PROXY_HEADERS	true	Enable proxy header parsing in uvicorn
SERVER_FORWARDED_ALLOW_IPS	127.0.0.1	Trusted proxy IPs/CIDRs for forwarded headers
SERVER_API_ONLY	false	Skip process autostart; serve API + WS bridge only (separate-engine boots). Multi-worker uvicorn (multiple FastAPI processes sharing a broker) is supported — see docs/architecture.md Deployment Modes for the AI-review fanout dedup contract under N>1. Set SNAPPER_COORDINATOR_INSTANCE_ID + SNAPPER_COORDINATOR_INSTANCE_COUNT per worker to enable the shared partitioning.
PROCESS_AUTOSTART_PROFILE	all	Select which enabled process configs this node starts: all for single-container/dev, api for backend-without-market-publishers, feed for the dedicated feed container.
TELEMETRY_RECORDING_ENABLED	false	Record pings, heartbeats, and GET reads to telemetry table. High volume — enable for debugging only

The Default column reflects the Pydantic defaults on BootstrapSettingsLoader (src/snapper/config/bootstrap.py) for the server / encryption / ZMQ / coordinator rows. The observability rows further down (SYSTEM_METRICS_*, RETENTION_*, DB_METRICS_*, DB_POOL_*, SNAPPER_*_PROBE) are NOT loaded through that class. application/system_metrics/snapshotter.py and application/db_stats/snapshotter.py read their env vars directly via os.environ. The retention env vars are read in two places: application/retention/scheduler.py (RETENTION_INTERVAL_SECONDS, RETENTION_DISABLED, RETENTION_OUTPUT_DIR) and application/retention/service.py (RETENTION_DRY_RUN), with fallback resolvers defined in application/retention/policies.py. The defaults shown for those rows reflect the per-module fallbacks.

The repo-root .env.example intentionally ships with Docker-friendly overrides for two server values: SERVER_HOST=0.0.0.0 (so the dev server binds inside containers) and SERVER_FORWARDED_ALLOW_IPS=127.0.0.1,172.17.0.1 (so requests forwarded by the default Docker bridge are accepted). Caveat: the make run-server / make dev-backend targets in the Makefile pass --host 0.0.0.0 on the command line, which wins over the .env value via Typer's host or s.server_host resolution — so editing .env only changes SERVER_HOST for direct snapper server invocations (no --host flag). SERVER_FORWARDED_ALLOW_IPS is honored on every server-start path because no CLI flag overrides it; make migrate-dev does not bind either value.

ZeroMQ

Variable	Default	Description
ZMQ_BROKER_XSUB	tcp:// 127.0.0.1:7500	XSUB endpoint (publishers connect here)
ZMQ_BROKER_XPUB	tcp:// 127.0.0.1:7501	XPUB endpoint (subscribers connect here)
ZMQ_BROKER_BIND_XSUB	empty	Optional broker bind endpoint for XSUB. Empty means bind ZMQ_BROKER_XSUB; cross-container deployments set this to tcp://0.0.0.0:7500.
ZMQ_BROKER_BIND_XPUB	empty	Optional broker bind endpoint for XPUB. Empty means bind ZMQ_BROKER_XPUB; cross-container deployments set this to tcp://0.0.0.0:7501.

Bind endpoints exist because ZMQ bind() requires a local interface while cross-container clients connect through the Docker service name. In the compose split, the backend broker binds 0.0.0.0 and both the backend and snapper-feed connect to tcp://snapper:7500/7501.

Database Engine Pool

These optional variables are read directly by the repository engine factory and only affect PostgreSQL engines.

Variable	Default	Description
DB_POOL_SIZE	SQLAlchemy default	Per-process pool size.
DB_MAX_OVERFLOW	SQLAlchemy default	Per-process overflow connections.

Set them on the dedicated feed container when publisher subprocesses would otherwise multiply the default pool toward PostgreSQL max_connections.

Coordinator Sharding

The trade runtime supports horizontal sharding via these variables. With SNAPPER_COORDINATOR_INSTANCE_COUNT > 1 each instance owns a partition of the active wallets / outbox rows, enabling multi-worker uvicorn deployments and parallel trade-zmq instances.

Variable	Default	Description
SNAPPER_COORDINATOR_INSTANCE_ID	0	Zero-based shard index for this instance
SNAPPER_COORDINATOR_INSTANCE_COUNT	1	Total number of instances in the cluster
SNAPPER_COORDINATOR_OUTBOX_MAX_SCAN_ROWS	1000	Per-tick upper bound on outbox rows scanned by this shard; set to unbounded, none, or an empty value to disable the cap. Positive integers are accepted.

Observability Pipeline

These variables tune the always-on background pipelines that emit system-metrics snapshots, run retention archive+purge sweeps, and sample DB stats. See [observability.md](#) for the underlying snapshots and the retention window math.

Variable	Default	Description
SYSTEM_METRICS_INTERVAL_SECONDS	5	Cadence for the in-process system-metrics snapshotter
SYSTEM_METRICS_HISTORY_CAP	17280	Maximum number of snapshots retained in memory ($\approx$ 24 h at 5 s cadence)
RETENTION_INTERVAL_SECONDS	3600	Cadence for the retention archive+purge tick
RETENTION_DISABLED	false	Disable the retention loop entirely (still allow on-demand archive via CLI)
RETENTION_DRY_RUN	false	Compute the window and counts but skip writes/purges
RETENTION_OUTPUT_DIR	data	Base directory for archive CSV writes (matches the CLI --output-dir default)
DB_METRICS_INTERVAL_SECONDS	60	Cadence for the per-table row-count / index-health sampler
DB_METRICS_DISABLED	false	Disable the DB stats sampler
SNAPPER_TICK_PROBE	unset	Enable per-stage tick hot-path histograms in publisher logs when truthy (1, true, yes)
SNAPPER_TRADE_PROBE	unset	

Variable	Default	Description
		Enable per-stage trade hot-path histograms in publisher logs when truthy (1, true, yes)

The two probe variables flush one log line roughly every 10 seconds with per-stage count, rate, p50, p95, p99, and max. Leave them unset outside targeted throughput investigations.

Database Settings

Sensitive data is stored encrypted in the settings table. Wallet-scoped exchange credentials (api keys, secrets, PEM material) live in a separate `wallet_credentials` table — see [Wallet Credentials](#) below. Shared market-data provider keys stay in settings.

Market Data API Keys

Key	Description
<code>polygon_api_key</code>	Polygon.io API key (shared market data — not wallet-scoped)

Wallet Credentials

Per-exchange trading credentials live in the `wallet_credentials` table and are loaded by `CredentialResolver` during per-wallet executor startup. Each row carries:

Column	Description
<code>wallet_public_id</code>	UUID of the owning wallet ((label, is_paper) unique in wallets table)
<code>exchange</code>	Exchange identifier (kraken, kraken_futures, walutomat, paper)
<code>credential_type</code>	Envelope shape: <code>api_key_secret</code> , <code>rsa_pem</code> , <code>oauth</code> , or <code>paper</code>
<code>encrypted_payload</code>	Fernet-encrypted JSON envelope (shape depends on <code>credential_type</code>)
<code>label</code>	Human-readable description of the credential row

Envelope shapes by `credential_type`:

- `api_key_secret` — `{"api_key": "...", "api_secret": "..."} (Kraken, Kraken Futures)`
- `rsa_pem` — `{"api_key": "...", "private_key_pem": "..."} (Walutomat)`
- `oauth` — `{"client_id": "...", "client_secret": "...", "refresh_token": "..."} (Kraken)`
- `paper` — `{"initial_balance": "10000.0"} (paper wallets)`

The shipped seed profile (dev.toml) bootstraps wallet_credentials on a fresh database; operators can add their own data/seed/{profile}.toml override for other environments. Note: the seed loader supports the api_key_secret, rsa_pem, and paper envelope shapes — the oauth shape is accepted at the credential-resolver layer but **not** by the current seed loader, so OAuth credentials must be inserted through the REST wallet_credentials create/rotate routes (POST /api/wallets/{wallet_public_id}/credentials, POST /api/wallets/{wallet_public_id}/credentials/{credential_public_id}/rotate) rather than via seed.

The credential management REST surface AND the matching frontend UI (frontend/src/features/admin/CredentialManagement/) are both shipped. Day-to-day operators use the dashboard's Credential Management view; headless callers can hit the REST routes directly (curl / Postman — the snapper-mcp bridge does NOT expose credential CRUD as MCP tools). Other recovery options are re-running the seed against a clean DB (seed is idempotent — it skips wallets that already exist) or direct SQL surgery on the encrypted payload as a last resort.

Seed file structure:

```
[[wallets]]
label = "default"
is_paper = false

[[wallets.credentials]]
exchange = "kraken"
credential_type = "api_key_secret"
api_key = "your-kraken-api-key"
api_secret = "your-kraken-api-secret"

[[wallets.credentials]]
exchange = "walutomat"
credential_type = "rsa_pem"
api_key = "your-walutomat-api-key"
private_key_pem_base64 = "your-pem-base64-encoded"
```

Two wallets named default — one with is_paper=true and one with is_paper=false — coexist via the (label, is_paper) unique index. The seed loader encrypts every payload with the master-password Fernet key before insert.

Trading Parameters

Key	Default	Description
instruments	{"kraken": ["*"], "kraken_futures": ["*"], "kraken_equities": ["*"], "walutomat": ["*"], "polygon": ["*"]}	Instruments per exchange (* = all instruments)
timeframes	["1m"]	Candle timeframes (list)
risk_max_leverage	1.0	Maximum leverage

Key	Default	Description
risk_r_per_trade	0.005	Risk per trade (0.5%)

The trade runtime always uses the durable (outbox-driven) dispatch path. The engine writes TradeCommand rows to the database; the OutboxDispatcher picks them up and publishes to ZMQ. Executor-side VenueEvent writes are fail-closed — a failed persist raises before the executor acknowledges the venue event.

Market Persist Policy

Market-data publishers always publish ZMQ frames, but DB persistence is controlled by these settings so high-volume feeds can be cache-first:

Key	Description
market_persist_ticks	Persistence mode for tick rows
market_persist_trades	Persistence mode for trade rows
market_persist_candles	Persistence mode for candle rows
market_persist_extra	Explicit extra instrument allowlist
market_persist_exclude	Explicit instrument denylist

MarketPersistPolicy refreshes from settings and admin bus events, then injects immutable allow/deny snapshots into publishers. Invalid edits are rejected for that refresh cycle and the previous policy remains active. Use `/api/market/cache/health` and `/api/market/coverage` to verify the effective persisted universe.

Publisher Micro-Batch

Key	Default	Description
write_buffer_flush_ms	50	Flush age threshold in ms (cached at start)
write_buffer_candle_max_rows	100	Candle batch size trigger
write_buffer_tick_max_rows	500	Tick batch size trigger
write_buffer_trade_max_rows	500	Trade batch size trigger

These settings are cached when the publisher starts and require a process restart to take effect.

Authentication

Key	Default	Description
auth_secret_key	change-me-in-production-use-openssl-rand-hex-32	JWT signing secret
csrf_secret_key	change-me-in-production-csrf-key	CSRF token signing secret
auth_algorithm	HS256	JWT signing algorithm
auth_access_token_expire_minutes	15	Access token lifetime
auth_refresh_token_expire_days	7	Refresh token lifetime
auth_refresh_token_expire_days_extended	30	Extended refresh token lifetime
ws_token_ttl_seconds	900	WebSocket token lifetime
csrf_token_expire_minutes	60	CSRF token lifetime
session_secure	false	Require HTTPS for session cookies
session_same_site	lax	Session cookie SameSite mode
session_domain	""	Session cookie domain override
ui_origin	""	Additional allowed UI origins for CORS and WS origin validation

.env.example File

The block below mirrors the variables documented in .env.example at the repo root. The inline comments are abbreviated here for readability; the canonical version of the file (with full per-variable rationale and rotation guidance) lives in .env.example itself.

```
# Database connection
DB_URL=sqlite+aiosqlite:///./data/snapper.db

# Encryption settings for sensitive data (API keys, secrets)
# IMPORTANT: Change this default in production!
MASTER_PASSWORD=snapper_default_master_password_v1

# Server settings
```

```
SERVER_HOST=0.0.0.0
SERVER_PORT=8000
SERVER_RELOAD=false
SERVER_API_ONLY=false
PROCESS_AUTOSTART_PROFILE=all
SERVER_PROXY_HEADERS=true
SERVER_FORWARDED_ALLOW_IPS=127.0.0.1,172.17.0.1

# Telemetry recording (pings, heartbeats, GET reads – high volume)
TELEMETRY_RECORDING_ENABLED=false

# ZMQ broker settings
ZMQ_BROKER_XSUB=tcp://127.0.0.1:7500
ZMQ_BROKER_XPUB=tcp://127.0.0.1:7501
# ZMQ_BROKER_BIND_XSUB=tcp://0.0.0.0:7500
# ZMQ_BROKER_BIND_XPUB=tcp://0.0.0.0:7501

# PostgreSQL pool clamps for multi-process feed deployments
# DB_POOL_SIZE=2
# DB_MAX_OVERFLOW=3

# Coordinator sharding (multi-instance trade-zmq / multi-worker uvicorn)
SNAPPER_COORDINATOR_INSTANCE_ID=0
SNAPPER_COORDINATOR_INSTANCE_COUNT=1
SNAPPER_COORDINATOR_OUTBOX_MAX_SCAN_ROWS=1000

# Observability pipeline (system-metrics snapshotter)
SYSTEM_METRICS_INTERVAL_SECONDS=5
SYSTEM_METRICS_HISTORY_CAP=17280

# Retention policy framework
RETENTION_INTERVAL_SECONDS=3600
RETENTION_DISABLED=false
RETENTION_DRY_RUN=false
RETENTION_OUTPUT_DIR=data

# DB stats sampler (per-table row-count snapshots)
DB_METRICS_INTERVAL_SECONDS=60
DB_METRICS_DISABLED=false

# Publisher hot-path probes, disabled by default
# SNAPPER_TICK_PROBE=1
# SNAPPER_TRADE_PROBE=1
```

Accessing Configuration in Code

Bootstrap settings (without database)

```
from snapper.config.settings import get_settings

settings = get_settings()
db_url = settings.db_url
host = settings.server_host
```

With database access

```
from snapper.config.settings import get_settings_with_service

settings = get_settings_with_service(settings_service)
polygon_key = settings.polygon_api_key # Decrypted from database
```

Wallet-scoped exchange credentials do NOT appear on AppSettings. They are loaded by CredentialResolver during per-wallet executor startup:

```
from snapper.config.credentials import CredentialResolver

resolver = CredentialResolver(repository)
envelope = await resolver.get_credentials(
    exchange="kraken",
    wallet_public_id="01975a8b-3c7d-7000-8000-0000000000a1",
)
# envelope == {"api_key": "...", "api_secret": "..."}
```

Managing Settings via API

List Settings

```
GET /api/settings
GET /api/settings?category=exchange
```

List Categories

```
GET /api/settings/categories
```

Set Setting

```
POST /api/settings/{key}/set
X-CSRF-Token: <csrf_token>
Content-Type: application/json

{
  "public_id": "<uuid7>",
  "session_id": "<client-session>",
  "sequence_id": 1,
  "timestamp": "2026-01-18T12:00:00Z",
  "payload": {
    "value": "new_value",
    "category": "system",
```

```
    "description": "optional description"
  }
}
```

Remove Setting

```
POST /api/settings/{key}/remove
X-CSRF-Token: <csrf_token>
Content-Type: application/json

{
  "public_id": "<uuid7>",
  "session_id": "<client-session>",
  "sequence_id": 2,
  "timestamp": "2026-01-18T12:00:01Z",
  "payload": {}
}
```

Environment Configuration

Development

```
# .env.development
DB_URL=sqlite+aiosqlite:///./data/snapper.db
SERVER_HOST=127.0.0.1
SERVER_PORT=8000
SERVER_RELOAD=true
SERVER_PROXY_HEADERS=true
SERVER_FORWARDED_ALLOW_IPS=127.0.0.1
```

Production

```
# .env.production
DB_URL=postgresql+asyncpg://user:pass@db.example.com/snapper
MASTER_PASSWORD=<strong_random_password>
SERVER_HOST=0.0.0.0
SERVER_PORT=8000
SERVER_RELOAD=false
SERVER_PROXY_HEADERS=true
SERVER_FORWARDED_ALLOW_IPS=127.0.0.1,172.17.0.1
```

Docker

Variables can be passed via docker-compose.yml:

```
services:
  snapper:
    env_file: .env
    environment:
      - DB_URL=postgresql+asyncpg://snapper:password@postgres/snapper
```

Settings Encryption

Sensitive settings are encrypted with Fernet (AES-128-CBC + HMAC-SHA256):

1. Salt is derived from MASTER_PASSWORD via SHA-256
2. Encryption key is derived via PBKDF2-HMAC-SHA256 (100,000 iterations)
3. Values are encrypted with Fernet and stored in database

If MASTER_PASSWORD changes, encrypted settings become inaccessible. Use snapper settings-rotate-encryption to rotate passwords safely.

Configuration Validation

The system validates configuration at startup via the shared .env allowlist check. The encrypt/decrypt round-trip is not a startup step; it runs as part of the snapper settings-rotate-encryption command.

The allowlist is the union of BootstrapSettingsLoader aliases and each subsystem's exported ENV_VARS set. Unknown keys fail startup with suggestions, so typos such as MASTER_PASWORD do not silently fall back to defaults. When adding a new env var, register it on the owning module's ENV_VARS or add it as a bootstrap settings field.

Development

Guidelines for developers working on the Snapper project.

Requirements

- Python 3.14+
- Poetry
- Node.js 26+ and pnpm 11+ (frontend engines, Docker UI build, and CI workflows all standardize on Node 26)
- Build tools for Python packages. TA-Lib is optional at runtime: the indicator adapter uses TA-Lib when importable and falls back to pure Python implementations otherwise.
- Pre-commit hooks

Environment Setup

```
# Install system dependencies (Linux/macOS)
make system-deps

# Install Python dependencies
make setup

# Install frontend dependencies
make ui-setup

# Synchronize pre-commit hooks
make pre-refresh

# Refresh dependencies, local tools, and Dockerfile tool pins
make update
```

Quality Gates

Main Command

```
make check-all
```

Executes the complete quality gate:

1. Formatting (ruff, black, isort)
2. Linting (ruff)

3. Type checking (mypy)
4. Docstring compliance
5. No # comments in Python
6. Canonical __main__ guards
7. Empty __init__.py files
8. No forbidden temporal mutations
9. Egress compose safety check (check-egress-compose)
10. No pragma/noqa/ignore exclusions
11. Vendor-neutrality check (check-vendor-neutral)
12. Pydantic-only FastAPI I/O (check-pydantic-routes)
13. Tests with 100% coverage
14. Frontend (ui-typecheck, ESLint, Prettier, dead code, i18n checks, tests with coverage)

Individual Steps

Command	Description
make fmt	Check formatting
make fmt-fix	Fix formatting
make lint	Ruff linting
make lint-fix	Fix linting issues
make typecheck	Mypy type checking
make test	Tests (parallel)
make test-serial	Tests (sequential, debug)
make cov	Tests with coverage
make check-egress-compose	Reject unsafe snapper-egress compose wiring

Automatic Fixes

```
make fix          # Backend: fmt-fix + lint-fix + move-imports
make fix-all      # Backend + frontend
```

Code Standards

Types (MANDATORY)

All functions must have type annotations:

```
def calculate_rsi(prices: pd.Series, period: int = 14) -> pd.Series:
    """Calculate RSI indicator."""
    ...

async def fetch_data(symbol: str) -> list[Candle]:
    """Fetch candle data."""
    ...
```

Test functions must have -> None:

```
async def test_rsi_calculation() -> None:
    """Test RSI calculation returns expected values."""
    ...
```

Docstrings (Google style)

```
def process_signal(signal: Signal, config: StrategyConfig) -> Order | None:
    """Process trading signal and generate order.

    Takes a trading signal and configuration, validates the signal,
    and generates an appropriate order if conditions are met.

    Args:
        signal: Trading signal to process.
        config: Strategy configuration dictionary.

    Returns:
        Generated order if valid, None otherwise.

    Raises:
        ValueError: If signal data is invalid.
    """
    ...
```

No # Comments

comments are forbidden. Use docstrings:

```
# WRONG
def foo():
    # This calculates something
    x = 1 + 1

# CORRECT
def foo():
    """Calculate the sum."""
    x = 1 + 1
```

Imports

Imports at the top of file, single lines:

```
from datetime import UTC
from datetime import datetime

import pandas as pd
from sqlalchemy import select

from snapper.config.settings import get_settings
from snapper.data.models import Candle
```

Forbidden:

- from __future__ import annotations
- if TYPE_CHECKING: blocks
- Re-exports in __init__.py

__init__.py

__init__.py files must be empty (or docstring only):

```
"""Snapper strategies module."""
```

Tests

Structure

```
tests/
├── conftest.py          # Shared fixtures
├── api/                 # API tests
├── application/         # Business logic tests
├── cli/                 # CLI tests
├── data/                # Data layer tests
├── indicators/          # Indicator tests
├── messaging/           # ZMQ tests
├── strategies/          # Strategy tests
└── ...
```

Writing Tests

```
import pytest
from snapper.indicators.rsi import rsi

class TestRSI:
```

```

"""Test suite for RSI indicator."""

def test_rsi_returns_series_with_correct_length(self) -> None:
    """
    Given a price series of length N
    When RSI is calculated with period P
    Then the result has length N.
    """
    prices = pd.Series([100, 102, 101, 103, 105])
    result = rsi(prices, period=14)
    assert len(result) == len(prices)

def test_rsi_values_in_valid_range(self) -> None:
    """
    Given any price series
    When RSI is calculated
    Then all values are between 0 and 100.
    """
    prices = pd.Series([100, 102, 101, 103, 105, 104, 106])
    result = rsi(prices, period=3)
    assert (result >= 0).all()
    assert (result <= 100).all()

```

Async Tests

```

import pytest

@pytest.mark.asyncio
async def test_fetch_candles() -> None:
    """Test fetching candles from repository."""
    repo = get_repository(db_url)
    candles = await repo.get_candles("BTC-USD", limit=10)
    assert len(candles) <= 10

```

Tests must produce zero pytest warnings. A common pitfall: AsyncMock() makes **every** attribute on the mock async, including synchronous methods such as session.add() or session.expunge(). The fix is to override the synchronous attributes explicitly:

```

from unittest.mock import AsyncMock, MagicMock

session = AsyncMock(add=MagicMock(), expunge=MagicMock())

```

Run `python -m pytest tests/ -W error::RuntimeWarning` to fail the suite on any unawaited-coroutine warnings.

Fixtures

```

@pytest.fixture
def sample_candles() -> list[Candle]:
    """Provide sample candle data for tests."""
    return [

```

```
Candle(open=100, high=105, low=95, close=102, volume=1000),  
Candle(open=102, high=110, low=100, close=108, volume=1200),  
]
```

100% Coverage

The project requires 100% code coverage (TDD). Check:

```
make cov
```

Coverage is configured under [tool.coverage.run] / [tool.coverage.report] in pyproject.toml. The fail_under = 100 threshold enforces the TDD requirement.

Frontend

Setup

```
make ui-setup
```

Development

```
make ui-dev
```

Frontend at <http://localhost:3000/> with backend proxy.

Quality Checks

Command	Description
make ui-lint	ESLint
make ui-format	Prettier check
make ui-typecheck	TypeScript type check
make ui-dead-code	Knip dead code
make ui-i18n-check	Frontend locale/catalog consistency
make ui-test	Vitest tests
make ui-cov	Tests with coverage

Fixes

```
make ui-fix    # lint-fix + format-fix + dead-code-fix
```

Type Generation

```
make ui-gen-types  # OpenAPI types, WebSocket types, Zod schemas, entity  
aliases, permissions
```

Internationalization

The frontend i18n catalogs in `frontend/src/locales/<lang>/*.json` are the source of truth for translated UI strings across 45 locales. iOS mirrors the subset of strings used on the native client by porting them into `ios/Snapper/Resources/Localization/Localizable.xcstrings`.

Market catalog (description / asset class / sector)

The `market.description.*`, `market.assetClass.*`, and `market.sector.*` namespaces are ported from the frontend JSON into the iOS xcstrings by `scripts/port_market_catalog.py`. Run after editing any `frontend/src/locales/<lang>/market.json` file under one of those namespaces:

```
python scripts/port_market_catalog.py          # write  
python scripts/port_market_catalog.py --check # CI parity gate
```

The `--check` mode is wired into `make ui-check` via `ui-i18n-check-market`, so a forgotten regeneration fails the backend + frontend quality gate. Placeholder tokens (`{{name}}` / `{{assetClass}}`) are rewritten to xcstrings positional codes (`%1$@`, `%2$@`) based on appearance order in the EN template.

iOS-side coverage of the resulting catalog is enforced by `CatalogParityTests` (45 locales × every key declared in `SnapperTests/i18n/ExpectedKeys.swift`); adding a new key to the catalog requires adding it to `ExpectedKeys.swift` in the same iOS PR.

Alerts catalog (legacy direction: xcstrings → JSON)

Historically the iOS alerts catalog was the source of truth and frontend caught up via `scripts/port_ios_alert_catalog.py`. That direction is preserved for back-compatibility; new namespaces should flow frontend → iOS via the market-catalog pattern above.

Database Migrations

Dev DB Location

The local server SQLite database lives at `./data/snapper.db`. Test and coverage Make targets use an isolated SQLite fixture at `./data/dev.db` by setting `DB_URL=sqlite+aiosqlite:///./data/dev.db` inside the Makefile. This keeps `make test`, `make cov`, and `make check-all` from mutating the database used by a running local server, even when `.env` points at `./data/snapper.db` or PostgreSQL.

Migration Strategy (Development)

The schema is built from a chain of sequential Alembic revisions (`0001_init.py`, `0002_instrument_feed_health.py`, ...), each of which revises the previous one. To change the schema, add a new numbered revision rather than editing an existing one, then apply and reseed:

```
make migrate-dev
```

`make migrate-dev` applies Alembic migrations (via `snapper db-init`) then seeds development data (via `snapper db-seed --profile dev`). There is no standalone `make migrate` target -- always use `make migrate-dev` or `make migrate-prod`.

Applying (Production)

```
make migrate-prod
```

Rollback

```
snapper db-downgrade
```

Docker

Build

```
make docker-build-dev    # Development image
make docker-build-prod   # Production image
make docker-migrate-dev  # Initialize and seed the Docker SQLite database
```

Run

```
make docker-migrate-dev  
make docker-run
```

Docker Compose

```
make docker-migrate-dev  
docker compose up -d
```

CI/CD

Pipeline executes the same consolidated gate used locally:

1. make check-all — Backend checks, frontend checks, exclusion scan, and tests with coverage

For debugging a failing pipeline locally, the equivalent steps are:

1. make check — Backend quality checks
2. make ui-check — Frontend lint, format, dead-code, type, and i18n catalog checks (ui-lint ui-format ui-dead-code ui-typecheck ui-i18n-check ui-i18n-check-alerts ui-i18n-check-market)
3. make check-exclusions — No pragma/noqa/ignore bypasses
4. make cov — Backend tests with coverage
5. make ui-cov — Frontend tests with coverage

Workflow

Before Commit

```
make check-all
```

Before PR

1. Ensure make check-all passes
2. Add/update tests
3. Update documentation if needed
4. Verify 100% coverage

Code Review Checklist

- [] Types for all functions
- [] Docstrings (Google style)
- [] No # comments
- [] Tests for new functionality
- [] 100% coverage
- [] make check-all passes

Project Structure

```
snapper/
├── src/snapper/           # Source code
│   ├── api/              # API schemas
│   ├── application/      # Business logic
│   ├── auth/             # Authentication
│   ├── cli/              # CLI interface
│   ├── config/           # Configuration
│   ├── core/             # Core types
│   ├── data/             # Data layer
│   ├── egress/           # snapper-egress sidecar entry point
│   ├── i18n/             # Backend i18n catalogs
│   ├── indicators/       # Technical indicators
│   ├── infrastructure/   # External integrations
│   ├── interface/        # WebSocket
│   ├── mcp/              # MCP server (delegate-facing tool surface)
│   ├── messaging/        # ZeroMQ
│   ├── server/           # FastAPI
│   ├── strategies/       # Trading strategies
│   └── utils/            # Utilities
├── tests/                # Tests
├── frontend/             # React dashboard
├── docs/                 # Documentation
├── scripts/              # Helper scripts
└── data/                 # Local data
```

Tools

Ruff

Linters and formatters. Configuration in pyproject.toml:

```
[tool.ruff.lint]
select = ["E", "F", "W", "I", "B", "UP", "C4", "SIM", "N", "PERF", "ISC",
"PLW", "PLC", "A", "TID", "D"]
```

Mypy

Type checker. Strict mode enabled:

```
[tool.mypy]
python_version = "3.14"
strict = true
```

Black

Formatter. Line length 100:

```
[tool.black]
line-length = 100
```

Pre-commit

Hooks run before commit:

```
pre-commit install
pre-commit run --all-files
```

Debugging

Logging

```
from loguru import logger

logger.info(f"Processing signal: {signal}")
logger.debug(f"Calculated RSI: {rsi_value}")
logger.error(f"Failed to execute order: {e}")
```

ZMQ Logger

```
snapper zmq-logger --payload
```

Serial Tests

```
make test-serial
```

pdb/ipdb

```
import ipdb; ipdb.set_trace()
```

Documentation

Docstrings

Every module, class, and public function must have a docstring.

docs/

Markdown documentation in docs/ directory.

PDF

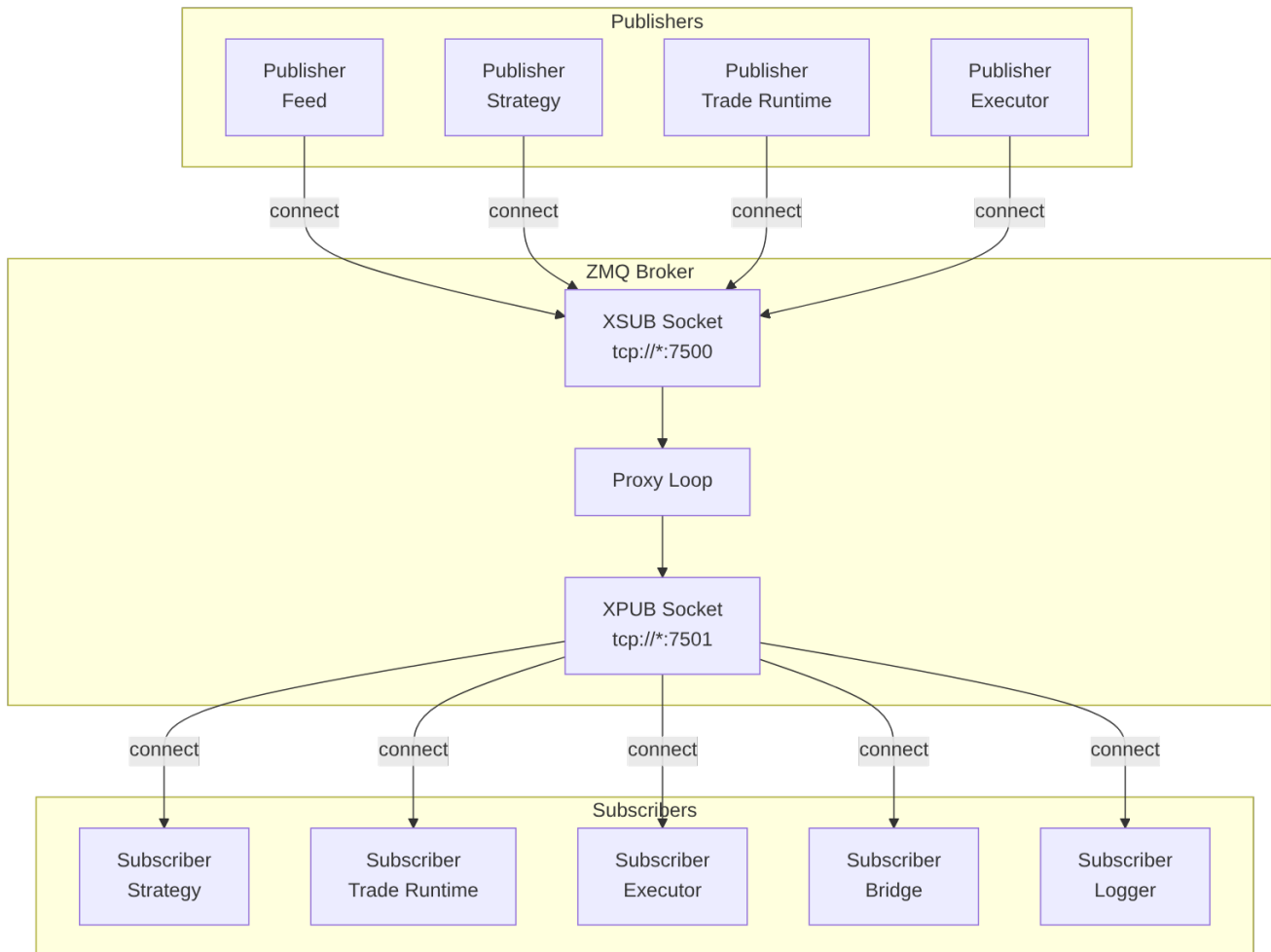
```
make docs-pdf
```

Generates frontend/public/snapper.pdf from README + docs/*.md so the docs ship with the frontend bundle and are linked from the login page.

Messaging (ZeroMQ)

Snapper uses ZeroMQ for inter-process communication in a pub/sub architecture. The central component is an XPUB/XSUB broker that routes messages.

Architecture



XPUB/XSUB Broker

The broker is the central message hub:

- **XSUB** (tcp://*:7500) — Publishers connect here
- **XPUB** (tcp://*:7501) — Subscribers connect here

The broker forwards messages in both directions:

- Data: XSUB → XPUB
- Subscriptions: XPUB → XSUB

Starting the Broker

```
snapper broker
```

Or programmatically:

```
from snapper.messaging.infrastructure.broker import ZmqBrokerThread

broker = ZmqBrokerThread()
broker.start()
# ...
broker.stop()
```

Topics

Topic format varies by category (see per-category tables below).

Market Data

Topic	Description
market.kraken.BTC-USD.candles.1h	Hourly BTC/USD candles from Kraken
market.kraken.BTC-USD.ticks	BTC/USD ticks from Kraken
market.kraken_equities.AAPL.candles.1d	Daily AAPL candles from Kraken Equities (note: live market.* topics exclude polygon — Polygon is replay-only via market.paper.polygon....)

Signals

Topic	Description
signals.paper.BTC-USD.rsi_btc_1h	Paper trading signals
signals.kraken.BTC-USD.live	Live signals from Kraken

Orders & Executions

Topic	Description
orders.commands.kraken.BTC-USD.submit	Order requests for Kraken
orders.events.kraken.BTC-USD.executed	Order executions from Kraken

Topic	Description
orders.events.kraken.BTC-USD.submitted	Order submitted events
orders.events.kraken.BTC-USD.accepted	Order accepted events
orders.events.kraken.BTC-USD.rejected	Order rejected events
orders.events.kraken.BTC-USD.cancelled	Order cancelled events
orders.events.kraken.BTC-USD.expired	Order expired events
orders.events.kraken.BTC-USD.replaced	Order replaced events

TradeCommand, VenueEvent, and TradeProjectionCheckpoint are durable database artifacts used by the trade runtime. They are not additional ZMQ topics.

System

Topic	Description
system.heartbeats.executor. {exchange}	Executor heartbeat for the single-wallet template path (4 segments)
system.heartbeats.executor. {exchange}.{wallet_short}	Per-wallet executor heartbeat (5 segments). wallet_short is the last 12 lowercase hex characters of the wallet UUID (see snapper.core.wallet_short)
system.heartbeats.strategy. {name}	Strategy heartbeat (e.g. strategy.rsi_btc_1h)
system.heartbeats.feed. {exchange}	Feed heartbeat (e.g. feed.kraken, feed.paper.kraken)
system.settings	Configuration change notifications
system.symbol_aliases	Symbol cache invalidation

Heartbeat component values use dot notation matching the topic path after system.heartbeats.: executor.kraken, executor.kraken.019d6ca45f2e, strategy.rsi_btc_1h, feed.kraken, feed.paper.kraken. The per-wallet executor heartbeat envelope additionally carries meta.wallet_public_id so subscribers that prefix-match the 4-segment parent topic can still disambiguate by reading the payload.

Admin

Cross-cutting administrative notifications fanned out to every backend subscriber that cares about user/operator state changes.

Topic	Description
admin.user_deactivated	A user (or AI delegate) was deactivated; revocation hook for in-process token caches + APNs push fan-out via the notify sidecar
admin.scope_granted	create_grant published a new active scope row (instrument-exclusive grant just inserted)
admin.scope_handed_over	handover finalized a scope handover (old row closed, new row open, both committed)
admin.scope_revoked	revoke_grant closed an active scope row; revoke hook for any subscriber holding cached principal state

Shape: admin.{resource} — exactly 2 segments. The validator only enforces the 2-segment shape; the resource token itself is not character-class-checked, so consumers must use the payload's type discriminator as the authoritative routing hint.

Accruals

Funding / rollover / borrow ledger entries that affect mark-to-market.

Topic	Description
accruals.kraken_futures.BTC-USD-PERP.funding	Perpetual funding
accruals.kraken.EUR-USD.rollover	FX rollover
accruals.kraken.BTC-USD.borrow	Borrow interest

Shape: accruals.{exchange}.{instrument}.{accrual_type} (4 segments). accrual_type must be one of funding, rollover, borrow.

Alerts

Per-user push-notification fanout consumed by the WS bridge and the APNs sidecar.

Topic	Description
alerts.<user_public_id>.order_fill_full	Order filled in full
alerts.<user_public_id>.order_rejected	Venue rejected the order
alerts.<user_public_id>.position_stop_loss_fired	Stop-loss fired on an open position
alerts.<user_public_id>.margin_warning	Margin warning
alerts.<user_public_id>.critical_system_error	Critical system error

Shape: alerts.{user_public_id}.{alert_type} (3 segments). user_public_id must be a UUID7; alert_type is the discriminated union in the AlertType Literal — the validator pulls the set via typing.get_args(AlertType) so adding a new alert type to the schema flows through automatically.

Plan Decisions

Bracket / trailing-stop / execution-plan decisions, emitted by the PlanExecutorService immediately after each ExecutionPlanDecision SCD2 insert. The notify sidecar subscribes to turn stop-loss / take-profit fires into iOS pushes.

Topic	Description
plans.decisions.<plan_public_id>	One topic per execution plan; payload carries the decision frame

Shape: plans.decisions.{plan_public_id} (3 segments) with plan_public_id in UUID7 form.

AI Reviews

Outbound delegate-consultation frames for the human-in-the-loop review queue. The bridge applies a per-frame scope filter so only the selected AI delegate receives frames for wallets and instruments covered by its live scope grant. Non-delegate WebSocket principals fail closed for this topic family even if they somehow request a subscription.

Topic	Description
ai_reviews.<user_public_id>.<strategy_public_id>.request	Strategy is asking for review
ai_reviews.<user_public_id>.<strategy_public_id>.decision_ack	Operator's decision acknowledged
ai_reviews.<user_public_id>.<strategy_public_id>.caps_violation	Caps violation after approval

Shape: ai_reviews.{user_public_id}.{strategy_public_id}.{suffix} (4 segments). Both ID segments are UUID7; suffix is one of the three frame discriminators above. The payload must include string wallet_public_id and instrument_public_id values so the bridge can check the delegate's grant before forwarding.

Backtest

Per-run lifecycle and progress events.

Topic	Description
backtest.<wallet_public_id>.<run_public_id>.started	Run kicked off
backtest.<wallet_public_id>.<run_public_id>.progress	Periodic progress tick
backtest.<wallet_public_id>.<run_public_id>.milestone	Engine progress milestone — payload's milestone field is one of 25pct/50pct/75pct (BacktestProgressMilestone Literal)
backtest.<wallet_public_id>.<run_public_id>.completed	Run finished
backtest.<wallet_public_id>.<run_public_id>.failed	Run failed
backtest.<wallet_public_id>.<run_public_id>.cancelled	Run cancelled

Shape: backtest.{wallet_public_id}.{run_public_id}.{event} (4 segments). wallet_public_id and run_public_id are UUID7; event is drawn from BacktestProgressEvent in [data.py](#) — the validator's BACKTEST_EVENTS set is derived via typing.get_args(BacktestProgressEvent) so the wire schema and the validator share one source of truth. Subscription-only prefixes (backtest., backtest.{wallet}., backtest.{wallet}.{run}.) are accepted on the subscribe.py / bridge.py paths.

Bus

Internal cross-service event bus over the shared broker.

Topic	Description
bus.delegate_offline	Delegate session torn down
bus.ai_review_decision	AI review decision recorded — fast-path fanout to the in-process AI review bus listener that resolves the strategy's await future (WS fanout of the operator's ack is the separate external ai_reviews.<user>.<strategy>.decision_ack topic). The matching bus.ai_review_request topic is intentionally NOT published today
bus.caps_violation_after_ai_approve	Caps violation post-approval

Shape: bus.{name} (2 segments) where name matches [a-z][a-z0-9_]* (snake_case). The payload's StrictDataSchema.type discriminator is the authoritative routing hint; the validator only enforces structural shape so the schema layer can reject unknown payload types.

Process Manager Events

Snapshot + per-run events emitted by ProcessLauncherService whenever the configured set of processes changes or a run transitions. Frontend subscribes via wsDispatcher and uses these to invalidate React Query caches (no REST polling).

Topic	Description
processes.events.summary.<instance_id>	Full snapshot of every configured process's current status (start/stop/crash/completion transitions)
processes.events.configured.<instance_id>	Snapshot of currently-known process names. Emits in two places: create_process_config (persisted-config-row create), and spawn_per_wallet_executors (per-wallet executor instances appear). It does NOT fire on persisted-row delete or on per-wallet teardown today. The snapshot's process_names is the union of persisted config names and instance_configs.keys().
processes.events.runs.<process_name>	Per-run lifecycle transitions: running / succeeded / failed / cancelled
strategies.events.list.<instance_id>	Snapshot of canonical class paths for every STRATEGY-role process config (drives the Strategies view)

Shape: 4 segments. The instance-id / process-name tail must match [A-Za-z][A-Za-z0-9_-]* (regex `_PROCESSES_TOPIC_NAME_PATTERN`). Numeric SNAPPER_COORDINATOR_INSTANCE_ID values are converted to topic-safe slugs by `ProcessLauncherService.coordinator_topic_slug()`; the default instance therefore emits on `coord-0`, and instance 7 emits on `coord-7`. The launcher emits these best-effort: if no MessagePublisher is wired the helper no-ops, and send failures are logged without corrupting process state.

Message Data Classes

All messages use typed Data classes (Pydantic) from `snapper.messaging.schemas.data`. Every Data class inherits from `StrictDataSchema` and carries:

Field	Type	Description
public_id	string	UUID7 external identifier, stable across REST and WS
type	string	Literal type discriminator

Field	Type	Description
timestamp	datetime	Message creation time
session_id	string	UUID7 of the producer process session (required at construction)
sequence_id	int	Monotonic counter per topic within the session (required at construction)

session_id and sequence_id are allocated by producers from a shared SequenceTracker before the payload is constructed. MessagePublisher.send() serializes and forwards the already-complete payload without modifying provenance. Consumers can use these fields to detect message loss and producer restarts.

The old messaging.schemas.messages module still exists but only contains parse_message() and MessageParseError.

TickData

```
from datetime import UTC, datetime

from snapper.messaging.schemas.data import TickData

tick = TickData(
    public_id="019e1a2b-0000-7000-8000-000000000101",
    session_id="019e1a2b-0000-7000-8000-000000000001",
    sequence_id=1,
    timestamp=datetime.now(UTC),
    instrument="BTC-USD",
    exchange="kraken",
    bid=41990.0,
    ask=42010.0,
    last=42000.0,
    volume=0.5,
)
```

Fields:

Field	Type	Description
type	string	"tick"
public_id	string	UUID7 external identifier
instrument	string	Instrument symbol
exchange	string	Exchange name
bid	float None	Best bid price
ask		Best ask price

Field	Type	Description
	float None	
last	float None	Last traded price
volume	float	Volume
is_delayed	bool	Whether the feed delivers exchange-delayed ticks (e.g. ~10 min for TradFi index futures); strategies must gate on this before treating the price as current
is_extended_hours	bool None	Whether the tick occurred during an extended-hours TradFi session; None means the feed does not distinguish extended-hours ticks
timestamp	datetime	Timestamp
session_id	string	Producer session (inherited)
sequence_id	int	Monotonic counter per topic or logical stream (inherited)

CandleData

```

from datetime import UTC, datetime

from snapper.messaging.schemas.data import CandleData

candle = CandleData(
    public_id="019e1a2b-0000-7000-8000-0000000000102",
    session_id="019e1a2b-0000-7000-8000-0000000000001",
    sequence_id=2,
    timestamp=datetime.now(UTC),
    instrument="BTC-USD",
    exchange="kraken",
    timeframe="1h",
    open_at=datetime(2026, 1, 18, 12, 0, tzinfo=UTC),
    open=42000.0,
    high=42500.0,
    low=41800.0,
    close=42300.0,
    volume=1234.56,
)

```

Fields:

Field	Type	Description
type	string	"candle"
public_id	string	UUID7 external identifier

Field	Type	Description
instrument	string	Symbol
exchange	string	Exchange
timeframe	string	Timeframe (1m, 5m, 1h, etc.)
open_at	datetime	Exchange-provided candle interval start time
open	float	Open price
high	float	High
low	float	Low
close	float	Close price
volume	float	Volume
vwap	float null	Volume-weighted average price (optional)
trades	int null	Number of trades in the candle (optional)
timestamp	datetime	Timestamp

SignalData

```

from datetime import UTC, datetime

from snapper.messaging.schemas.data import SignalData

signal = SignalData(
    public_id="019e1a2b-0000-7000-8000-0000000000201",
    session_id="019e1a2b-0000-7000-8000-000000000010",
    sequence_id=1,
    timestamp=datetime.now(UTC),
    instrument="BTC-USD",
    exchange="paper",
    side="buy",
    strength=0.85,
    reason="RSI <= 30",
    strategy_name="rsi_btc_1h",
    price=42000.0,
    fired_at=datetime.now(UTC),
)

```

Fields:

Field	Type	Description
type	string	"signal"
public_id	string	UUID7 external identifier

Field	Type	Description
instrument	string	Symbol
exchange	string	Target exchange
side	string	"buy" or "sell"
strength	float	Signal strength 0.0-1.0
reason	string	Reason
strategy_name	string	Strategy name (required for paper exchange signals)
price	float	Price
fired_at	datetime	Domain time when signal was generated
timestamp	datetime	System timestamp

Paper signals (exchange == "paper") require strategy_name to be set. The schema enforces this invariant at construction time so invalid paper signals cannot be created.

OrderRequestData

```
from datetime import UTC, datetime

from snapper.messaging.schemas.data import OrderRequestData

order = OrderRequestData(
    public_id="019e1a2b-0000-7000-8000-0000000000301",
    session_id="019e1a2b-0000-7000-8000-000000000020",
    sequence_id=1,
    timestamp=datetime.now(UTC),
    instrument="BTC-USD",
    exchange="kraken",
    client_order_id="ord_123",
    side="buy",
    order_type="limit",
    price=42000.0,
    quantity=0.1,
    strategy_id="rsi_btc_1h",
    mode="live",
)
```

ExecutionData

```
from datetime import UTC, datetime

from snapper.messaging.schemas.data import ExecutionData

execution = ExecutionData(
    public_id="019e1a2b-0000-7000-8000-0000000000401",
```

```

    session_id="019e1a2b-0000-7000-8000-000000000030",
    sequence_id=1,
    timestamp=datetime.now(UTC),
    instrument="BTC-USD",
    exchange="kraken",
    client_order_id="ord_123",
    exchange_order_id="KRAKEN-456",
    side="buy",
    price=42000.0,
    size=0.1,
    last_size=0.1,
    last_price=42000.0,
    fee=0.001,
    fee_asset="USD",
    status="filled",
    executed_at=datetime.now(UTC),
)

```

OrderData

OrderData is the canonical event for order state. It carries the request-time margin metadata (leverage, reduce_only) all the way to subscribers and the REST /orders endpoint, so the frontend can render shorts and reduce-only intent.

```

from datetime import UTC, datetime

from snapper.messaging.schemas.data import OrderData

order_status = OrderData(
    public_id="019e1a2b-0000-7000-8000-0000000000501",
    session_id="019e1a2b-0000-7000-8000-000000000020",
    sequence_id=2,
    timestamp=datetime.now(UTC),
    instrument="BTC-USD",
    exchange="kraken",
    client_order_id="ord_123",
    exchange_order_id="KRAKEN-456",
    side="sell",
    order_type="limit",
    size=0.1,
    filled_size=0.0,
    status="accepted",
    price=42000.0,
    created_at=datetime.now(UTC),
    leverage=3,
    reduce_only=False,
)

```

HeartbeatData

```

from datetime import UTC, datetime

from snapper.messaging.schemas.data import HeartbeatData

```

```
heartbeat = HeartbeatData(  
    public_id="019e1a2b-0000-7000-8000-0000000000601",  
    session_id="019e1a2b-0000-7000-8000-000000000001",  
    sequence_id=1,  
    timestamp=datetime.now(UTC),  
    component="zmq_broker",  
    status="healthy",  
    sequence=1,  
    lag_ms=5,  
)
```

ProcessSummaryEventData

Full snapshot published on `processes.events.summary.<instance_id>` whenever the launcher detects a status transition. Frontend replaces its React Query cache entry wholesale — no diff reconciliation.

Fields:

Field	Type	Description
type	string	"process_summary_event"
coordinator	string	Topic-safe slug of the emitting node (default coord-0); mirrors the topic suffix so consumers can attribute rows to the coordinator that sampled them
processes	list[ProcessSummaryItem]	Unordered snapshot of per-process rows (configured first in repository row order, then runtime per-wallet instances — consumers must not rely on this order)
snapshot_at	datetime	Bus time the snapshot was assembled

ProcessSummaryItem fields: name, running, enabled, role, lifecycle, active_public_id, rss_bytes (int | None, subprocess resident set size in bytes; None for thread-mode/unsampled processes), and cpu_percent (float | None, whole-process CPU% which can exceed 100 on multi-threaded children; None under the same conditions and 0.0 on the first sample after a restart). PID / command / exit_code are intentionally absent — they only exist for native subprocesses, so detailed status is fetched via REST when needed.

ProcessConfiguredEventData

Published on `processes.events.configured.<instance_id>` from two launcher sites only: `create_process_config` (persisted-config-row create) and `spawn_per_wallet_executors` (per-wallet executor instances appear). The launcher does NOT emit on persisted-row delete or on per-wallet teardown today. `process_names` is the sorted union of persisted config names AND `instance_configs.keys()` at snapshot time.

Fields:

Field	Type	Description
type	string	"process_configured_event"
process_names	list[string]	All configured process names at snapshot time
snapshot_at	datetime	Bus time the snapshot was assembled

ProcessRunEventData

A single process-run lifecycle transition. Per-run (not snapshot) so consumers can append to their run-history view without re-fetching.

Fields:

Field	Type	Description
type	string	"process_run_event"
process_name	string	Owning process
run_id	string	Stable identifier for this run
status	string	running / succeeded / failed / cancelled (mirrors ProcessRunStatusEnum.value)
started_at	datetime	Bus time the run kicked off
completed_at	datetime None	Bus time the run finished; None while in flight
exit_code	int None	Native subprocess exit code; None for async-task processes (the launcher folds non-zero exits into the run record's error field) and None while in flight

StrategyListEventData

Published on `strategies.events.list.<instance_id>` whenever a STRATEGY-role process config is created or its start/stop state transitions. The launcher derives the payload from **persisted** process configs filtered by `role == STRATEGY` — NOT from the in-memory `StrategyFactory.STRATEGY_CLASSES` registry (which is static after import time). Frontend uses this to refresh the Strategies view without REST polling.

Fields:

Field	Type	Description
type	string	"strategy_list_event"
strategy_classes	list[string]	Sorted canonical class paths of every active STRATEGY-role process config
snapshot_at	datetime	Bus time the snapshot was assembled

Publisher

MessagePublisher (recommended)

MessagePublisher sends fully constructed payloads to an explicit stream_key. Producers allocate session_id and sequence_id from a shared SequenceTracker before constructing the payload, then pass both the topic and payload to MessagePublisher.send().

```
import zmq.asyncio
from snapper.messaging.infrastructure.validated_socket import
ValidatedPublisher, apply_hwm, HWM_MARKET_DATA
from snapper.messaging.infrastructure.publisher import MessagePublisher,
SequenceTracker
from snapper.messaging.topics.builders import topic_for_message

ctx = zmq.asyncio.Context()
raw_socket = ctx.socket(zmq.PUB)
apply_hwm(raw_socket, sndhwm=HWM_MARKET_DATA)
raw_socket.connect("tcp://127.0.0.1:7500")

tracker = SequenceTracker()
publisher = MessagePublisher(ValidatedPublisher(raw_socket), tracker)

topic = topic_for_message(candle_data)

await publisher.send(topic, candle_data)
publisher.close()
```

For brevity, the example assumes candle_data was already constructed with session_id and sequence_id allocated from tracker.

One SequenceTracker per component process; all MessagePublisher instances within that component share it. Counters survive socket reconnects — only a full component restart creates a new session.

Per-Topic Counters

SequenceTracker maintains monotonic counters keyed by **ZMQ topic** (e.g. market.kraken.BTC-USD.candles.1m, orders.events.kraken.BTC-USD.executed). Each topic has its own independent counter. Producers allocate provenance using the final ZMQ topic string and then pass that same topic to MessagePublisher.send().

For DB-only writes that do not flow over ZMQ (instruments, users, settings seed data), the counter key is the destination table name as a logical identifier.

For WS control and REST middleware paths, logical channel names are used: server.control, server.telemetry, rest.control, rest.telemetry.

For paper market data or other cases where the topic cannot be derived from the payload alone, pass an explicit topic override:

```
topic = "market.paper.kraken.BTC-USD.ticks"
await publisher.send(topic, tick_data)
```

ValidatedPublisher (low-level)

Direct access to the socket wrapper without StrictDataSchema serialization helpers:

```
import zmq.asyncio
from snapper.messaging.infrastructure.validated_socket import
ValidatedPublisher, apply_hwm, HWM_MARKET_DATA

ctx = zmq.asyncio.Context()
raw_socket = ctx.socket(zmq.PUB)
apply_hwm(raw_socket, sndhwm=HWM_MARKET_DATA)
raw_socket.connect("tcp://127.0.0.1:7500")
publisher = ValidatedPublisher(raw_socket)

topic = "market.kraken.BTC-USD.candles.1h"
payload = candle_data.publish_to(topic)
await publisher.send_multipart(topic, payload)
publisher.close()
```

Subscriber

Subscribing to messages via validated socket wrapper around a raw ZMQ SUB socket:

```
import zmq.asyncio
from snapper.messaging.infrastructure.validated_socket import
ValidatedSubscriber, apply_hwm, HWM_MARKET_DATA
from snapper.messaging.schemas.messages import parse_message

ctx = zmq.asyncio.Context()
raw_socket = ctx.socket(zmq.SUB)
apply_hwm(raw_socket, rcvhwm=HWM_MARKET_DATA)
raw_socket.connect("tcp://127.0.0.1:7501") # broker XPUB
```

```
subscriber = ValidatedSubscriber(raw_socket)
subscriber.subscribe("market.kraken.BTC-USD.")

while True:
    topic, payload = await subscriber.recv_multipart()
    envelope = parse_message(payload.decode())
    print(f"Received {envelope.type} on {topic}")
```

Market Data Publisher

Built-in publisher for exchange data:

```
snapper feed --symbols "BTC-USD,ETH-USD"
```

Programmatically:

```
from snapper.messaging.publishers.kraken import KrakenMarketDataPublisher

async def run_feed():
    publisher = KrakenMarketDataPublisher(symbols=["BTC-USD", "ETH-USD"])
    await publisher.start()
```

Candle Identity Cache

On startup the publisher loads the latest candle `public_id` per (instrument, timeframe) pair from the database into an in-memory cache. When a new tick arrives for an existing candle interval (`open_at` matches), the cached `public_id` is reused so the same logical candle keeps a stable identity across ZMQ, WebSocket, and the database. When `open_at` advances to a new interval, a fresh UUID7 is minted and the cache entry is replaced. This avoids any DB reads on the hot path.

Micro-Batch DB Persistence

DB writes are decoupled from the ZMQ publish path via per-loop micro-batching. Each producer loop (candles, ticks, trades) accumulates rows in a local batch and flushes to the database on a size or age threshold:

- **Size trigger:** candles flush at 100 rows, ticks/trades at 500 rows
- **Age trigger:** all streams flush after 50 ms since the first batch item
- **Publish-first:** ZMQ delivery happens before DB persistence; subscribers receive data without waiting for the database

Thresholds are configurable via `write_buffer_flush_ms`, `write_buffer_candle_max_rows`, `write_buffer_tick_max_rows`, and `write_buffer_trade_max_rows` settings (cached at publisher start, require process restart to change).

On shutdown (launcher cancellation or direct stop), each loop flushes its remaining batch in a finally block. The `asyncio.wait` age trigger uses a persistent future that is never cancelled on timeout, preserving the underlying exchange subscription iterator.

Candle flushes that hit an `IntegrityError` (e.g. out-of-order timestamps) fall back to row-by-row retry, isolating the bad row without losing the rest of the batch. The repository `upsert_candles()` contract is unchanged.

Per-stream flush error counters (`_flush_errors`) drive the heartbeat status field: "warning" if any stream has errors, "healthy" otherwise. Counters reset to zero on the next successful flush of that same stream.

Order Executor

Service executing orders:

```
snapper executor -e kraken
```

Executor:

1. Subscribes to `orders.commands.{exchange}.*` topics
2. Receives `OrderRequestData` from the trade runtime via the durable outbox dispatcher
3. Executes order via exchange API
4. Persists `VenueEvent` rows for accepted, fill, and terminal observations
5. Publishes `ExecutionData` or `OrderData`

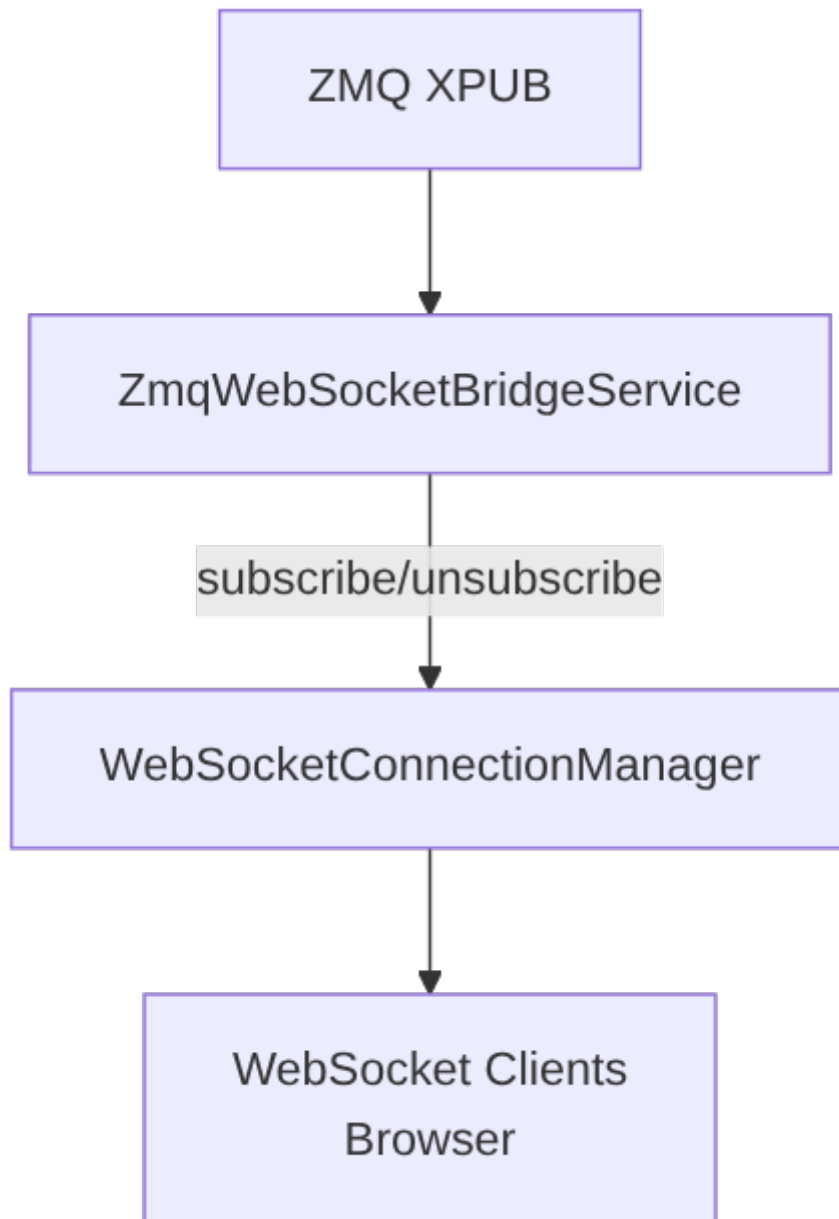
Failed `VenueEvent` persistence is treated as fail-closed on the accepted/fill paths and aborts downstream publish from the executor.

The trade runtime subscribes to `orders.events.*` to keep `TradeService` and `BalanceService` in sync during normal operation. `VenueEvent` rows are used as the durable recovery and reconciliation backbone.

`PlanExecutorService` (see `docs/architecture.md` → Execution Plans) also subscribes to `orders.events.` and `market.` on the broker `XPUB`, routes `ExecutionData` → fill propagation, `OrderData` → plan status transitions, and `TickData` → evaluator dispatch. Plan cancels flow through the same `TradeCommand` outbox as submits, with `command_type="cancel"` branching `_outbox_publish` to emit `OrderCancelData` on `orders.commands.{exchange}.{instrument}.cancel`.

ZMQ-WebSocket Bridge

Bridge between ZMQ and WebSocket for frontend:



Bridge automatically:

- Subscribes to ZMQ topics when WebSocket client subscribes
- Validates each inbound ZMQ message as JSON and runs `GapDetector.check()` before forwarding
- Drops messages with malformed JSON (logged as WARNING; counted in `invalid_messages` per topic)
- Forwards messages via `_forward_to_clients` with per-subscription backpressure
- Throttles market data per subscriber (configurable `throttle_ms`)
- Drops market data when a client exceeds `MAX_PENDING_MESSAGES_MARKET` (100)

- Disconnects slow clients on trade topics when exceeding `MAX_PENDING_MESSAGES_TRADE` (1000)
- Unsubscribes when last client disconnects
- Records control events (subscribe errors, client disconnects) to the control table

Gap Detection

GapDetector tracks `session_id` and `sequence_id` partitioned by the tuple (`received_topic`, `wallet_public_id`) — wallet scoping avoids false gaps when interleaved per-wallet streams share a topic — and emits log messages when it finds sequence gaps or producer session resets.

```
from snapper.messaging.infrastructure.gap_detector import GapDetector

detector = GapDetector()
detector.check(received_topic, session_id, sequence_id)
```

Rules:

- Messages without provenance (`session_id == ""` or `sequence_id == 0`) are **rejected** with a `WARNING` log and a `rejected_unstamped` counter increment. This makes unstamped traffic observable rather than silently passing through.
- First message on a topic: sets baseline. If `sequence_id > 1` the subscriber joined mid-stream (logged `INFO`).
- Same topic, new `session_id`: producer restarted (logged `INFO`, counter resets).
- In-order message (`sequence_id == expected`): accepted silently.
- Gap (`sequence_id > expected`): logs `WARNING` with the missing range, then advances.
- Duplicate / reorder (`sequence_id < expected`): logs `DEBUG`, state unchanged.

`GapDetector.check()` returns `True` when the message carries valid provenance and was processed, `False` when rejected as unstamped.

GapDetector is wired into the ZMQ-WebSocket bridge and into subscriber loops inside the executor, trade runtime coordinator, and `strategy_listen_loop()`.

Message Logger

Debug tool for monitoring ZMQ traffic:

```
snapper zmq-logger --payload --audit-file logs/zmq.jsonl
```

Logs all messages passing through the broker.

Configuration

Endpoints in .env:

```
ZMQ_BROKER_XSUB=tcp://127.0.0.1:7500
ZMQ_BROKER_XPUB=tcp://127.0.0.1:7501
```

Throttling

TopicSchema defines throttling per topic:

```
TopicSchema(
    pattern="market.",
    category="market",
    throttle_ms=100, # Max 10 msg/s
)
```

Message Validation

ValidatedPublisher and ValidatedSubscriber validate **topic strings**, not message payloads. ValidatedPublisher.send_multipart() calls validate_topic() before sending; ValidatedSubscriber.subscribe() calls validate_subscription_pattern() before subscribing. Both raise TopicValidationError on invalid topics.

High Water Mark (HWM) Policy

ZMQ sockets use explicit high water marks via apply_hwm() from validated_socket.py to bound queue depth and prevent silent message loss:

Tier	Constant	Value	Used by
Order flow	HWM_ORDER_FLOW	0 (unlimited)	Executors, trade runtime coordinator
Broker	HWM_BROKER	50 000	XSUB (rcvhwm), XPUB (sndhwm)
Market data	HWM_MARKET_DATA	20 000	Publishers, strategies, bridge, settings, symbol updaters
Audit	HWM_AUDIT	10 000	Message logger

apply_hwm() must be called **before** connect() or bind().

Retry and Resilience

Sockets automatically:

- Reconnect on connection loss
- Buffer messages when broker unavailable
- LINGER=0 on close (discard pending)
- HWM applied before connect/bind (see above)

Topic Derivation

`topic_for_message()` in `snapper.messaging.topics.builders` derives the canonical ZMQ topic from a typed Data class. Producers typically call it before `MessagePublisher.send()`.

```
from snapper.messaging.topics.builders import topic_for_message

topic = topic_for_message(candle_data)
await publisher.send(topic, candle_data)
```

Paper market data topics include a `source_exchange` segment that is not carried in the payload, so they must always be published with an explicit topic override.

WebSocket Message Provenance

All WebSocket protocol messages (authentication, subscription management, ping/pong, errors) inherit directly from `StrictDataSchema`, so they carry the same provenance fields as ZMQ data payloads: `public_id`, `session_id`, `sequence_id`, and `timestamp`. Every payload item in the system — whether it flows over ZMQ, REST, or WebSocket — has a uniform provenance envelope.

Audit Tables: Control and Telemetry

Two destination tables provide always-available observability for non-domain traffic:

- **control** — Always-on audit for commands, authentication events, subscribe/unsubscribe messages, and REST mutation requests. Recording is wired in three places:
- **WS handlers** — `_record_ws_control()` records auth, subscribe, unsubscribe, and error events with `transport="ws"`. Client causation linkage is extracted from inbound messages via `_extract_client_provenance()` and stored as `client_session_id` and `client_public_id` on the control row.

- **REST middleware** — `ClientProvenanceMiddleware._record_control()` records every mutation (POST / PUT / DELETE / PATCH per `_MUTATION_METHODS`) with `transport="rest"`, redacted payload, outcome (ok/error/exception), and server-side provenance.
- **ZMQ bridge** — `_record_bridge_control()` records subscribe errors and client disconnects with `transport="zmq"`.

All control writes use a finally block so the record is persisted regardless of whether the request succeeded or failed. The write is non-blocking: any DB failure is logged and swallowed so the response already sent to the client is never invalidated.

- **telemetry** — Toggleable high-volume table for pings, heartbeats, pongs, and GET read requests. Recording is gated by the `TELEMETRY_RECORDING_ENABLED` environment variable (default false). When disabled, `SequenceTracker` counters still increment — only the DB write is skipped. Recording is wired in:
- **WS handlers** — `_record_ws_telemetry()` records ping/pong/heartbeat events
- **REST middleware** — `_record_telemetry()` records GET reads (health, status, entity endpoints)

The non-blocking audit invariant applies to both tables: audit writes never reject, delay, or invalidate the primary request/message flow.

Best Practices

1. **One broker per system** — All components connect to the same broker
2. **Topic hierarchy** — Use hierarchy for filtering (market.kraken.*)
3. **Data types** — Always use typed Data classes from `messaging.schemas.data`
4. **Provenance** — Use `MessagePublisher` (not `ValidatedPublisher` directly) so every message carries `session_id` and `sequence_id` for gap detection
5. **One SequenceTracker per component** — Create it once at `start()` and share across all publisher instances; restart creates a new session
6. **Heartbeats** — Keep component-specific heartbeat cadences small and regular
7. **Graceful shutdown** — Close sockets with `LINGER=0`

Observability

Snapper exposes operator-facing health metrics so a dashboard can see the gradient toward exhaustion before a crash. The current observability surface is **Cluster A** of the three-cluster monitoring initiative: process-level counters sampled into an in-memory ring buffer with a REST query API.

Cluster B (per-table SCD2 DB stats — /api/metrics/db/tables) and Cluster C (retention policy framework — /api/metrics/retention) are both live alongside Cluster A. Cluster C defines what "archivable" means, which Cluster B's archivable counter then mirrors.

Endpoints

All four routes are gated by Permission.READ_SYSTEM_STATUS (held by AI_DELEGATE, VIEWER, OPERATOR, ADMIN). The two POST routes also require CSRF (cookie-auth path); Authorization: Bearer requests bypass CSRF per the project-wide auth contract.

Method	Path	Purpose
GET	/api/metrics/system	Latest snapshot
GET	/api/metrics/system/history? since&until&limit	Windowed slice of the ring buffer
POST	/api/metrics/system/tracemalloc/start? duration_s	Arm Python tracemalloc with auto-stop deadline
POST	/api/metrics/system/tracemalloc/stop	Disarm tracemalloc + cancel pending deadline

Failure contract

If the snapshotter singleton failed to start at lifespan time, the app.state.system_metrics_snapshotter attribute stays None and every metrics route returns HTTP 503 with body {"detail": "system metrics snapshotter not available"}. The rest of the application continues to serve other endpoints.

Cold-start contract

SystemMetricsSnapshotter.start() takes one eager synchronous sample BEFORE returning, so the first request after lifespan startup completes hits a populated buffer.

Snapshot fields

Each sample captures eight nested groups plus two top-level flags. All field names are stable; downstream consumers (frontend, iOS) regenerate from the backend OpenAPI / JSON-Schema export and pin the wire shape.

process

Field	Type	Description
pid	int	Process ID.
uptime_seconds	float	Monotonic-clock seconds since the snapshotter was constructed.
status	str	psutil process status (running, sleeping, ...).
num_threads	int	Live thread count.
num_fds	int	Open file descriptor count.
num_connections	int	Open TCP/UDP socket count.

cpu

Field	Type	Description
process_percent	float	psutil.Process.cpu_percent reading at sample time.
user_time_seconds	float	Cumulative user-mode CPU.
system_time_seconds	float	Cumulative system-mode CPU.
cgroup_quota_microseconds	int null	cgroup CPU quota microseconds per period; null when absent or unlimited.
cgroup_throttled_count	int null	Cumulative throttle event count from cpu.stat nr_throttled; null when cgroup absent.

memory

Field	Type	Description
rss_bytes	int	Resident set size (RSS).
rss_peak_bytes	int	Peak RSS from /proc/self/status VmHWM; falls back to rss_bytes on non-Linux.

Field	Type	Description
vms_bytes	int	Virtual memory size.
python_traced_bytes	int null	Bytes tracked by tracemalloc; null when tracemalloc is OFF.
native_bytes	int null	max(0, rss - python_traced) when tracemalloc is active; null otherwise — exposes "native dark matter" (numpy / pandas / zmq / aiosqlite native cache / pydantic-core Rust).
cgroup_limit_bytes	int null	cgroup memory cap; null when absent or unlimited.
cgroup_current_bytes	int null	Current cgroup memory consumption.
saturation_pct	float null	cgroup_current / cgroup_limit when both available; null otherwise.

asyncio

Field	Type	Description
active_tasks	int	len(asyncio.all_tasks()).
pending_tasks	int	Subset of all_tasks whose done() is False.

gc

The fields below describe the API/wire shape returned by `/api/metrics/system`. Internally, the snapshotter stores the generation counters as a tuple (collections_per_gen: tuple[int, int, int] on SystemMetricsSnapshot); the route mapper flattens that tuple into the three collections_gen{N} fields below before serializing.

Field	Type	Description
collections_gen0	int	Cumulative GC collection count for generation 0.
collections_gen1	int	Generation 1.
collections_gen2	int	Generation 2.
uncollectable	int	Cumulative count of objects GC could not free.
current_objects	int	Sum of gc.get_count() across all generations at sample time.

limits

Field	Type	Description
rlimit_nproc	int	Soft RLIMIT_NPROC.
rlimit_nofile	int	Soft RLIMIT_NOFILE.
rlimit_as_bytes	int	Soft RLIMIT_AS (address space).

saturation

Percentage toward exhaustion (0.0 - 1.0). The gradient operators actually need to see (vs raw counts) — the rate of increase here is the leading indicator of a thread leak or fd leak.

Field	Type	Description
threads_pct	float null	num_threads / rlimit_nproc; null when rlimit_nproc is RLIM_INFINITY or zero.
fds_pct	float null	num_fds / rlimit_nofile; null under the same conditions.

db_internal

Field	Type	Description
aiosqlite_live_connections	int	len(_live_aiosqlite_connections) — atomic read; each entry corresponds to one OS thread under NullPool.
pool_size	int null	Reserved for future queue-pool instrumentation. Currently always null in Cluster A; will populate once the SQLAlchemy pool reflection helper lands alongside the DB_POOL_MODE=queue rollout.
pool_checked_out	int null	Reserved for future queue-pool instrumentation. Currently always null in Cluster A.

Top-level flags

Field	Type	Description
bus_time	datetime (UTC)	Sample timestamp; window queries match against this field.

Field	Type	Description
tracemalloc_active	bool	True iff tracemalloc.is_tracing() was True at sample time.
cgroup_version	"v1" "v2" null	cgroup detection result; null on hosts without cgroup.

Configuration

The snapshotter reads two environment variables directly (no AppSettings extension this iteration). Defaults match the in-code constants in [snapper.application.system_metrics.snapshotter](#).

Variable	Default	Effect
SYSTEM_METRICS_INTERVAL_SECONDS	5	Seconds between sampler ticks. Empty / unparseable / non-positive falls back to the default.
SYSTEM_METRICS_HISTORY_CAP	17280	Ring buffer cap (snapshot count). 17280 $\approx$ 24h at the default 5s interval. Empty / unparseable / non-positive falls back to the default.

Operators can drop the ring buffer cap to reduce resident memory ($720 \approx 1h \approx 360$ KB).

Publisher hot-path probes are separate, opt-in diagnostics:

Variable	Default	Effect
SNAPPER_TICK_PROBE	unset	Logs per-stage tick publisher timing histograms every ~ 10 seconds when truthy.
SNAPPER_TRADE_PROBE	unset	Logs per-stage trade publisher timing histograms every ~ 10 seconds when truthy.

Use these only during targeted throughput investigations; when unset the hot path pays only a branch and return.

Sampling cadence + ring buffer

- Sample interval: configurable, default **5 seconds**.
- Per-snapshot byte budget: ~ 500 bytes (≈ 30 metrics, ~ 15 bytes each).
- Default cap 17280 snapshots $\approx$ **8.6 MB resident**.
- Eviction: oldest-first via `collections.deque(maxlen=N)` semantics.

- All buffer access goes through an `asyncio.Lock` so concurrent route reads see a consistent view (single-writer / multi-reader).

cgroup detection

Runs at every sample and degrades silently when paths are absent:

- v2 default in modern Docker (kernel 5.0+, Docker Engine 20.10+): `/sys/fs/cgroup/memory.max`, `cpu.max`, `cpu.stat`, `memory.current`.
- v1 fallback: `memory/memory.limit_in_bytes`, `memory/memory.usage_in_bytes`, `cpu,cpuacct/cpu.cfs_quota_us` (or the legacy `cpu/cpu.cfs_quota_us` mount), and the matching `cpu.stat` file for `nr_throttled`.
- `cgroup_version` reports the detected layout; null on dev hosts without cgroup (macOS, non-containerised Linux without unified hierarchy).

Tracemalloc

Off by default — enabling `tracemalloc` costs **5-10% CPU** while active and holds extra metadata in process memory. Operators arm it briefly to capture the `python_traced_bytes` byte counter for the `native_bytes` diagnostic, then auto-stop fires after a bounded window.

Knob	Default	Cap
<code>duration_s</code> (POST query)	600s	3600s (clamped)

Calling `start` while already armed REPLACES the deadline (cancels the previous timer, starts a fresh one with the new duration). The route response carries the clamped `requested_duration_seconds`, so the operator sees what was actually applied.

Example invocations

Latest snapshot:

```
curl -fsS \
  -H "Authorization: Bearer ${SNAPPER_ACCESS_TOKEN}" \
  https://snapper.example.com/api/metrics/system \
  | jq '.payload | {threads_pct: .saturation.threads_pct,
rss: .memory.rss_bytes, aiosqlite: .db_internal.aiosqlite_live_connections}'
```

Last hour, capped at 60 samples:

```
SINCE=$(date -u -d '-1 hour' +%FT%TZ)
curl -fsS \
  -H "Authorization: Bearer ${SNAPPER_ACCESS_TOKEN}" \
```

```
"https://snapper.example.com/api/metrics/system/history?since=${SINCE}
&limit=60" \
| jq '.count, [.payload[].memory.rss_bytes]'
```

Arm tracemalloc for 5 minutes to inspect the native gap:

```
curl -fsS -X POST \
-H "Authorization: Bearer ${SNAPPER_ACCESS_TOKEN}" \
  "https://snapper.example.com/api/metrics/system/tracemalloc/start?
duration_s=300" \
| jq
sleep 30
curl -fsS \
-H "Authorization: Bearer ${SNAPPER_ACCESS_TOKEN}" \
  https://snapper.example.com/api/metrics/system \
| jq '.payload.memory | {rss: .rss_bytes, traced: .python_traced_bytes,
native: .native_bytes}'
```

Multi-instance note

The ring buffer is per-process. With $N > 1$ instances each hosts its own buffer; aggregation across instances is out of scope for Cluster A. Operators who need cross-instance views should poll each instance directly until a downstream collector lands (Cluster B/C territory).

Retention

Cluster C ships a **policy-driven retention loop** that periodically archives + purges old rows from event tables, complementing the existing manual snapper archive CLI. Policies are declarative (`RetentionPolicy(table, retain_days, backlog_lookback_days)`) and evaluated by `RetentionService` on every scheduler tick.

Endpoint

Method	Path	Purpose
GET	/api/metrics/retention	Most recent scheduler tick's per-policy summary

The route is gated by `Permission.READ_SYSTEM_STATUS` (same gate as the system metrics surface).

Failure / cold-start contract

The route returns HTTP 503 with one of three details:

- "retention scheduler not available" — singleton failed to start at lifespan time (the lifespan hook leaves `app.state.retention_scheduler` absent so the route falls through to 503).
- "retention scheduler disabled" — the operator set `RETENTION_DISABLED=true`; the scheduler is parked.
- "retention scheduler not yet run" — eager run did not populate the summary before the first request (defensive; should not happen in production because `start()` awaits the eager `run_once`).

Policy semantics

For each scheduler tick, with `today_utc = datetime.now(UTC).date()`:

```
oldest_kept_day      = today_utc - retain_days      # rows on this day stay in DB
last_eligible_day    = oldest_kept_day - 1d         # day_end (inclusive)
earliest_scanned_day = last_eligible_day - backlog_lookback_days # day_start (inclusive)
```

The archiver call covers the inclusive day-range `[day_start, day_end]`, matching `EventArchiver.export(...)` semantics. Steady state after the backlog drains processes one new day per tick.

Concrete example with `today_utc = 2026-05-01`, `retain_days = 1`, `backlog_lookback_days = 30`:

- `oldest_kept_day = 2026-04-30` (rows with timestamp on 2026-04-30 stay in DB).
- `last_eligible_day = 2026-04-29` (last day archived + purged this tick).
- `earliest_scanned_day = 2026-03-30` (oldest day touched this tick).
- The archiver query covers `[2026-03-30 00:00 UTC, 2026-04-30 00:00 UTC)` per the repository semantics.

The shipped default policy:

Table	retain_days	backlog_lookback_days
telemetry	1	30

Configuration

Variable	Default	Effect
RETENTION_INTERVAL_SECONDS	3600	Scheduler loop period. Parsed as a positive float; empty / unparseable / non-positive values silently fall back to the default.
RETENTION_DISABLED	false	Truthy ("1"/"true"/"yes", case-insensitive) parks the scheduler entirely.
RETENTION_DRY_RUN	false	Truthy forces purge=False on every archiver call regardless of policy. Operators flip to true for one cycle when adding a NEW high-volume policy to verify the window before any DB row is deleted.
RETENTION_OUTPUT_DIR	data	Filesystem root passed to EventArchiver; matches the snapper archive CLI default.

Roll-out checklist for a new high-volume policy

1. Add the new RetentionPolicy entry to RETENTION_POLICIES in src/snapper/application/retention/policies.py.
2. Set RETENTION_DRY_RUN=true for one full interval and inspect GET /api/metrics/retention — check that the day_start / day_end window matches expectations and archived_rows is non-zero.
3. If the new policy targets a deeply backlogged table, run a one-time snapper archive --table=<...> --from=<old> --to=<recent> --purge to drain backlog before flipping RETENTION_DRY_RUN=false.
4. Flip RETENTION_DRY_RUN=false. Subsequent ticks will both archive AND purge.

Multi-instance note

The retention scheduler is single-instance only for v1. With $N > 1$ instances, RETENTION_DISABLED=true on $N-1$ instances avoids double-archive / double-purge races. A coordinator-based extension is a future iteration.

Cluster B — per-table SCD2 stats (DB metrics)

GET /api/metrics/db/tables returns the latest sampled per-table row counters for every registered event + state SCD2 table. The endpoint is operator telemetry: at a glance it answers "is telemetry growing as expected", "are there orders accumulating closed versions", "how many rows are eligible for the next retention cycle".

Cluster B endpoint

GET /api/metrics/db/tables — READ_SYSTEM_STATUS permission gate. Returns a DbStatsResponse envelope wrapping a DbStatsData payload with tables: list[TableStatsItem] in the sampler's deterministic order (STATE-table block first alphabetical, EVENT-table block second alphabetical).

Cluster B failure / cold-start contract

State	Status	Detail	Retry-After
Snapshotter helper failed before assigning	503	DB metrics snapshotter not initialized	(none)
DB_METRICS_DISABLED=true (operator opt-out)	503	DB metrics snapshotter disabled via DB_METRICS_DISABLED	(none)
Started but no sample yet (cold-start)	503	DB metrics snapshotter has not completed a sample yet	<interval_seconds>
Healthy	200	—	—

The cold-start window lasts up to DB_METRICS_INTERVAL_SECONDS (default 60s) — by design, the sampler does NOT block lifespan startup on the first sample. Frontend dashboards must poll-with-backoff using the Retry-After header.

Snapshot fields

TableStatsItem

Field	Type	Notes
table	str	Table name (key in EVENT_TABLES or STATE_TABLES).
table_kind	"event" "state"	Discriminates wire semantics.
total	int \ null	Total row count. On PostgreSQL this is a pg_class.reltuples planner estimate (GREATEST(reltuples, 0) — fast, accurate within ANALYZE/autovacuum drift, intended for growth-trend monitoring); on the SQLite dev fixture it is an exact COUNT(*). null only on per-table query failure with no prior sample to clone.
current	int \ null	

Field	Type	Notes
		Active SCD2 versions (rows whose known_to equals the SCD2 sentinel) for state tables; null for event tables (no SCD2 lifecycle — reporting 0 would imply the dimension exists).
closed	int \ null	Superseded SCD2 versions for state tables; null for event tables.
archivable	int \ null	Row count in the policy retention window when a RETENTION_POLICIES entry applies; null when no policy applies (semantically distinct from 0).
is_stale	bool	True when the row was reused from a prior sample after a per-table query timeout or exception.
last_sampled_at	datetime	UTC timestamp of the row's source sample. On a stale clone this is the original timestamp, NOT the current cycle's clock.

DbStatsData

Field	Type	Notes
snapshot_started_at	datetime	When the sampler began the cycle.
snapshot_completed_at	datetime	When the sampler finished the cycle.
interval_seconds	int	Echo of the configured cadence.
tables	list[TableStatsItem]	One entry per registered table.

Per-kind semantics

- **EVENT tables** (append-only — ticks, trades, signals, executions, telemetry, control): total comes from the dialect-aware row-count primitive — a pg_class.reltuples planner estimate on PostgreSQL, an exact COUNT(*) on the SQLite fixture. The current / closed axes are null because event tables have no SCD2 lifecycle. archivable is non-null only for tables with a registered policy (currently only telemetry).
- **STATE tables** (SCD2-versioned — orders, positions, instruments, etc.): current = COUNT(known_to == sentinel) (exact) counts active SCD2 versions; total comes from the dialect-aware row-count primitive (PG planner estimate / SQLite exact); closed = max(0, total - current) is derived and clamped at 0 to absorb stale PostgreSQL planner estimates where the exact current temporarily exceeds the estimated total. archivable is null until a policy is registered for the table.

Cluster B/C alignment

The archivable counter computes the same window as Cluster C's `compute_retention_window(today_utc, policy)`: the half-open timestamp $\geq$ `day_start` midnight UTC AND timestamp $<$ `day_end + 1d` midnight UTC predicate, with the policy's (`retain_days`, `backlog_lookback_days`) knobs. This means dashboard archivable and the next retention cycle's `archived_rows` MUST agree exactly — if they drift, one of the two has a boundary bug. The `tests/application/db_stats/test_cluster_alignment.py` integration test pins the equality contract.

Cluster B configuration

Variable	Default	Effect
DB_METRICS_INTERVAL_SECONDS	60	Sampler loop period. Empty / unset $\rightarrow$ default; a non-integer or out-of-range value raises <code>ValueError</code> at startup.
DB_METRICS_DISABLED	false	Truthy ("1"/"true"/"yes", case-insensitive) parks the snapshotter entirely. PG operators with deployments lacking <code>psycopg2</code> in deps boot cleanly with this flag (the underlying repo factory is never called).

`PER_TABLE_TIMEOUT_SECONDS` is a module constant (30s, not env- configurable for v1). On per-table timeout, the snapshotter clones the prior `TableStats` with `is_stale=True`; if no prior sample exists the row carries all-null counters. Per-table failures NEVER abort the sampler tick.

Sampling order + cold-start

STATE tables sample first (alphabetical by name), EVENT tables second. Atomic swap of `_latest_snapshot` happens at the END of the cycle — readers wait one full `interval_seconds` before the first 200, regardless of order. STATE-first ordering is for resilience to EVENT timeouts: if a heavy EVENT table (e.g. ticks at 50M rows) times out, STATE counters were already computed and the snapshot still publishes with EVENT entries marked `is_stale=True` from a prior run.

telemetry.timestamp index

Cluster B's archivable query on telemetry filters on Telemetry.timestamp (the bus-time column inherited from TemporalMixin). Without an index on that column, the query is a full table scan over a high-volume audit table. The ix_telemetry_timestamp index over Telemetry.timestamp is part of the consolidated 0001_init migration (src/snapper/data/migrations/versions/0001_init.py), so both Cluster B's counter and Cluster C's existing retention scan run in $O(\log n)$ from the first deployed schema.

Operations runbook — multi-instance trade coordinator

The trade runtime supports static-hash shard partitioning. At `SNAPPER_COORDINATOR_INSTANCE_COUNT=1` (the default) a single `TraderCoordinator` owns every shard. Scaling to $N \geq 2$ deploys multiple coordinators against the same DB and ZMQ broker, each owning $\sim 1/N$ of the shards deterministically via SHA-256 of the shard key.

This document covers operating the N-instance trade runtime: scale-up, scale-down, and crash recovery. It is intentionally opinionated about systemd because systemd template units are the stable recipe. A contract for alternative orchestrators (Docker Compose, Kubernetes, Nomad) is also specified so operators can adopt them with empirical verification.

Prerequisites: systemd template unit

All subsequent recipes assume a template unit `snapper-trade-zmq@.service` that consumes `%i` as the instance id:

```
# /etc/systemd/system/snapper-trade-zmq@.service
[Unit]
Description=Snapper Trade Coordinator (instance %i)
After=network.target snapper-broker.service

[Service]
Type=simple
User=snapper
EnvironmentFile=/etc/snapper/coordinator.env
Environment=SNAPPER_COORDINATOR_INSTANCE_ID=%i
ExecStart=/opt/snapper/bin/snapper trade-zmq
Restart=always
RestartSec=2

[Install]
WantedBy=multi-user.target
```

With this template, `snapper-trade-zmq@0.service` and `snapper-trade-zmq@1.service` are distinct systemd units with separate cgroups and separate journal streams on the same host. Per-instance PID is queried via `systemctl show -p MainPID snapper-trade-zmq@0.service`.

`/etc/snapper/coordinator.env` holds the shared bootstrap env, including `SNAPPER_COORDINATOR_INSTANCE_COUNT=2` during $N=2$ operation:

```
DB_URL=postgresql+asyncpg://snapper:...@db.internal/snapper
ZMQ_BROKER_XSUB=tcp://broker.internal:7500
ZMQ_BROKER_XPUB=tcp://broker.internal:7501
MASTER_PASSWORD=...
SNAPPER_COORDINATOR_INSTANCE_COUNT=2
```

Scale up N=1 → N=2 (systemd)

```
# 1. Edit /etc/snapper/coordinator.env – set
SNAPPER_COORDINATOR_INSTANCE_COUNT=2.

# 2. Stop the currently-running single-instance coordinator.
sudo systemctl stop snapper-trade-zmq@0.service

# 3. Verify graceful shutdown.
sudo systemctl status snapper-trade-zmq@0.service | grep -E "Active:|Main PID:"

# 4. Start both instances.
sudo systemctl start snapper-trade-zmq@0.service
sudo systemctl start snapper-trade-zmq@1.service

# 5. Verify both are live.
sudo systemctl is-active snapper-trade-zmq@0.service # expect: active
sudo systemctl is-active snapper-trade-zmq@1.service # expect: active
sudo journalctl -u snapper-trade-zmq@0.service -n 20 | grep "instance 0/2"
sudo journalctl -u snapper-trade-zmq@1.service -n 20 | grep "instance 1/2"
```

Scale down N=2 → N=1 (systemd)

```
# 1. Stop BOTH coordinator instances first – full cutover is required.
sudo systemctl stop snapper-trade-zmq@0.service
sudo systemctl stop snapper-trade-zmq@1.service

# 2. Verify nothing remains.
sudo systemctl list-units 'snapper-trade-zmq*' --state=active --no-legend
# (expect empty output)

# 3. Edit /etc/snapper/coordinator.env – set
SNAPPER_COORDINATOR_INSTANCE_COUNT=1.

# 4. Start instance 0 only.
sudo systemctl start snapper-trade-zmq@0.service
```

Why full cutover is required

Scale up and scale down both require stopping ALL old-N coordinators BEFORE starting any new-N coordinator. Overlapping old-N and new-N instances is the failure mode to prevent.

Scenario: improper restart with overlap. If an operator skipped the cutover step and left instance 0 running as count=1 while starting instance 1 as count=2:

- Instance 0 believes it owns ALL shards (count=1 → every owns() returns True).
- Instance 1 believes it owns shards where $\text{hash}(\text{shard_key}) \% 2 == 1$.
- Both coordinators dispatch the same TradeCommand for shards where $\text{hash}() \% 2 == 1 \rightarrow$ duplicate venue orders OR a race on status='created' → 'dispatched'.

The ~10s dispatch pause for owned shards during the cutover window is the explicit no-HA trade-off. HA was excluded from scope by design.

Crash recovery (systemd)

If instance K crashes:

```
# 1. Diagnose.
sudo journalctl -u snapper-trade-zmq@1.service -n 100 --no-pager

# 2. Check whether Restart=always brought it back.
sudo systemctl is-active snapper-trade-zmq@1.service

# 3a. If active: confirm the coordinator resumed with its original ownership.
sudo journalctl -u snapper-trade-zmq@1.service -n 200 | grep "instance 1/2"
sudo journalctl -u snapper-trade-zmq@1.service -n 200 | grep "recovery complete"

# 3b. If inactive (Restart=always gave up): manually restart.
sudo systemctl restart snapper-trade-zmq@1.service
```

During the outage window, shards owned by the crashed instance pause until the coordinator comes back. Typical outage is under 10s if Restart=always is configured.

Recovery rebuilds engines from:

1. Checkpoints (carry shard_key directly, filtered by ownership).
2. Live executions (filtered by recovered shard_key; paper rows are EXCLUDED under N>1 because ExecutionRow has no strategy_tag).
3. Active orders (same treatment as executions).

Under N>1 paper mode, state that never produced a checkpoint is not recovered by the restarted coordinator. Paper strategies that need to survive restart MUST persist checkpoints.

Known limitation — REST orders with wallet_public_id under N>1

The REST order endpoints at `src/snapper/server/order_routes.py` build the `TradeCommand.shard_key` without the wallet segment, while the signal-driven engine's `shard_key` appends `.w{wallet_short}` via `_compute_shard_key` when `wallet_public_id` is non-empty.

At $N=1$ this is dormant because every shard is owned by the single coordinator, the outbox ownership filter is a no-op at $N=1$ (the trader always passes `ShardOwnership(0, 1)`, whose `owns()` returns `True` for every shard when `instance_count == 1`), and the CID guard is gated on `instance_count > 1`.

Under $N>1$, a REST order with a non-empty `wallet_public_id` writes a `TradeCommand` whose `shard_key` hashes to a different instance than the engine for the same (exchange, instrument, wallet). The owning coordinator dispatches the command to the venue correctly, but the venue ACK is dropped by the CID guard on every coordinator because the CID was never registered in `_order_shard_keys` (the REST path does not populate it).

Consequence at $N>1$: REST orders with `wallet_public_id` reach the venue, but the coordinator's `TradeService` projection does not reflect fills/cancellations. Monitoring via the orders and executions tables still works — only the in-memory coordinator state drifts.

Mitigation until a forward fix lands: for $N>1$ deployments that need REST orders, either (a) set `wallet_public_id=""` on the REST request, or (b) stay on $N=1$ for workflows that mix signal-driven + REST-driven orders. Signal-only workflows are unaffected.

Non-systemd deployments (contract)

The operational recipes are anchored on the systemd template above. For non-systemd platforms, the deployment author writes a recipe that satisfies the same contract:

1. Two distinct supervised processes on the same host (or across hosts) with `SNAPPER_COORDINATOR_INSTANCE_ID=0` and `=1` respectively, and `SNAPPER_COORDINATOR_INSTANCE_COUNT=2` in both.
2. Per-instance lifecycle (stop / start / restart / logs) via the orchestrator's native mechanism — never by process pattern match (`pkill -f` is too broad for multi-instance).
3. Scale-up and scale-down follow the same stop-all → verify-empty → start-new cutover flow as the systemd recipes above.
4. At least one concrete recipe per chosen orchestrator, verified empirically, with a provenance line:

Verified on YYYY-MM-DD against <platform-version>

The regex enforced by tests/meta/test_operations_runbook.py is `^Verified on \d{4}-\d{2}-\d{2} against \S+$` — exact casing, anchored to the start and end of the line. The meta-test searches the entire file body for at least one matching line; there is no required section placement.

Acceptable orchestrators for the non-systemd recipe are docker compose, kubectl (Kubernetes), or nomad — the meta-test rejects bare docker because docker run alone does not provide the per-instance lifecycle the contract requires.

Verified environments

Verified on 2026-04-18 against docker-compose-v2.29.7

Kraken Equities (TradFi) market data

TradFi index futures (Kraken FCM: MNQ/ES/YM/RTY/NKD/M2K) are shipped in market-data-only mode. The symbol updater remains disabled by default at code level, while the publisher is registered as an enabled market-data process. Persisted DB Setting rows still control runtime enable/autostart state, so rollout + rollback happen without code deploys.

Contracts used by this flow:

- REST: `iapi.kraken.com/api/internal/markets/all/futures-contracts` for instrument metadata; `.../{ws_symbol}/ticker/history` for historical candles. Requires Origin: `https://pro.kraken.com` + Referer: `https://pro.kraken.com/` headers.
- WS: `wss://ws-equities.kraken.com` for live delayed (~10 min) ticks + trades. The outer envelope's delayed flag propagates into every TickData message as `is_delayed`.
- No order API — every submit route (POST `/api/orders`, POST `/api/execution-plans`, POST `/api/trailing-stops`) calls `snapper.server._capability_guard.require_tradable` and rejects TradFi instruments with HTTP 422 `error_code=instrument_market_data_only`.

Market diagnostics

Use these REST endpoints during feed rollout and incident response:

- GET `/api/market/feed-health` — current per-symbol publisher health. Filter with `exchange=kraken_equities` and optionally `fresh_within_seconds=<n>` to hide stale rows from stopped publishers.

- GET /api/market/coverage — per-exchange counts for active instruments, fresh ticks, fresh candles, gated-off instruments, and dark instruments. Tune tick_window_seconds and candle_window_seconds to match the venue's expected cadence.
- GET /api/market/cache/health — cache and persist-policy snapshot: number of cached instruments, cached pair-stat entries, and persisted instrument universe size.
- GET /api/market/cache/stats/configured and /api/market/cache/stats/{exchange_a}/{symbol_a}/{exchange_b}/{symbol_b} — cached Pearson/cointegration diagnostics for configured pairs.

The feed-health table is current-state rather than SCD2 history. Its natural key is (coordinator, exchange, channel, symbol), where channel encodes the stream kind and timeframe together (e.g. ohlc:1m) and symbol is the wire-format product id, so split-feed deployments can show which coordinator is publishing a stale or missing stream.

Per-wallet executors

Executor process configs are templates named executor_<exchange>. At runtime, ProcessLauncherService spawns one instance per active wallet_credentials row, named executor_<exchange>_w<wallet_short>. Those instances load credentials through CredentialResolver and ignore commands for other wallets. Operators should start/stop the generated wallet instances, not the bare templates.

Enable the feed

```
# 1. Populate Symbol + SymbolExchangeCapability rows from iapi.
#   (kraken_equities entry must be present for is_tradeable
#   cache hits; default deny on missing row.)
snapper update-kraken-equities-symbols --force

# 2. Verify the rows landed. SCD2 active rows have known_to set to the
#   canonical sentinel (9999-12-31 23:59:59.000000 on SQLite – see
#   _KNOWN_TO_ACTIVE_SQLITE in src/snapper/data/models.py).
sqlite3 data/snapper.db \
  "SELECT COUNT(*) FROM symbol_exchange_capabilities
  WHERE exchange='kraken_equities'
  AND known_to='9999-12-31 23:59:59.000000';"
# expect >= 38 (indices only; full catalog up to ~180 contracts)

# 3. Trigger an immediate runtime launch of the symbol-updater +
#   feed-publisher processes. `POST /api/processes/<name>/start`
#   invokes `ProcessLauncherService.start_process_by_name`, which
#   reads the existing persisted Setting row and applies the
#   request payload as RUNTIME-ONLY overrides – it does NOT
#   persist a new config row, and it does NOT go through
#   `ProcessRegistrySyncer`. (Persistent enable/autostart state
#   lives on the create / configure path: `POST /api/processes` or
#   the Settings page; the registry syncer reconciles that
```

```

# persisted state onto the launcher.)
# The body is the full request envelope: `json_body` calls
# `ProcessStartRequest.model_validate_json` on the raw body, and the
# envelope (StrictDataSchema, extra="forbid") REQUIRES session_id,
# sequence_id, public_id, and timestamp. A bare `{"payload":{}}` is
# rejected with HTTP 422 before the handler runs. `payload` may carry
# optional `mode` / `parameters` runtime overrides.
curl -X POST http://localhost:8000/api/processes/kraken_equities_symbol_updater/
start \
  -H "Authorization: Bearer <token>" \
  -H "X-CSRF-Token: <csrf>" \
  -H "Content-Type: application/json" \
  -d
'{"type":"process_start_request","session_id":"<sid>","sequence_id":1,"public_id":"<uuid7>"}'

curl -X POST http://localhost:8000/api/processes/kraken_equities_feed_publisher/
start \
  -H "Authorization: Bearer <token>" \
  -H "X-CSRF-Token: <csrf>" \
  -H "Content-Type: application/json" \
  -d
'{"type":"process_start_request","session_id":"<sid>","sequence_id":1,"public_id":"<uuid7>"}'

# To stop a running process later:
# curl -X POST http://localhost:8000/api/processes/<name>/stop ...

# 4. `/start` returns once the process is launched in the runtime;
# there is no "wait one syncer interval" step on this path.
# If you need the new state persisted across restarts, do that
# via `POST /api/processes` (create / configure) rather than
# via `/start`.

```

Backfill historical candles

```

# 30-day 1-hour backfill of the configured KRAKEN_EQUITIES instruments.
# The default setting is the wildcard ["*"], which expands at runtime to
# the full mapped venue universe (all ~180 contracts).
snapper kraken-equities-backfill-candles -t 1h -d 30

# Or the Makefile wrapper (equivalent)
make backfill-kraken-equities-candles

# Single symbol, daily candles, 90 days
snapper kraken-equities-backfill-candles -s MNQM6-CME -t 1d -d 90 --no-resume

```

The endpoint is undocumented/internal — upstream application-layer failures (HTTP 200 with result=null or non-empty errors) raise RuntimeError so they are distinguishable from legitimately-empty windows. See `src/snapper/infrastructure/exchanges/implementations/kraken_equities.py:get_ohlcv` for the error contract.

Quarterly rotation

The default KRAKEN_EQUITIES instruments setting is the wildcard ["*"], which expands at runtime to the full mapped venue universe, so the default scope tracks every available contract automatically. Rotation discipline applies when an operator narrows the setting to an explicit allowlist: the chosen TradFi symbols are quarterly expiry contracts (MNQM6-CME = Jun 26, etc.) and must be rotated before the InstrumentSpec.expiry_at timestamp on any listed symbol drops below 14 days. Rotation cadence:

1. Pull the current expiry list. native_symbol lives on the symbols table (joined to instruments via symbol_public_id); expiry data lives on instrument_specs keyed by instrument_public_id:

```
sqlite3 data/snapper.db \ "SELECT sym.native_symbol, datetime(spec.expiry_at) FROM
instruments AS i JOIN symbols AS sym ON sym.public_id = i.symbol_public_id AND
sym.known_to = '9999-12-31 23:59:59.000000' JOIN instrument_specs AS spec ON
spec.instrument_public_id = i.public_id AND spec.known_to = '9999-12-31
23:59:59.000000' WHERE i.exchange = 'kraken_equities' AND i.known_to = '9999-12-31
23:59:59.000000';" 2. Identify the next quarterly (e.g. MNQU6-CME Sep 26 when MNQM6-
CME Jun 26 drops below 14 days). 3. Add the new nearest contract by either changing the
default KRAKEN_EQUITIES entry inside the AppSettings.instruments property default
literal in src/snapper/config/app.py (code path — requires a deploy; roll back via git
revert), OR overriding the instruments DB Setting row via the Settings page or POST /api/
settings (immediate, no deploy). instruments is a read-only @property, so the DB Setting
row is the only runtime override.
```

Troubleshooting

- **Zero instruments after update-kraken-equities-symbols --force:** iapi reachability / Origin header mismatch. Verify `curl -H 'Origin: https://pro.kraken.com' -H 'Referer: https://pro.kraken.com/' \ 'https://iapi.kraken.com/api/internal/markets/all/futures-contracts?delayed=true' | head -c 200` returns JSON with `result.data[...]`, not `result:null,errors:[...]`.
- **WS disconnect loops on ws-equities.kraken.com:** inspect Kraken status page; no operator action required — the WS client backs off on reconnect. If the `symbol_updater` keeps returning zero rows across two consecutive runs, fire a monitoring alert.
- **Submit returns 422 instrument_market_data_only:** expected. TradFi is observation-only. Point the strategy at a `can_trade=True` instrument (crypto/xStocks) for execution.

snapper-egress sidcar — operator runbook

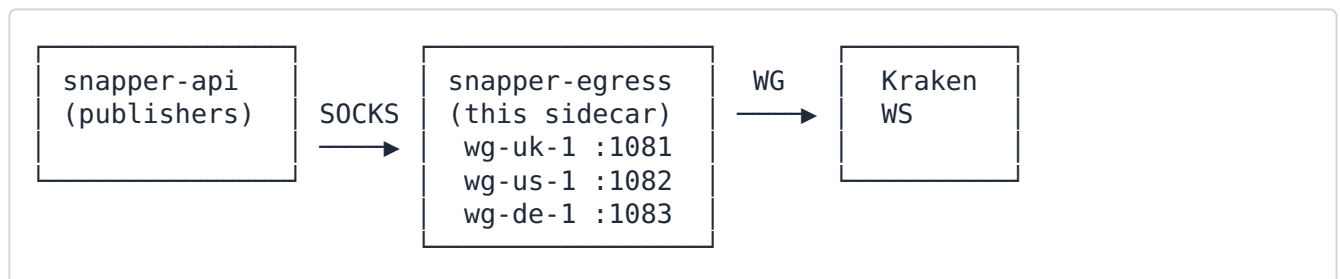
The snapper-egress sidcar runs N WireGuard tunnels plus N matching SOCKS5 listeners inside one container, so the Kraken Spot publisher (and any future market-data publisher) can route its WebSocket traffic out through an alternate egress IP without changing source code.

This sidcar is part of the egress-multiplexer subsystem.

Prerequisites

- **Docker Compose v2.** depends_on.condition: service_healthy is Compose v2 syntax. Compose v1 silently ignores it.
- **Kernel WireGuard on the host when tunnels are configured.** Verify with `lsmod | grep wireguard` or `modprobe wireguard`. The sidcar bind-mounts `/lib/modules` and uses the host kernel module via netlink (no userspace WireGuard). Empty/default deployments with no declared tunnels still expose `/ready` without probing WireGuard. Use `/readyz` when you need strict "every declared tunnel is up" semantics.
- **MASTER_PASSWORD** in the same .env the snapper-api container uses. The sidcar reads encrypted `egress_tunnel_*_private_key` settings via the same Fernet path; mismatched master passwords silently mean "no tunnels load".

Architecture (short)



Per tunnel: 1. WireGuard interface `wg-<id>` lives in the sidcar container. 2. Source-based routing rule sends packets from the tunnel's IP through the matching WG interface. 3. SOCKS5 listener binds outbound sockets to the tunnel's IP → kernel routes them via the tunnel.

The snapper-api's EgressPool (`socks5h://snapper-egress:<port>`) reserves a route per Kraken WS handshake. On HTTP 429 the active route is quarantined and the next handshake uses an alternate.

Adding a tunnel

Preferred path: run the provisioning helper from a checkout that has scripts/ available. The production Docker image does not copy scripts/ into /app, so run it on the host or bind-mount the checkout into a maintenance container with the same DB_URL.

```
poetry run python scripts/provision_egress_tunnel.py \  
  --tunnel-id wg-uk-1 \  
  --interface wg-uk-1 \  
  --address 10.64.12.34 \  
  --prefix-length 32 \  
  --private-key-file ./secrets/wg-uk-1.key \  
  --peer-pubkey "<peer-public-key>" \  
  --peer-endpoint vpn.example.com:51820 \  
  --socks5-listen-port 1081 \  
  --priority 10 \  
  --dry-run
```

Remove --dry-run after validation. The helper writes the descriptor, encrypted private/ optional preshared key settings, and merges the SOCKS5 route into egress_pool. It can also restart the sidecar with --restart-sidecar; snapper-api still needs a restart after egress_pool changes so the pool is rebuilt during lifespan startup.

Manual path:

Three DB settings per tunnel:

Key	Encrypted?	Type	Example
egress_tunnel_<id>	no (JSON descriptor)	text	{"interface":"wg-uk-1", "address":"10.64.12.34", "prefix_length":32, "peer_pubkey":"...", "peer_endpoint":"vpn.example.com:51820", "socks5_listen_port":1081, "priority":10}
egress_tunnel_<id>_private_key	yes (auto-encrypted because the key contains the private_key substring)	text	base64 WG private key
egress_tunnel_<id>_preshared_key	yes, optional	text	base64 WG PSK (or absent / empty)

Constraints:

- id MUST NOT contain underscores (it would collide with the _private_key / _preshared_key naming convention). Use hyphens.

- interface **MUST** start with `wg-` and be ≤ 15 chars total (Linux `IFNAMSIZ`).
- `socks5_listen_port` **MUST** be 1024-65535, unique across tunnels. Convention: 1081 for the first tunnel, then 1082, 1083,
- address must parse as IPv4 or IPv6; IPv4 prefix ≤ 32 , IPv6 ≤ 128 .

Once the three keys exist in the DB, restart the sidecar:

```
docker compose restart snapper-egress
```

The container's lifespan calls `load_declared_tunnels(service) → wg_control.bring_up(...) → Socks5Server.start()` for each tunnel. Failures are best-effort: one bad tunnel does NOT block the others.

Wiring the route into the EgressPool

Edit the `egress_pool` setting to add the new route:

```
{
  "enabled": true,
  "on_all_quarantined": "wait",
  "routes": [
    {"id": "default", "kind": "direct", "priority": 100, "enabled": true},
    {"id": "wg-uk-1", "kind": "socks5",
      "proxy_url": "socks5h://snapper-egress:1081",
      "priority": 10, "enabled": true}
  ]
}
```

Priority semantics: `EgressPool.reserve()` sorts by (priority, `in_use_count`) and picks the **lowest** number first. For SOCKS5 to actually win selection, its priority **MUST** be less than the default route's priority. Bootstrap configurations that ship default: `priority=0` and a new tunnel at `priority=10` would silently keep using direct egress because $0 < 10$. Set the direct route to a high number (e.g. 100) when you want it to act as fallback only, or set `"enabled": false` on it to disable direct egress entirely.

Per-exchange routing — pinning a tunnel to one exchange

To restrict a SOCKS5 route to specific exchanges, add an `allowed_exchanges` field listing the exchange names (matching `ExchangeEnum values` — `"walutomat"`, `"kraken"`, `"kraken_futures"`, `"kraken_equities"`, `"polygon"`):

```
{
  "id": "wg-eu-1", "kind": "socks5",
  "proxy_url": "socks5h://snapper-egress:1084",
  "priority": 10,
```

```
"allowed_exchanges": ["walutomat"],  
"enabled": true  
}
```

A non-empty `allowed_exchanges` restricts the pool: when `pool.reserve(exchange=...)` is called with an exchange name not in the list, this route is filtered out before the (priority, `in_use_count`) sort. An empty `allowed_exchanges` (the default) means the route serves any exchange — the back-compatible path for existing pool entries.

The direct fallback path IGNORES `allowed_exchanges` by design. When every preferred route is quarantined, `EgressPool` returns the direct route as a last resort regardless of any exchange pin. If you want to deny direct egress for a specific exchange, set `"enabled": false` on the direct route instead. Unknown exchange names (typos) are rejected by `EgressPoolConfig` at config-load time, so a "krakeen" typo never silently makes a route unreachable.

Then restart `snapper-api` so the lifespan re-runs the `egress-pool` preflight:

```
docker compose restart snapper
```

The lifespan logs:

```
egress_pool: configured with 2 route(s), on_all_quarantined=wait
```

Verification

- Sidecar healthy?** `docker compose exec snapper-egress \ python -c "import urllib.request; print(urllib.request.urlopen('http://127.0.0.1:8081/ready', timeout=2).read())"` Expect a 200 with the running + failed tunnel id arrays. This is the Compose health endpoint and intentionally stays 200 after bootstrap even if an individual tunnel failed.
- All tunnels up?** `docker compose exec snapper-egress \ python -c "import urllib.request; print(urllib.request.urlopen('http://127.0.0.1:8081/readyz', timeout=2).read())"` Expect 200 only when every declared tunnel is up; otherwise it returns 503 with the failed tunnel ids.
- Tunnel status detail?** `docker compose exec snapper-egress \ python -c "import urllib.request; print(urllib.request.urlopen('http://127.0.0.1:8081/tunnels', timeout=2).read())"` Each tunnel id maps to `{"status": "up", "reason": null}`.
- WireGuard handshake established?** (operator debugging — uses `iproute2` shipped in the image) `docker compose exec snapper-egress ip -d link show wg-uk-1`
`docker compose exec snapper-egress ip rule show table 5000` The interface should be UP and the rule should pin from `<tunnel_addr> → table N`.

5. **Kraken WS uses the tunnel?** Quarantine the direct route from the snapper-api Python REPL: `python` from `datetime` import `UTC`, `datetime`, `timedelta` from `snapper.infrastructure.network.egress_pool` import `get_egress_pool` `pool = get_egress_pool()` `pool._quarantine_route( "default", datetime.now(UTC) + timedelta(seconds=600), "http-429", )` Then force a Kraken Spot reconnect (restart the publisher process) and observe the next handshake. The log line should show the tunnel route id; Cloudflare's view of the source IP should match the VPN exit.
6. **Stable tick flow?** `sql SELECT i.exchange, COUNT(*) AS n FROM ticks t JOIN instruments i ON i.public_id = t.instrument_public_id WHERE t.timestamp > NOW() - INTERVAL '60 seconds' GROUP BY i.exchange ORDER BY n DESC;` Kraken should show consistent counts (~1500-2000 ticks/s in steady state). Drops indicate either the tunnel is flapping or the SOCKS5 listener is mis-bound.

Threat-model + security notes

- The SOCKS5 listener runs WITH NO AUTHENTICATION. Isolation comes entirely from the Docker snapper-internal network. The CI lint hook scripts/`check_egress_compose.py` rejects any ports: `entry` or `network_mode: host` on the `snapper-egress` service so an operator cannot accidentally publish the listener to the host.
- All traffic on `snapper-internal` is trusted equally. Do NOT attach untrusted containers (other tenants, debugging shells from unknown sources) to that network. If you must, add SOCKS5 username/password auth in `socks5_server.py` (~10-line change).
- For extra defense-in-depth, run `docker compose config | grep -A5 snapper-egress` after any compose edit to verify the effective config still has no host port mapping. The lint hook catches the static cases but not every override pattern.
- WG private keys are stored Fernet-encrypted in the Postgres settings table. The encryption key is derived from `MASTER_PASSWORD` (env var). Loss of master password = loss of every encrypted setting (not just WG keys).

Decommissioning a tunnel

1. Remove the route from `egress_pool` (set `enabled: false` on that route OR delete the entry); restart `snapper-api`.
2. Delete the three settings (`egress_tunnel_<id>`, `egress_tunnel_<id>_private_key`, `egress_tunnel_<id>_preshared_key`).
3. Restart `snapper-egress`. The orchestrator's startup `bring_down` pass (idempotent cleanup before `bring_up`) removes any stale interface / ip-rule / routing-table entry left over.

Rotating private keys

1. Generate a new private key + matching peer config on the VPN side.
2. Update the egress_tunnel_<id>_private_key setting (SettingsService re-encrypts on write).
3. Restart snapper-egress to pick up the new key. The orchestrator's bring_up is idempotent and recreates the WG interface from scratch.

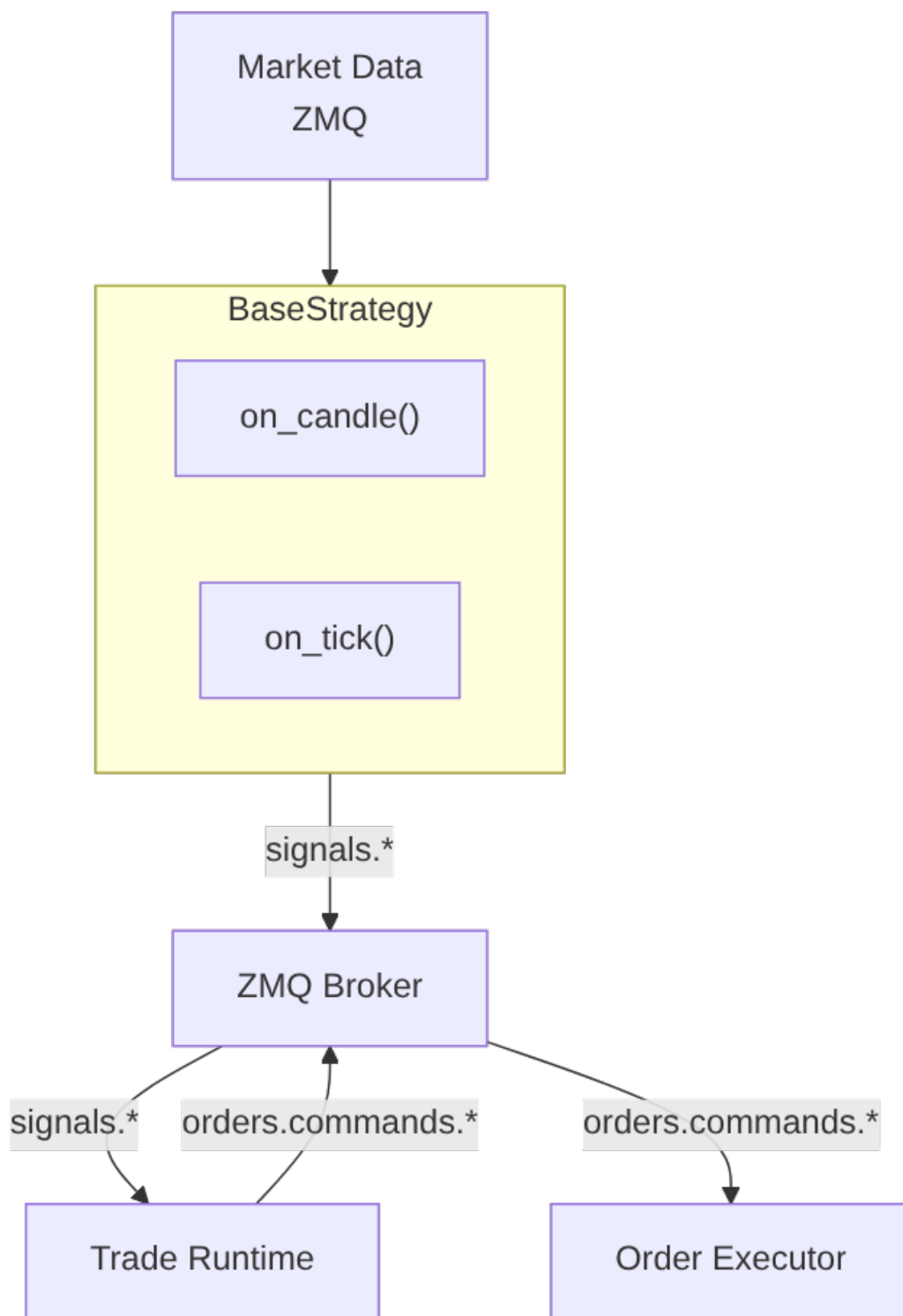
Operational expectations

- Sidecar startup: ~5-15 s (DNS resolution of peer endpoints is the dominant cost).
- HTTP 429 failover: ~1-2 s (route quarantined → next handshake picks alternate).
- Tunnel handshake age: WireGuard re-keys every 2 minutes; if wg show <iface> latest-handshakes returns > 180 s ago, the peer is unreachable.
- SOCKS5 listener latency: ≤ 1 ms additional vs direct route (single process, single async loop, no auth handshake).

Trading Strategies

Snapper provides a framework for creating trading strategies based on ZeroMQ messaging. Strategies subscribe to market data and publish signals. Strategies produce intent only; the trade runtime owns order lifecycle, positions, and balance projections.

Strategy Architecture



Creating Strategies

Basic Structure

```
from snapper.strategies.base import BaseStrategy, StrategySignal, StrategyConfig
from snapper.strategies.decorators import register_strategy,
create_strategy_process
from snapper.messaging.schemas.data import CandleData

@register_strategy("MyStrategy")
@create_strategy_process(
    process_name="my_strategy_btc",
    default_config={
        "name": "my_strategy_btc",
        "inputs": ["market.kraken.BTC-USD.candles.1h"],
        "outputs": ["BTC-USD"],
        "exchange": "paper",
        "params": {
            "threshold": 0.5,
            "period": 14,
        },
    },
)
class MyStrategy(BaseStrategy):
    """Strategy description in docstring."""

    def __init__(self, config: StrategyConfig) -> None:
        """Initialize strategy."""
        super().__init__(config)
        self.threshold = self.params.get("threshold", 0.5)
        self.period = self.params.get("period", 14)

    async def on_candle(self, instrument: str, candle: CandleData) ->
StrategySignal | None:
        """Process candle and generate signal."""
        candles = self.candle_buffer.get(instrument, [])
        if len(candles) < self.period:
            return None

        closes = [c.close for c in candles]
        current_price = closes[-1]

        if self._should_buy(closes):
            return StrategySignal(
                instrument=instrument,
                side="buy",
                strength=1.0,
                price=current_price,
                reason="Buy condition met",
            )

        if self._should_sell(closes):
            return StrategySignal(
                instrument=instrument,
                side="sell",
                strength=1.0,
                price=current_price,
                reason="Sell condition met",
            )
```

```

        return None

    def _should_buy(self, closes: list[float]) -> bool:
        """Buy condition logic."""
        return False

    def _should_sell(self, closes: list[float]) -> bool:
        """Sell condition logic."""
        return False

    async def reset(self) -> None:
        """Reset strategy state (for replay)."""
        pass

```

Decorators

@register_strategy(name)

Registers strategy class in factory under given name.

```

@register_strategy("RSIReversion")
class RSIReversion(BaseStrategy):
    ...

```

@create_strategy_process(process_name, default_config)

Creates process wrapper for strategy and registers in process manager.

```

@create_strategy_process(
    process_name="strategy_rsi_eth_1h",
    default_config={
        "name": "rsi_eth_1h",
        "inputs": ["market.kraken.ETH-USD.candles.1h"],
        "outputs": ["ETH-USD"],
        "exchange": "paper",
        "params": {"period": 14, "upper": 70.0, "lower": 30.0, "cooldown": 2},
    },
)
class RSIReversion(BaseStrategy):
    ...

```

Strategy Configuration

StrategyConfig

Field	Type	Description
name	string	Unique strategy instance name
strategy_class	string	Strategy class name
inputs	list[str]	List of ZMQ topics to subscribe
outputs	list[str]	List of instruments for signals
exchange	string	Target exchange (paper, kraken, kraken_futures, walutomat)
params	dict	Strategy-specific parameters
wallet_public_id	string	Wallet that will execute orders for this strategy. Still defaults to empty and is NOT validated by StrategyConfig itself (the dataclass keeps an empty default pending the NOT NULL tightening migration). The non-empty requirement applies only at runtime via the caps guard for any strategy that uses <code>create_ai_review_and_await()</code> — see "AI delegate consultation" below.
operator_public_id	string	Trading-identity operator that owns this strategy instance. Empty default; validated against the launching principal's <code>operator_public_ids</code> when populated.

AI delegate consultation

Strategies that consult an AI delegate via `create_ai_review_and_await()` and then forward the approved outcome to `emit_signal(outcome=...)` MUST set `StrategyConfig.wallet_public_id` to a non-empty value. The trader-coordinator's caps gate (`TradingCapsEnforcer.guard_with_ai_review_attribution`) fail-closes early when the engine submission's `wallet_public_id` is empty:

```
CapsViolationError: missing_wallet_for_ai_review_attribution
  submission.wallet_public_id required for the strategy AI gate;
  engine state must populate wallet on the submission before
  invoking this guard
```

The error is loud and operator-actionable: it identifies the exact missing config field. Without `wallet_public_id`, the engine cannot map the AI delegate's approved trade to a wallet for cap evaluation (per-user notional / open-orders / quantity caps key on `wallet+user`). Pure non-AI strategies that never invoke `create_ai_review_and_await()` continue to run without a wallet; they bypass caps via `guard_service_principal()` per the existing audit-bypass contract.

The `ai_review_public_id` carried on the published `SignalData` is transport-only end-to-end through the ZMQ wire. The companion `ai_review_dispatch_version` (dedup key) is also transport-only — the strategy citation validator does not compare it; the bus publisher reads `dispatch_version` from the cited row at publish time so downstream caps-violation fanout events are always tagged with the row-of-record version.

Input Topics

Common formats:

- `market.{exchange}.{instrument}.candles.{timeframe}` — Candle streams
- `market.{exchange}.{instrument}.ticks` — Tick streams
- `market.{exchange}.{instrument}.trades` — Trade streams

Examples:

- `market.kraken.BTC-USD.candles.1h` — Hourly BTC/USD candles from Kraken
- `market.kraken.ETH-USD.ticks` — ETH/USD ticks from Kraken
- `market.polygon.AAPL.trades` — AAPL trade tape from Polygon

Output Topics (Signals)

Generated automatically:

- Paper: `signals.paper.{instrument}.{strategy_name}`
- Live: `signals.{exchange}.{instrument}.live`

StrategySignal Structure

```
@dataclass
class StrategySignal:
    """Trading signal."""

    instrument: str
    side: TradeSide
    strength: float
    reason: str
    price: float
    timestamp: datetime | None = None
```

Built-in Strategies

RSIReversion

Mean-reversion strategy based on RSI.

Parameters:

Parameter	Default	Description
period	14	RSI period
upper	70.0	Overbought threshold (sell)
lower	30.0	Oversold threshold (buy)
cooldown	0 (class fallback) / 2 (registered process preset)	Candles between signals

Logic:

- Buy when $RSI \leq \text{lower}$ **or** RSI was at or below lower on the previous candle ($r_{\text{prev}} \leq \text{lower}$) and price has since reversed upward.
- Sell when $RSI \geq \text{upper}$ **or** RSI was at or above upper on the previous candle ($r_{\text{prev}} \geq \text{upper}$) and price has since reversed downward.
- The previous-threshold + reversal branch lets the strategy capture exits that miss the strict edge-touch but confirm via price action.

```
from snapper.strategies.rsi import RSIRversion
```

MACDCrossover

Trend-following strategy based on MACD.

Parameters:

Parameter	Default	Description
fast	12	Fast EMA
slow	26	Slow EMA
signal_period	9	StrategySignal line period

Logic:

- Buy when MACD histogram crosses from negative to positive
- Sell when histogram crosses from positive to negative

```
from snapper.strategies.macd import MACDCrossover
```

Cointegration

Pairs trading strategy based on cointegration. Trades the spread between two cointegrated instruments, entering when the z-score crosses `entry_threshold` and exiting on mean reversion past `exit_threshold`.

```
from snapper.strategies.cointegration import CointegrationPairs
```

`CointegrationPairs` is the canonical 2-leg consumer of `MultiLegSpreadMixin` — its 2-leg invariants are enforced by `self._init_legs(expected_count=2)` and `instrument1` / `instrument2` are backwards-compatible aliases for `self.legs[0]` / `self.legs[1]`. See the next section for the generalized N-leg API.

Parameters:

Parameter	Default	Description
<code>beta</code>	0.05	Hedge ratio between legs (long leg – beta × short leg)
<code>entry_threshold</code>	2.0	Z-score threshold for entry (in standard deviations)
<code>exit_threshold</code>	0.5	Z-score threshold for mean-reversion exit
<code>lookback_window</code>	50	Window length for rolling spread statistics
<code>min_data_points</code>	30	Minimum buffered candles before any signal is emitted

Multi-leg / N-leg basket strategies

Snapper ships a reusable mixin for strategies that emit synchronized signals across `N` legs (pair-trades, baskets, calendar spreads, cross-exchange arb). The 2-leg cointegration strategy is the canonical consumer; the same mixin generalizes to $N \geq 2$.

```
from snapper.strategies.multi_leg import MultiLegSpreadMixin, resolve_legs
```

Two invocation shapes

`resolve_legs(config, expected_count=None)` accepts either:

1. **Live ZMQ process** — inputs is a length-`N` list of market-data candle topics, one per leg. Leg instruments are parsed from each topic via `parse_market_topic`.

```
config = StrategyConfig(
    name="basket_btc_eth_sol",
    inputs=[
        "market.paper.kraken.BTC-USD.candles.1h",
        "market.paper.kraken.ETH-USD.candles.1h",
        "market.paper.kraken.SOL-USD.candles.1h",
```

```

    ],
    outputs=["BTC-USD", "ETH-USD", "SOL-USD"],
    ...
)
legs = resolve_legs(config, expected_count=3)

```

2. **Direct-DB backtest** — DirectDbEngine synthesises a single candles. `{exchange}.synthetic.{timeframe}` input and passes the leg instruments through outputs.

```

config = StrategyConfig(
    name="basket_btc_eth_sol",
    inputs=["candles.kraken.synthetic.1h"],
    outputs=["BTC-USD", "ETH-USD", "SOL-USD"],
    ...
)
legs = resolve_legs(config, expected_count=3)

```

`resolve_legs` raises `ValueError` if neither shape matches or if `expected_count` is set and the resolved tuple has the wrong length.

Mixin API

Subclassing both `BaseStrategy` and `MultiLegSpreadMixin` gives you:

Attribute / method	Purpose
<code>self.legs: tuple[str, ...]</code>	Resolved leg instruments in declaration order. Populated by <code>self._init_legs(expected_count=N)</code> from <code>__init__</code> .
<code>self._partner_legs(current)</code>	Tuple of leg names other than current, in declaration order.
<code>self._partner_prices(current)</code>	<code>{leg: last_close}</code> for partners with at least one buffered candle. Partners with empty buffers are omitted (same warmup gate as 2-leg cointegration).
<code>self._emit_partner_signals(current, builder)</code>	Calls <code>builder(leg, last_close)</code> for each buffered partner and queues the result via <code>self.emit_paired_signal</code> . The backtest batch processor (<code>process_time_batch</code>) drains every queued entry in the same timestep; the live <code>_listen_loop</code> does NOT drain the queue (see "Signal pairing semantics").

Worked example — 3-leg equal-weight basket

The partner-leg emission below (`_emit_partner_signals`) is drained only by the backtest engine. Run this strategy through `process_time_batch` (direct-DB or ZMQ replay) to exercise the synchronized N-leg rebalance; the live `_listen_loop` would drop the partner legs (see "Signal pairing semantics").

```
from snapper.messaging.schemas.data import CandleData
from snapper.core.types import TradeSide
from snapper.strategies.base import BaseStrategy, StrategyConfig, StrategySignal
from snapper.strategies.decorators import register_strategy,
create_strategy_process
from snapper.strategies.multi_leg import MultiLegSpreadMixin

@register_strategy("EqualWeightBasket")
@create_strategy_process(
    process_name="basket_btc_eth_sol",
    default_config={
        "name": "basket_btc_eth_sol",
        "inputs": [
            "market.paper.kraken.BTC-USD.candles.1h",
            "market.paper.kraken.ETH-USD.candles.1h",
            "market.paper.kraken.SOL-USD.candles.1h",
        ],
        "outputs": ["BTC-USD", "ETH-USD", "SOL-USD"],
        "exchange": "paper",
        "params": {"lookback_window": 50, "z_entry": 2.0, "z_exit": 0.5},
    },
)
class EqualWeightBasket(BaseStrategy, MultiLegSpreadMixin):
    """Z-score each leg vs the basket mean and rebalance when it diverges."""

    def __init__(self, config: StrategyConfig) -> None:
        super().__init__(config)
        self.lookback_window = int(self.params.get("lookback_window", 50))
        self.z_entry = float(self.params.get("z_entry", 2.0))
        self.z_exit = float(self.params.get("z_exit", 0.5))
        self._init_legs(expected_count=3)

    async def on_candle(self, instrument: str, candle: CandleData) ->
StrategySignal | None:
        partner_prices = self._partner_prices(instrument)
        if len(partner_prices) < len(self.legs) - 1:
            return None

        basket_mean = (candle.close + sum(partner_prices.values())) /
len(self.legs)
        z = (candle.close - basket_mean) / basket_mean

        if abs(z) < self.z_entry:
            return None

        side: TradeSide = "sell" if z > 0 else "buy"
        primary = StrategySignal(
            instrument=instrument,
            side=side,
            strength=min(1.0, abs(z) / self.z_entry),
```

```

        price=candle.close,
        reason=f"basket_z={z:+.2f}",
    )
    opposite: TradeSide = "buy" if side == "sell" else "sell"
    self._emit_partner_signals(
        instrument,
        lambda leg, price: StrategySignal(
            instrument=leg,
            side=opposite,
            strength=primary.strength,
            price=price,
            reason=f"basket_partner_of_{instrument}",
        ),
    )
    return primary

```

Signal pairing semantics

BaseStrategy.emit_paired_signal(signal) enqueues partner-leg signals on the same timestep as the primary signal returned from on_candle. The drain is **backtest-only**: the batch processor (process_time_batch, used by the direct-DB and ZMQ replay backtest paths) drains the queue immediately after the primary signal, so on those paths the full N-leg basket is processed as one synchronized rebalance — no race between legs.

The live/paper production BaseStrategy._listen_loop does NOT drain the pending-signal queue: it emits only the single signal returned by on_candle. Paired live emission is not yet wired, so partner-leg signals queued via emit_paired_signal / _emit_partner_signals are dropped when a multi-leg strategy runs live. Treat the basket/N-leg pairing API as backtest-only until live draining is implemented.

on_tick callbacks do not use the pairing queue; tick-level pairing must be implemented explicitly by the strategy.

When NOT to use the mixin

- Single-leg strategies (RSI, MACD, VWAP) — there are no partners to pair with; subclassing the mixin adds no value.
- Strategies whose legs use heterogeneous timeframes — legs assumes one timeframe per strategy process. Run one process per timeframe and coordinate via signals.
- Calendar spreads where one leg trades on a derived feed (e.g. funding-rate vs spot) — the input topic shape doesn't fit resolve_legs; subclass BaseStrategy directly.

Data Access

Candle Buffer

```
async def on_candle(self, instrument: str, candle: CandleData) ->
StrategySignal | None:
    candles = self.candle_buffer.get(instrument, [])

    recent = candles[-20:]

    import pandas as pd
    closes = pd.Series([c.close for c in candles])
```

CandleData

```
from snapper.messaging.schemas.data import CandleData
```

Fields:

Field	Type	Description
type	string	"candle"
instrument	string	Trading pair symbol (e.g. BTC-USD)
exchange	string	Source exchange
timeframe	string	Candle duration (1m, 1h, 1d, ...)
open_at	datetime	Exchange-provided candle interval start time
open	float	Opening price
high	float	High price
low	float	Low price
close	float	Closing price
volume	float	Total traded volume
vwap	float None	Volume-weighted average price (optional)
trades	int None	Number of trades in the candle (optional)

Technical Indicators

RSI

```
from snapper.indicators.rsi import rsi
import pandas as pd

closes = pd.Series([c.close for c in candles])
rsi_values = rsi(closes, period=14)
current_rsi = rsi_values.iloc[-1]
```

MACD

```
from snapper.indicators.ta_lib_adapter import macd

macd_line, signal_line, histogram = macd(closes, fast=12, slow=26, signal=9)
```

TA-Lib (via adapter)

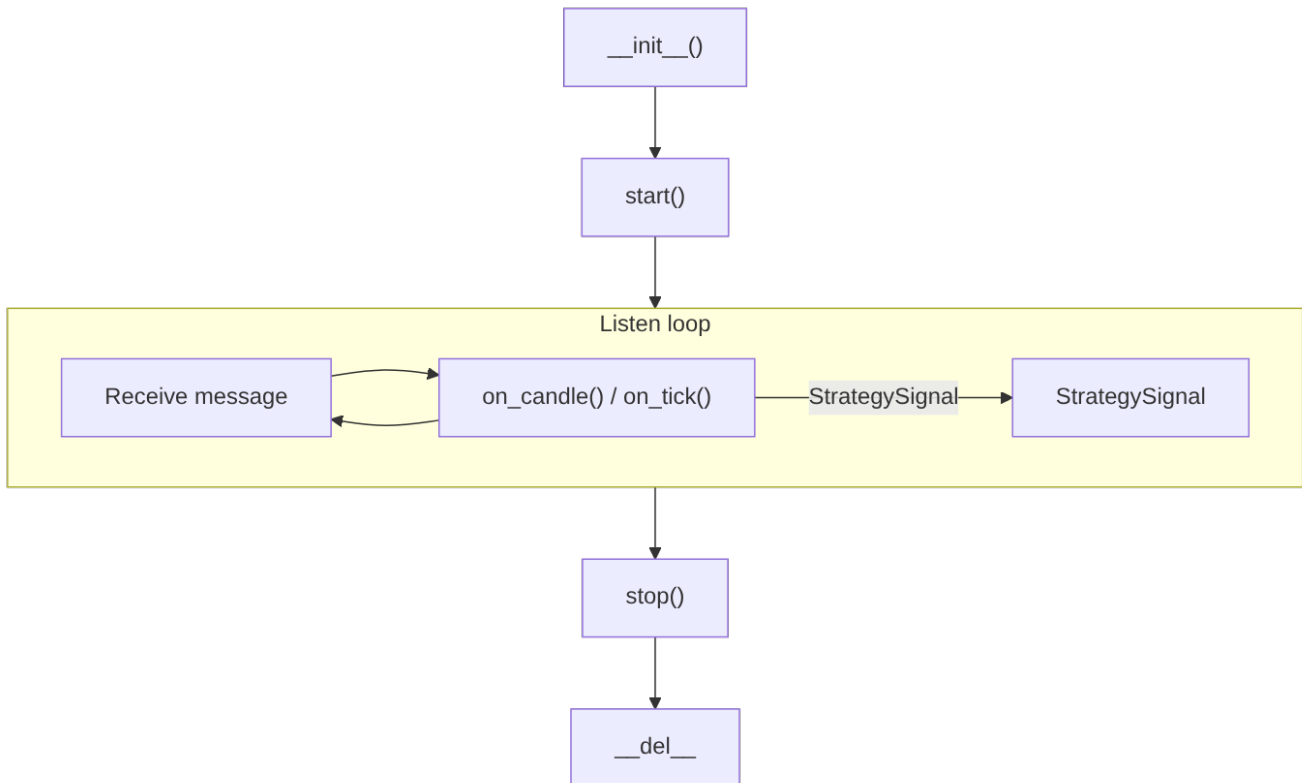
```
from snapper.indicators.ta_lib_adapter import rsi, macd
```

Configuration Validation

Framework automatically validates:

1. **Strategy name** — cannot be empty
2. **Inputs** — at least one required
3. **Outputs** — at least one instrument must be defined
4. **Exchange** — must be one of: paper, kraken, kraken_futures, walutomat
5. **Instruments** — must be available on selected exchange
6. **Paper/live mixing** — mixing paper inputs with live exchange not allowed

Strategy Lifecycle



BaseStrategy does not expose a separate `cleanup()` hook — teardown happens in `stop()` (cancel heartbeat + listen tasks, unsubscribe inputs, close the ZMQ publisher + context) and the implicit `__del__` (cancel any surviving tasks, close subscriber/publisher sockets, terminate the ZMQ context).

Running Strategies

Via Process Manager (recommended)

Strategies are automatically managed by the server:

```
snapper server
```

Strategies can be enabled/disabled in the dashboard.

Standalone (for testing)

```
import asyncio
from snapper.strategies.factory import StrategyFactory
from snapper.strategies.base import StrategyConfig

# Importing the strategy module fires the @register_strategy
```

```

# side effect that populates the factory's registry. Snapper's
# strategies/__init__.py is intentionally empty (no re-exports
# per project convention), so each concrete strategy must be
# imported by hand before the factory can resolve it by name.
import snapper.strategies.rsi # noqa: F401 # register RSIReversion

config = StrategyConfig(
    name="test_rsi",
    strategy_class="RSIReversion",
    inputs=["market.kraken.BTC-USD.candles.1h"],
    outputs=["BTC-USD"],
    exchange="paper",
    params={"period": 14, "upper": 70, "lower": 30},
)

factory = StrategyFactory()
strategy = factory.create_strategy(config)

async def main() -> None:
    # `BaseStrategy.start()` is non-blocking – it wires sockets and
    # spawns the listen + heartbeat tasks, then returns. Keep the
    # event loop alive while the background tasks process data.
    await strategy.start()
    try:
        await asyncio.Event().wait() # block until Ctrl-C / cancel
    finally:
        await strategy.stop()

asyncio.run(main())

```

Testing Strategies

Unit Tests

```

import pytest
from snapper.strategies.rsi import RSIReversion
from snapper.strategies.base import StrategyConfig
from snapper.messaging.schemas.data import CandleData

@pytest.fixture
def rsi_strategy() -> RSIReversion:
    config = StrategyConfig(
        name="test_rsi",
        strategy_class="RSIReversion",
        inputs=["market.kraken.BTC-USD.candles.1h"],
        outputs=["BTC-USD"],
        exchange="paper",
        params={"period": 14, "upper": 70, "lower": 30},
    )
    return RSIReversion(config)

async def test_buy_signal_on_low_rsi(rsi_strategy: RSIReversion) -> None:
    # Prepare data with low RSI

```

```
...
signal = await rsi_strategy.on_candle("BTC-USD", candle)
assert signal is not None
assert signal.side == "buy"
```

Backtesting

Use replay data from the dedicated paper.{source_exchange} market sub-prefix — the strategy listens to the same market.* shape as in live mode, only the source exchange is swapped for paper:

```
default_config={
    "inputs": ["market.paper.kraken.BTC-USD.candles.1h"],
    "exchange": "paper",
    ...
}
```

replay.* is not a valid ZMQ topic — BaseStrategy._listen_loop filters incoming frames to market.* and routes them through BaseStrategy._dispatch_market_data. The canonical builder is market_topic(...) in src/snapper/messaging/topics/builders.py; paper topics are built by passing exchange="paper" together with the venue under source_exchange=... (there is no separate paper_market_topic symbol).

Trailing Stop Plans

Trailing stops are server-side evaluated execution plans that ratchet a stop price as the market moves favorably and fire a market close when the stop is breached. Attach a trailing stop to an open position_cycle via POST /api/trailing-stops (see docs/api.md).

Parameters

Field	Type	Bounds	Meaning
trailing_pct	float	$0 < x < 100$	Distance from peak price, as percentage. 5.0 means "trigger 5 % below the highest favorable price seen".
min_lock_pct	float	$0 \leq x < 100$ (default 0)	Profit threshold the market must cross before the trailing stop arms. 0 means "arm immediately on attach"; 5.0 means "wait until price has moved 5 % in your favor before tracking a stop".

Behavior

- **Peak ratchet:** every tick where the last price moves favorably updates the peak; the stop level recomputes as $\text{peak} * (1 - \text{trailing_pct}/100)$ for longs or $\text{peak} * (1 + \text{trailing_pct}/100)$ for shorts.
- **Stop monotonicity:** for longs, the stop only moves UP (never down); for shorts the stop only moves DOWN. An adverse tick never loosens the stop.
- **min_lock dead zone:** while $\text{min_lock_pct} > 0$ and the market has not yet crossed that threshold, the trailing stop is attached and armed but NOT tracking — a close is NOT emitted no matter how far the price moves against you. A decision warning is logged on the `trailing_stop_created` row at create time.
- **Single trailing stop per cycle:** enforced by the partial unique index `uq_ep_active_trailing_stop_per_cycle`. A second create on the same open cycle returns HTTP 409.
- **Cycle close sweeps the plan:** if the cycle closes externally (reconciliation flat, manual close, bracket fill), `_sweep_cycle_closures()` cancels the trailing stop on the next clock tick.
- **Checkpoint persistence:** `peak_price` and `current_stop` are persisted every 10 s via the plan-executor checkpoint loop. On restart, state is restored and `peak_price` is floored at `entry_price` (long: $\max(\text{peak}, \text{entry})$; short: $\min(\text{peak}, \text{entry})$).
- **Downtime semantics:** if the service is down when the true market price crosses the stop, the stop fires on the first tick received after restart. There is no placed order on the exchange — the evaluator is purely server-side.

Interaction with brackets

A position cycle may carry both a bracket and a `trailing_stop` plan at the same time (the two partial unique indexes are disjoint per `plan_type`). Whichever fires first closes the cycle; the other is swept by the cycle-close handler.

Limitations

- `entry_price` and `total_quantity` are frozen at attach time. Dollar-cost averaging (scaling into a position) requires POST /cancel + re-attach after the new average is known.
- No exchange-side native trailing stops — every decision is made by `PlanExecutorService` against live ticks.
- Futures only (same capability gate as brackets: `supports_reduce_only`). Not supported on Kraken spot.
- No breakeven-move feature (ratchet stop to entry after N % profit).
- No time-based activation (arm after N minutes regardless of price).

- Live rollout gated on the same follow-ups (wallet-safe routing + orphan-cycle admin) that brackets depend on.

Cross-asset market-data pattern

Snapper enables strategies to subscribe to a market-data-only feed (`SymbolExchangeCapability.can_trade=False`) and emit signals whose target is an execution-capable instrument on a different venue. Kraken FCM index futures (`kraken_equities`) are the primary driver: they publish ~10-minute-delayed candles + ticks but have no order API.

Rules

- Strategies MAY subscribe to any combination of `can_market_data=True` topics regardless of the `can_trade` state of the source instrument.
- Every emitted `StrategySignal.instrument` MUST point at an `can_trade=True` instrument on the strategy's configured exchange. Snapper's order-entry capability guard (`snapper.server._capability_guard.require_tradable`) rejects submits against `can_trade=False` pairs with HTTP 422 `error_code='instrument_market_data_only'` — the signal would be dropped before reaching the venue.
- Tick-driven strategies that consume delayed feeds MUST gate on `TickData.is_delayed=True` before treating the price as current. Candle-driven cross-asset strategies inherit the source-feed latency implicitly; document the lag in the live runbook.
- Output-coverage enforcement at startup is conditional: when both `operator_public_id` and `wallet_public_id` are populated on the process config, `_enforce_strategy_outputs_covered` checks that every instrument in `StrategyConfig.outputs` falls within the operator's active scope grants. The empty-default path (neither field set) is permitted for backwards compatibility, and operator-only launches skip the output-coverage check.

Reference implementation

See `src/snapper/strategies/examples/tradfi_observe_crypto_execute.py` for a minimal EMA-crossover strategy that observes MNQM6-CME on `kraken_equities` and targets BTC-USD on kraken. The file is NOT registered with the process registry — it ships as a copy-paste starting point. Activation steps live in the module docstring; `tests/meta/test_reference_strategy_not_registered.py` enforces the non-registration invariant so future maintainers cannot accidentally promote the illustration into a live process.